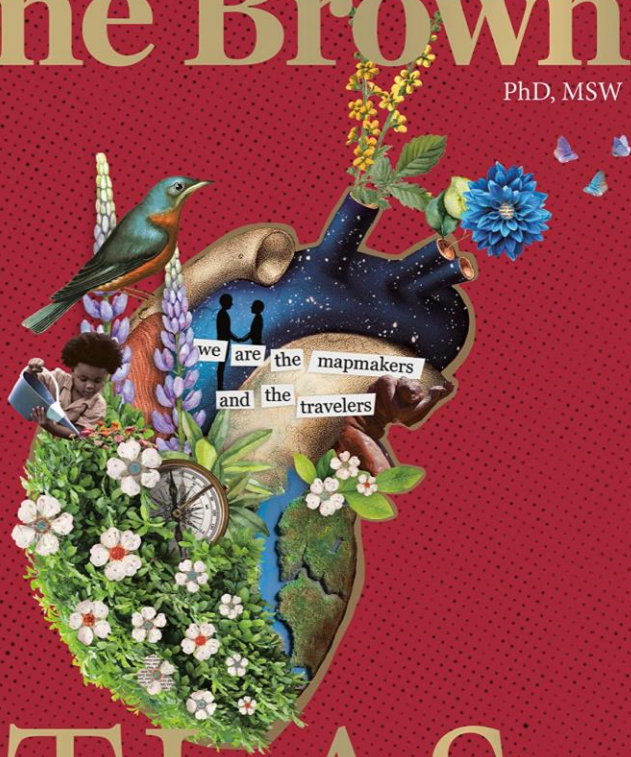


#1 *New York Times* bestselling author

# Brené Brown

PhD, MSW



# ATLAS OF THE HEART

Mapping Meaningful Connection and  
the Language of Human Experience

## HEART and LUNGS

OMFS 571 Clinical Anatomy

Karisa Yamamoto, DDS

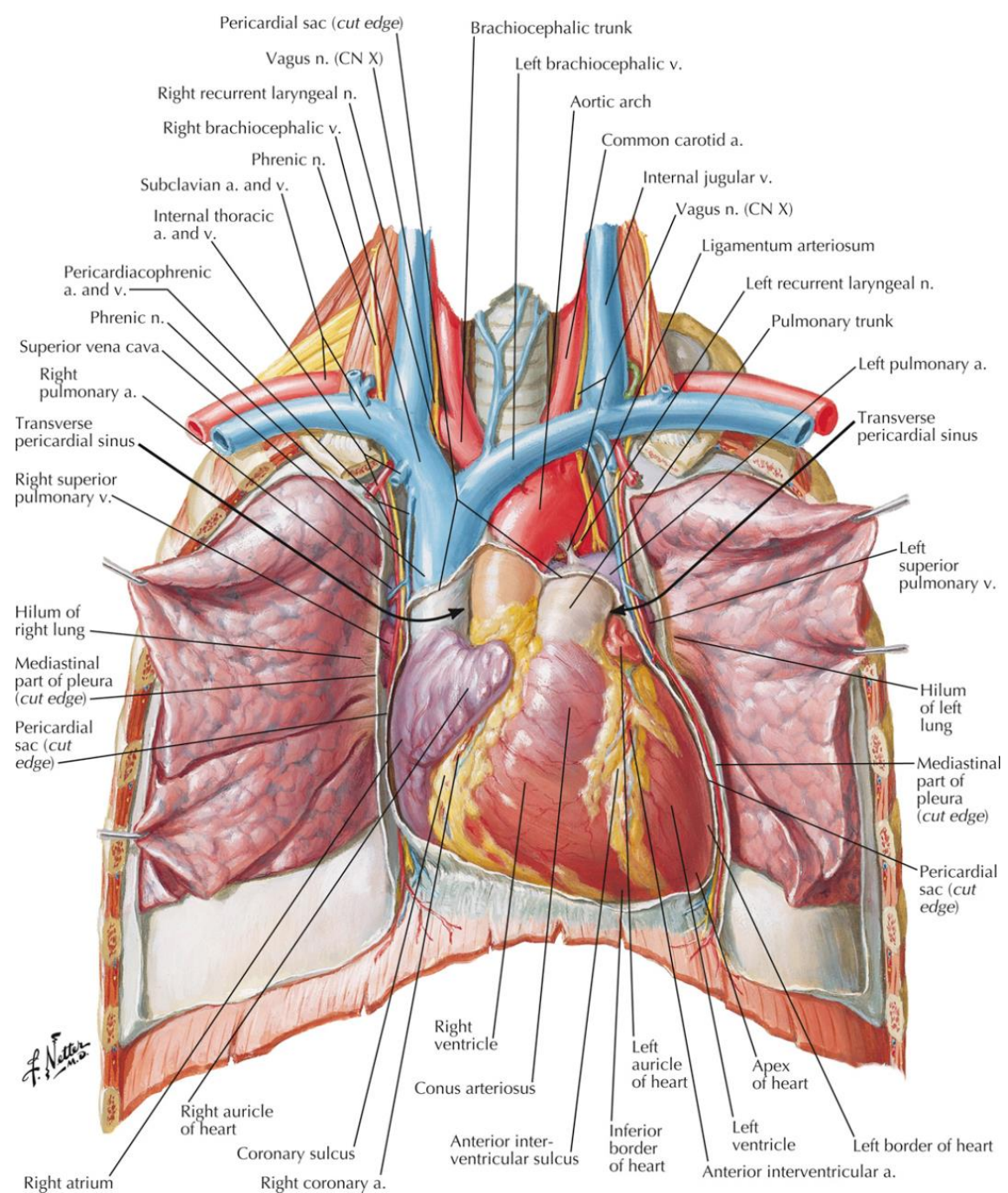
# Heart: Learning Outcomes

- Describe the blood flow physiology of the heart and the anatomical structures involved
- Describe the structures of the heart that are significant in embryonic development
- Describe the electrophysiology of the heart
- Describe the coronary circulation
- Describe the clinical anatomical correlations of valvular heart disease, infective endocarditis, cardiac referred pain and myocardial infarction

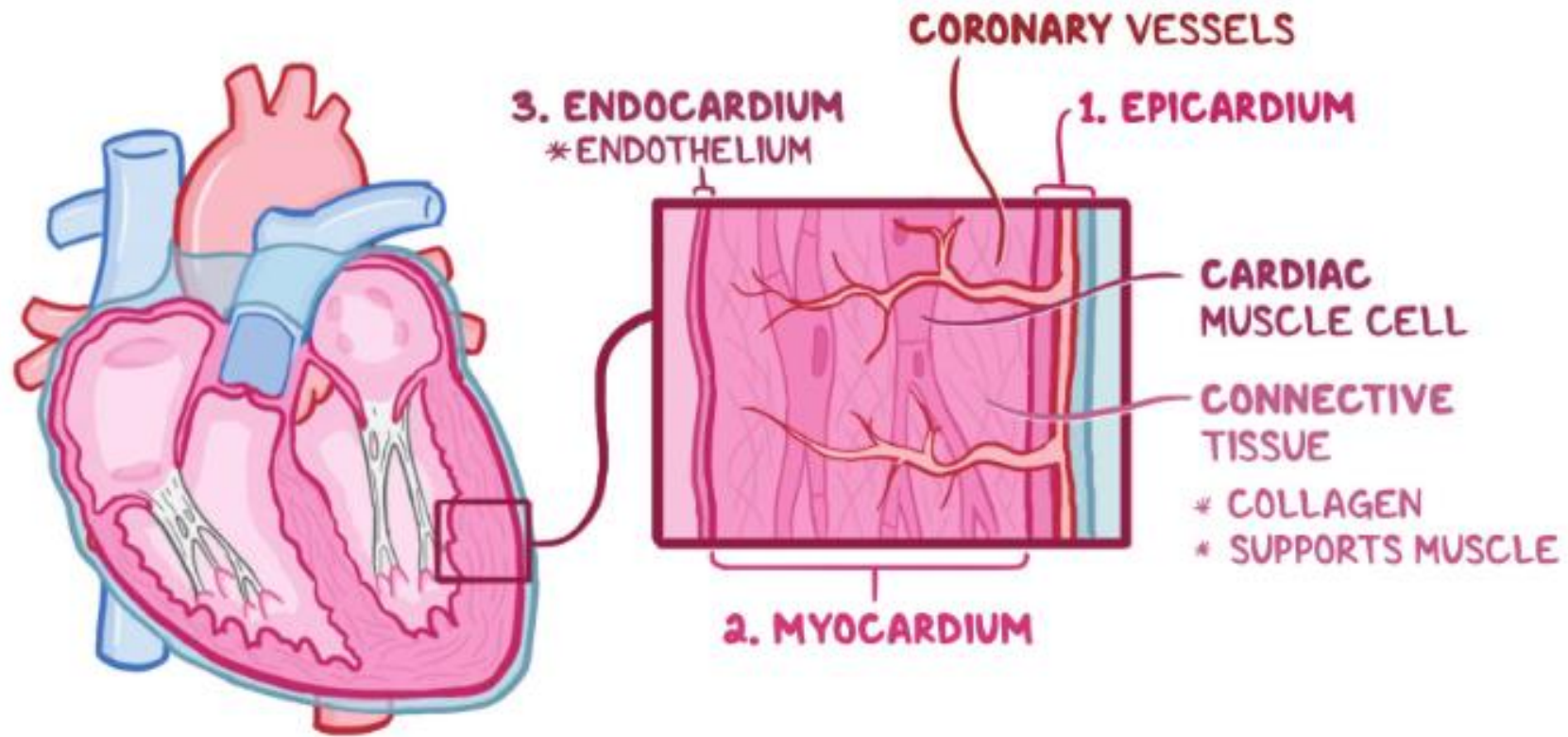


# ANATOMY OF THE HEART

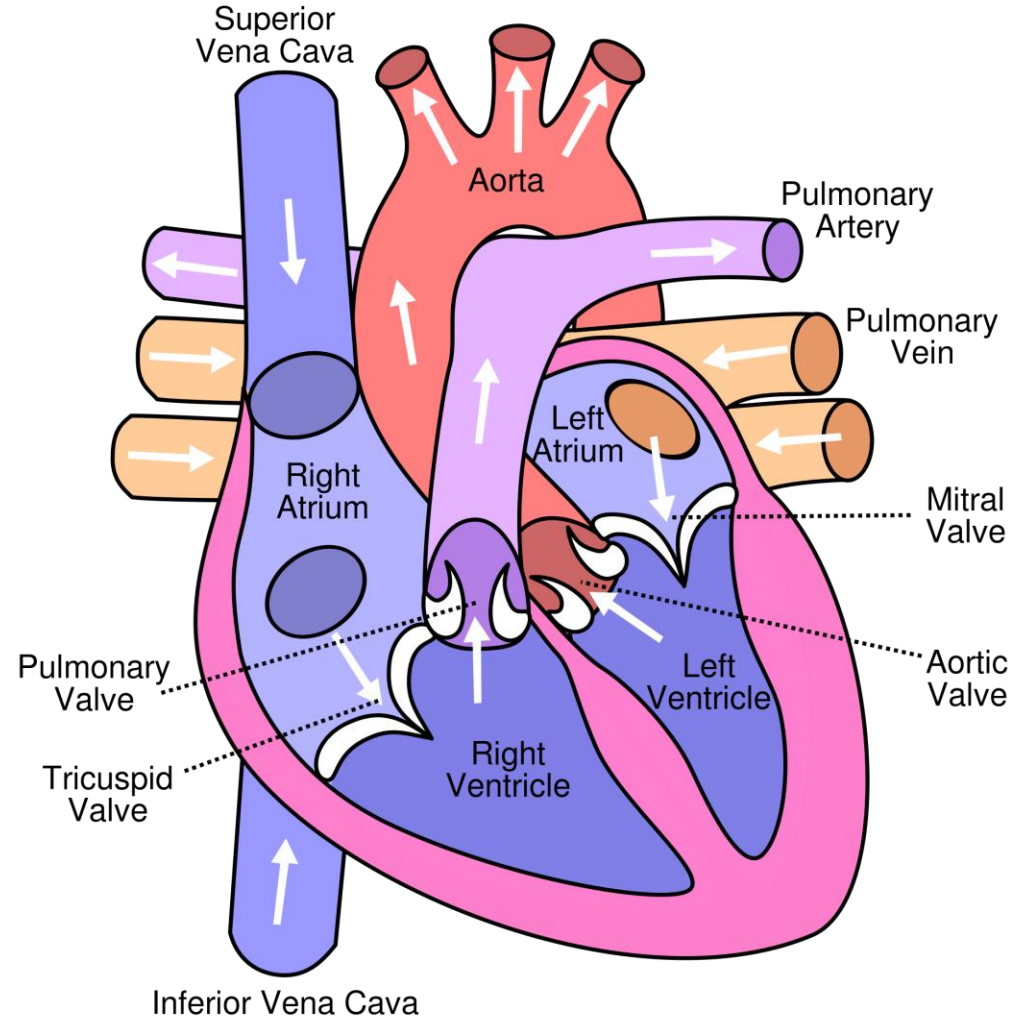
# The Heart



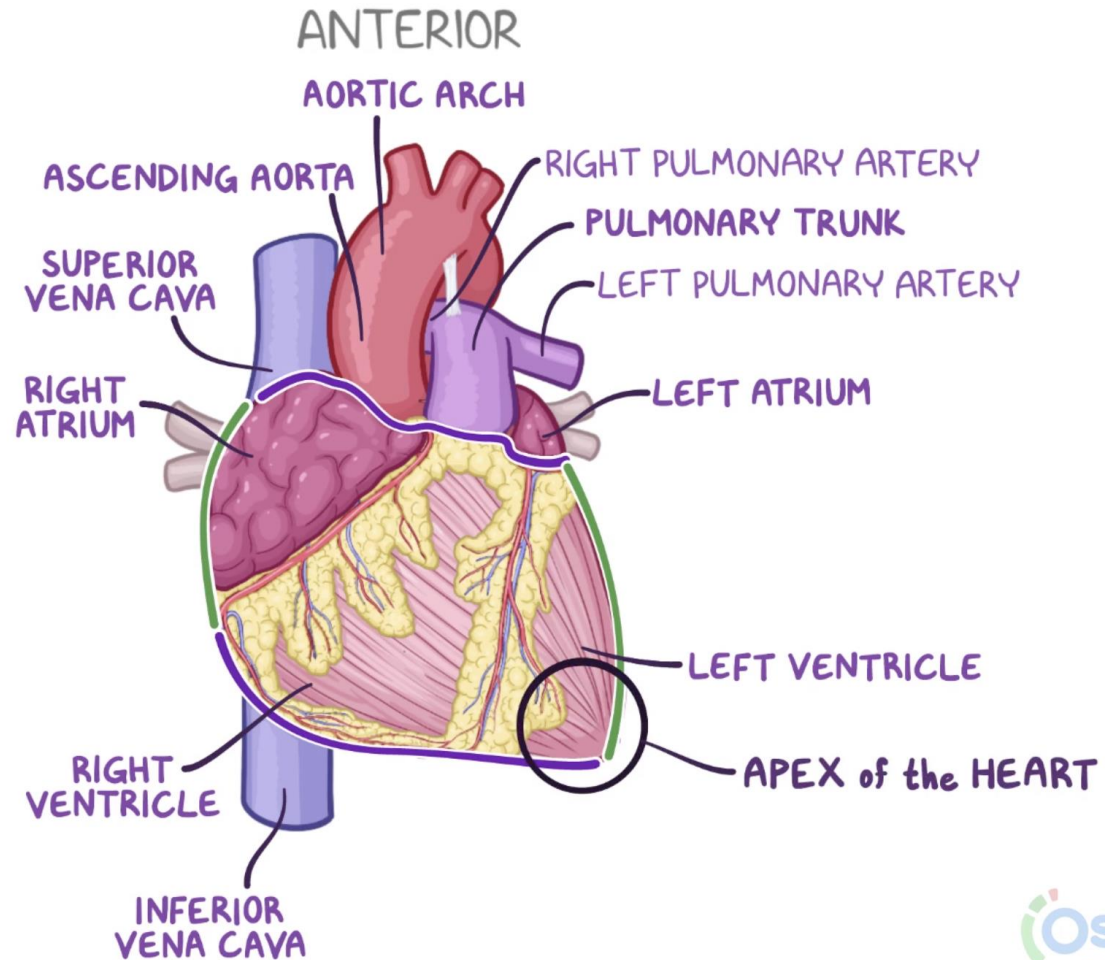
# Layers of the Heart



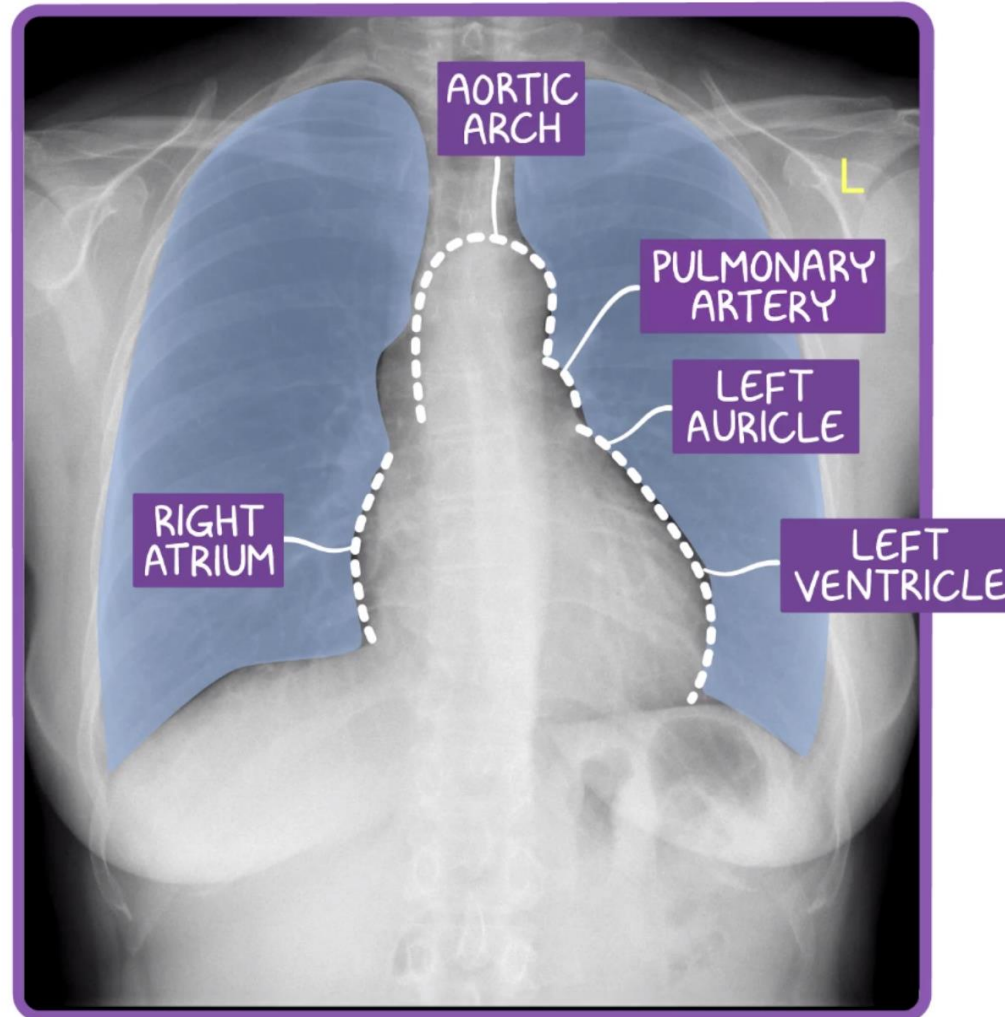
# Flow of Blood Through Heart



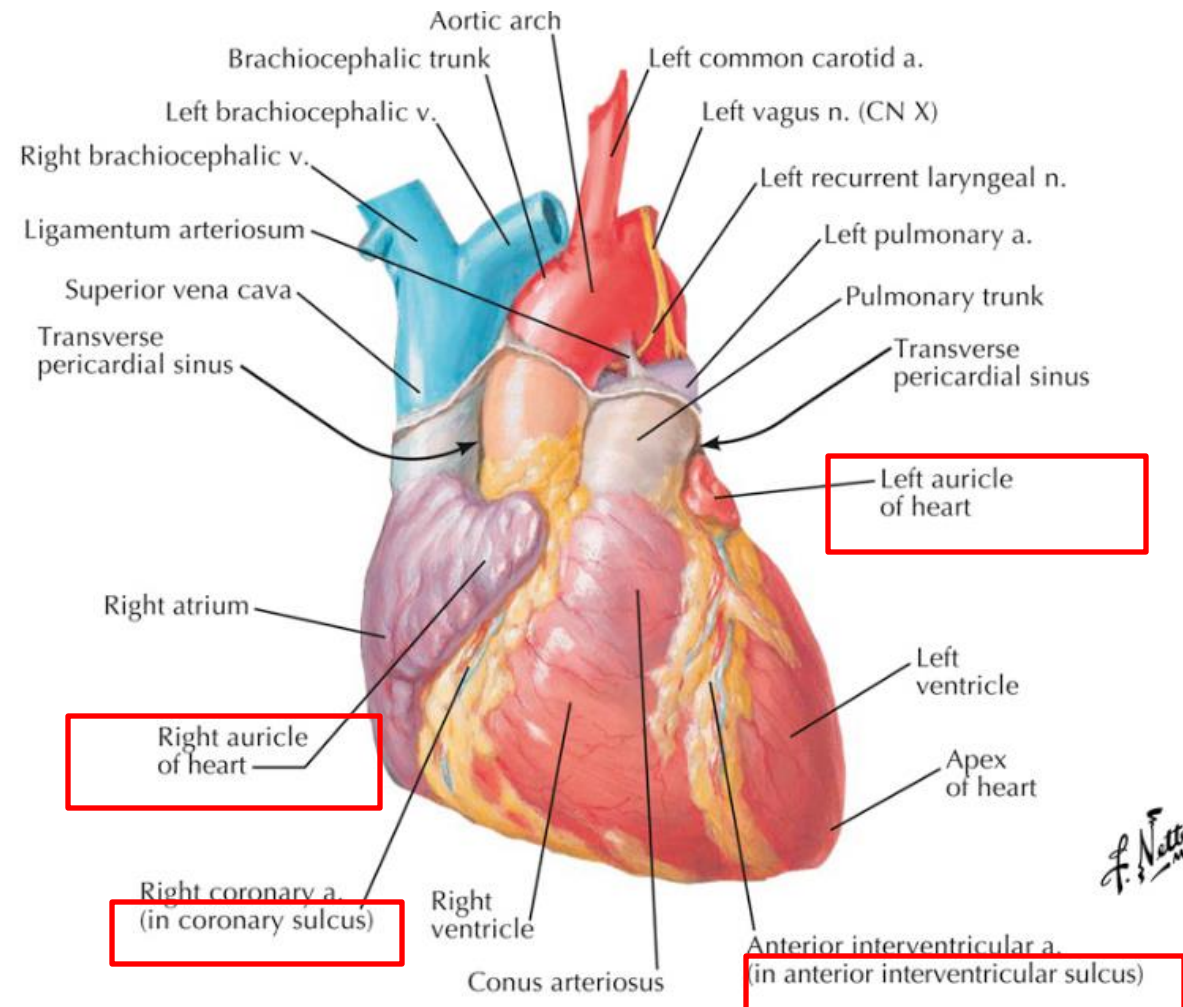
# Borders of the Heart



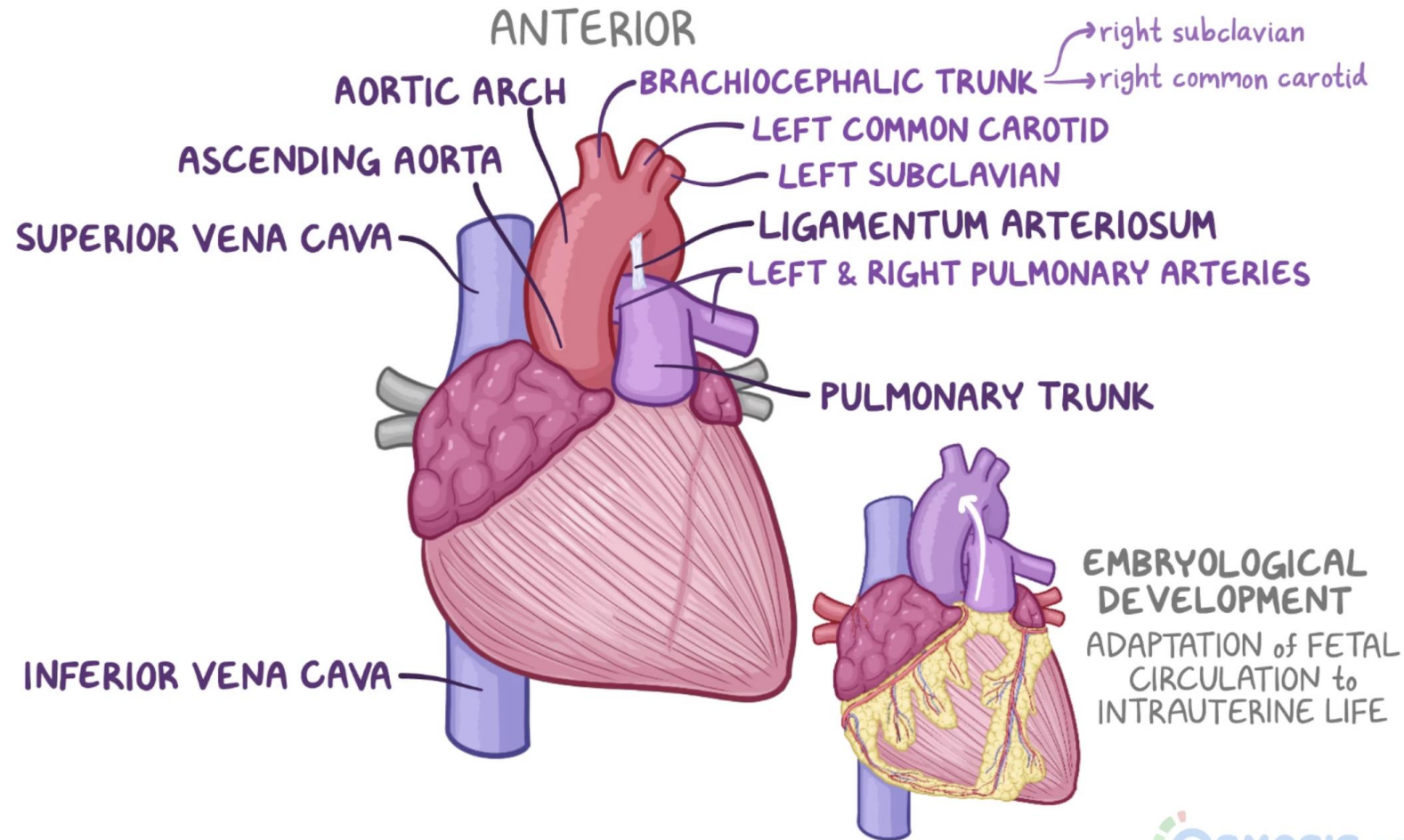
# Chest X-Ray: Borders of Heart



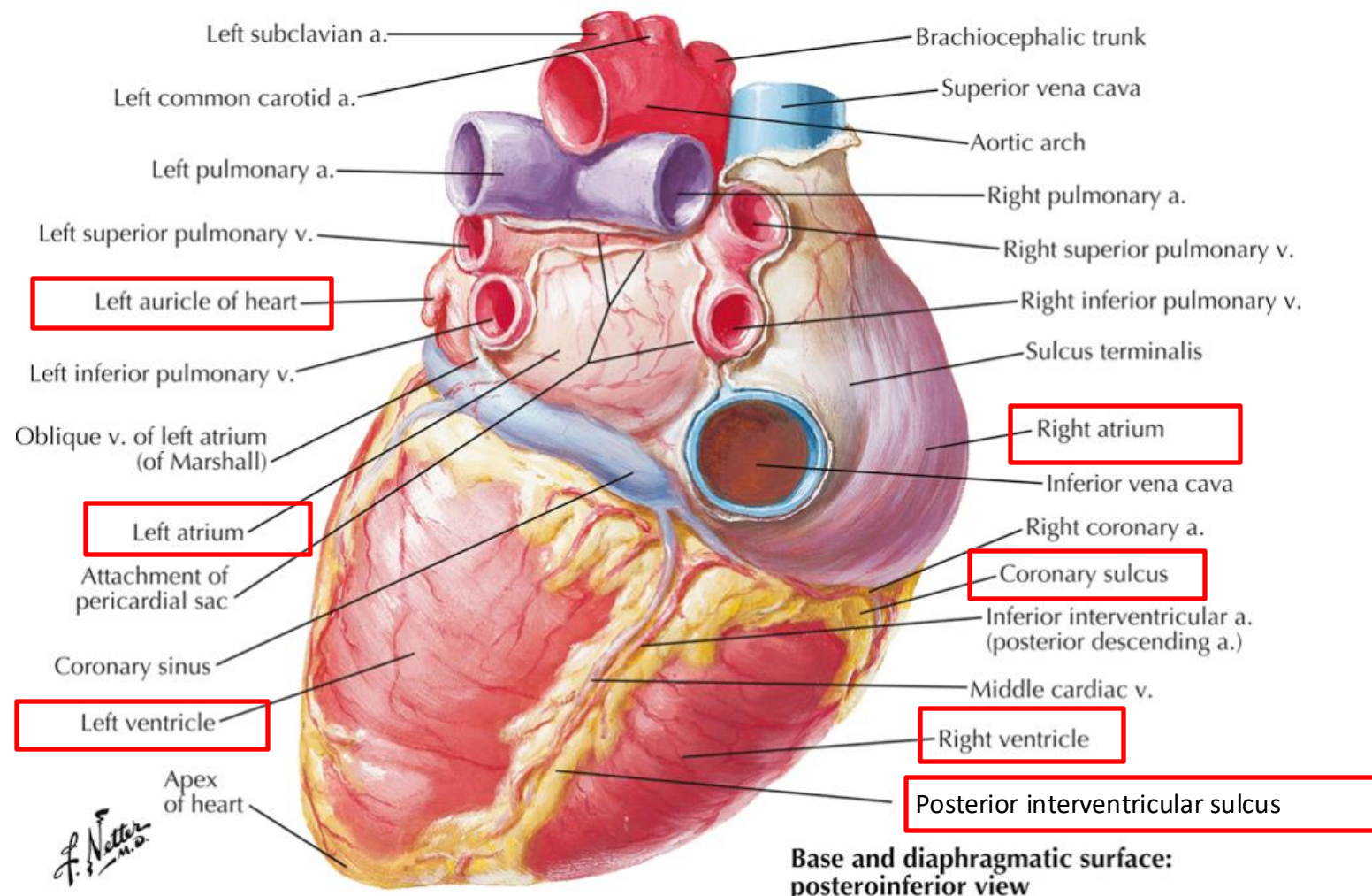
# Surface Anatomy of Heart: Anterior View



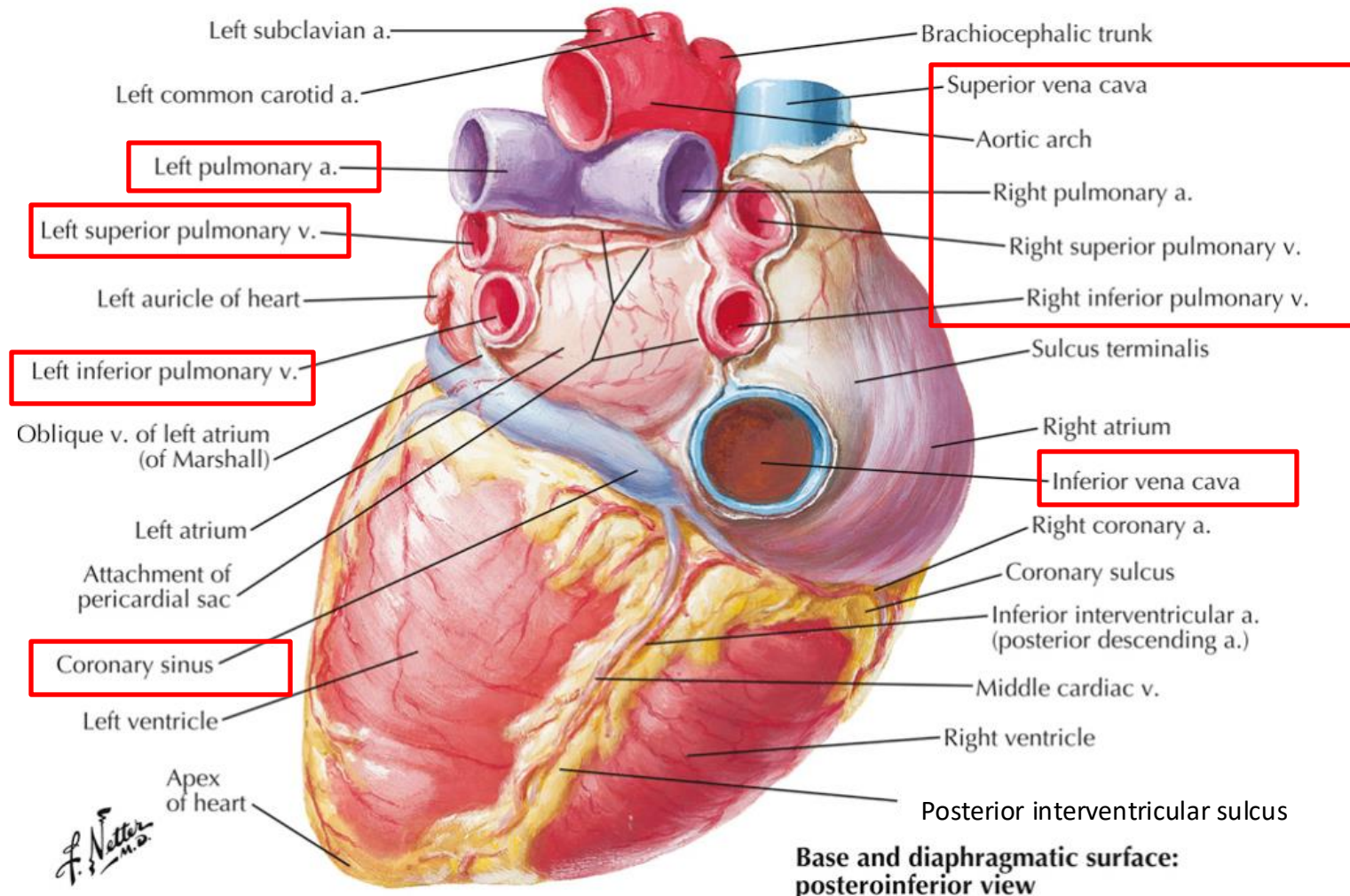
# Great Vessels of the Heart



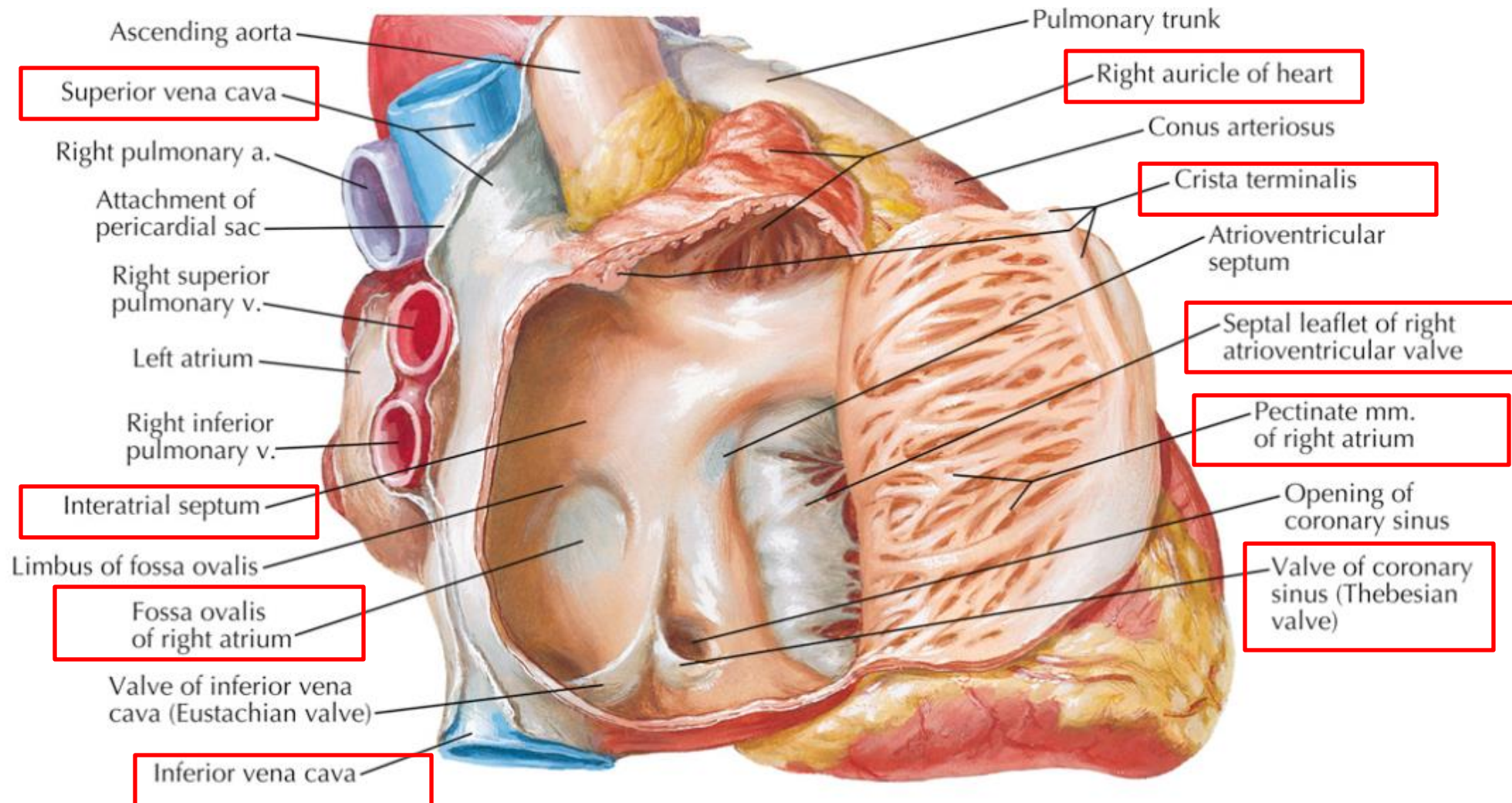
# Surface Anatomy of Heart: Posterior View

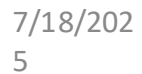


# Pulmonary Vessels

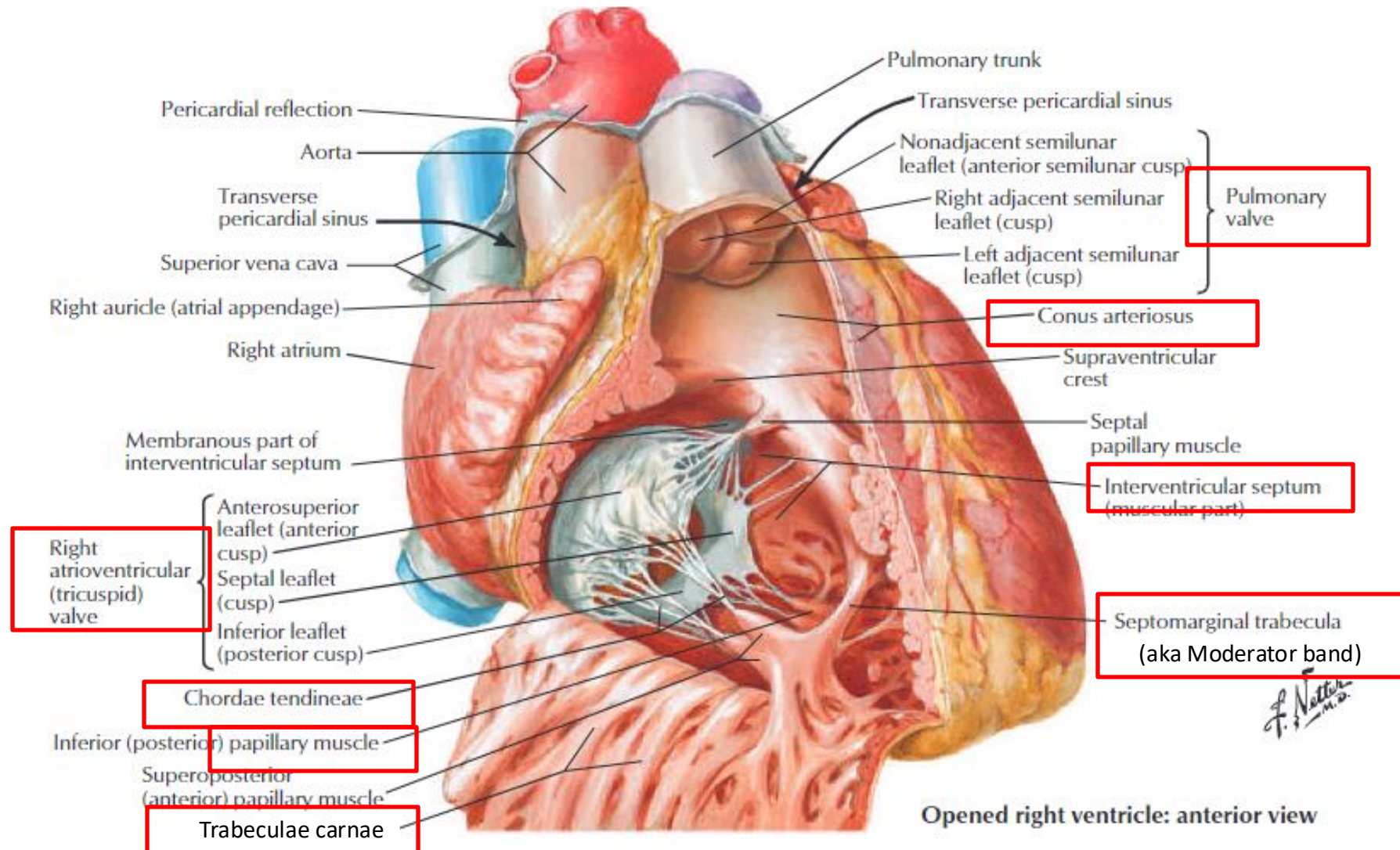


# Internal View of Right Atrium





# Internal View of the Right Ventricle



# Ventricle Dissection

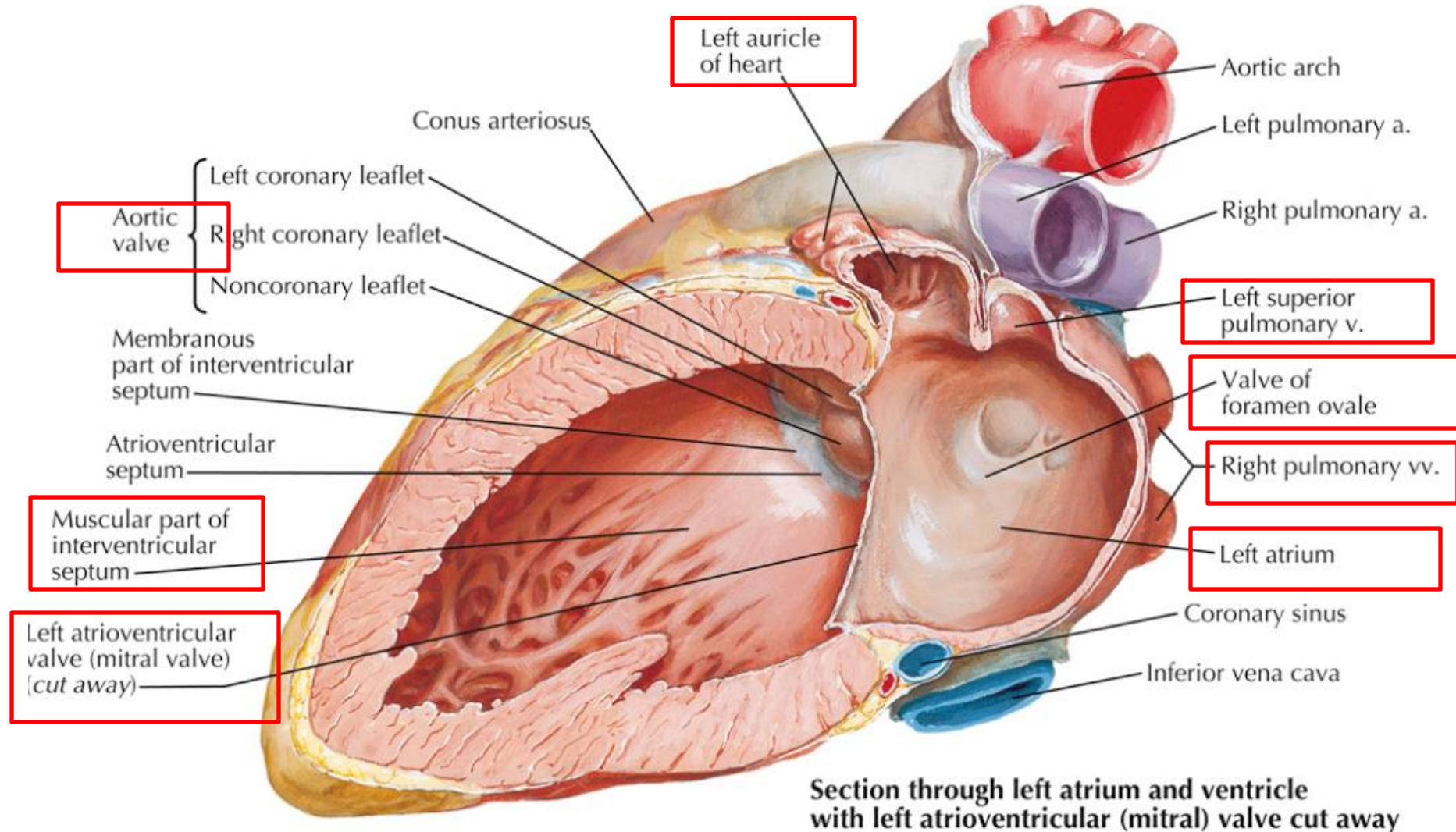


**Chordae tendineae**

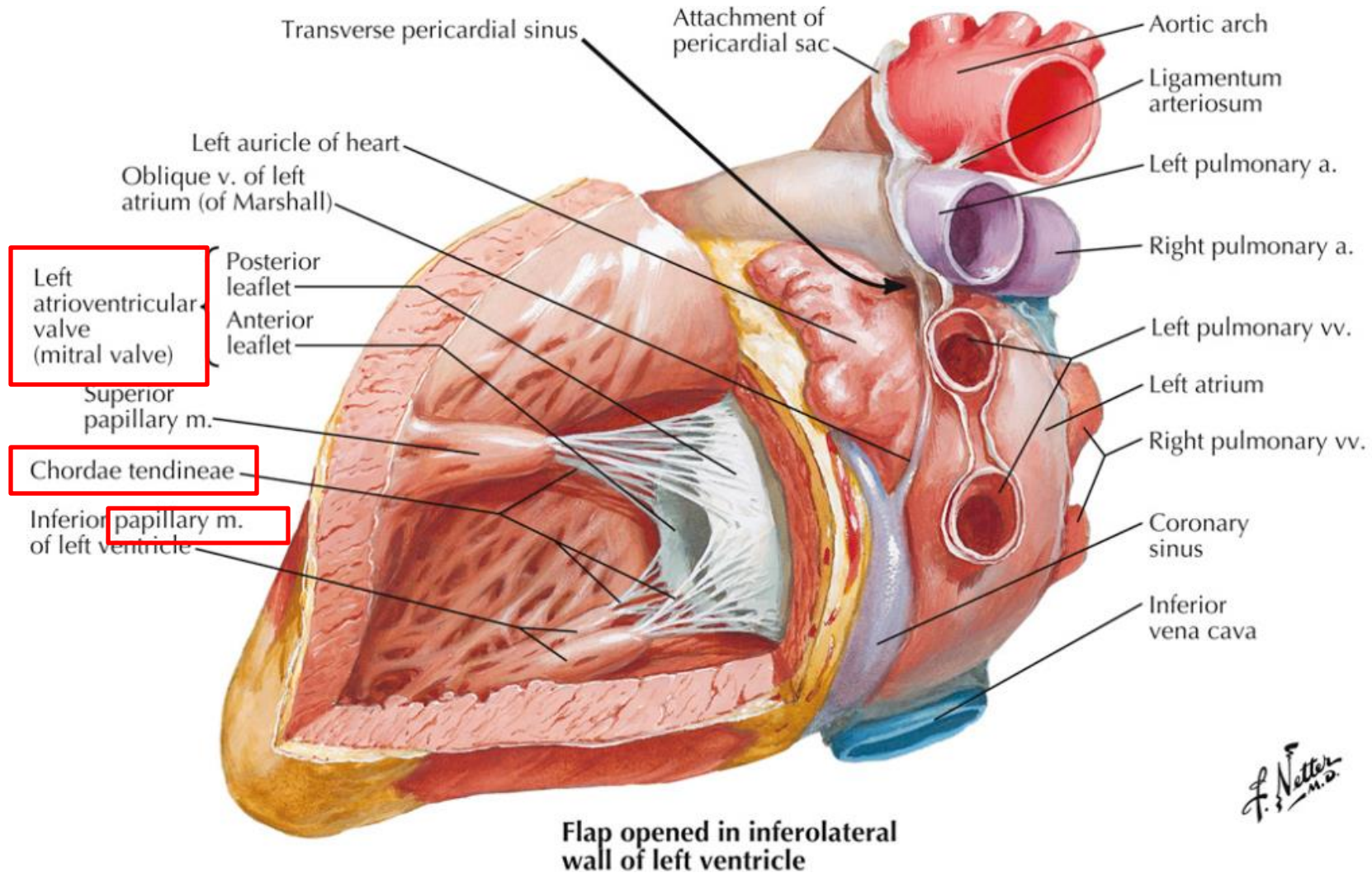
**Papillary muscles**

**Trabeculae carnae**  
*(ridges)*

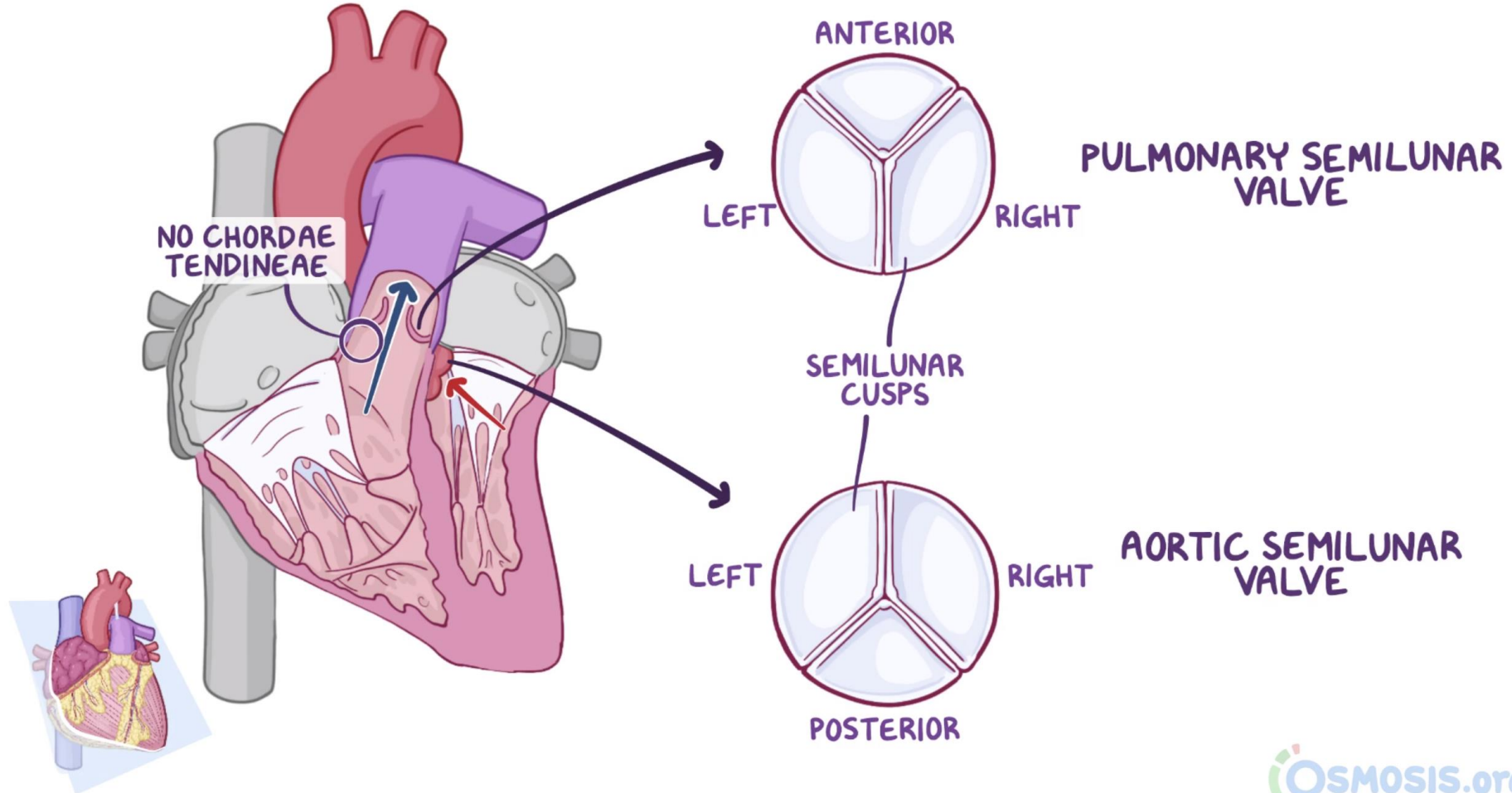
# Internal View of Left Atrium and Ventricle



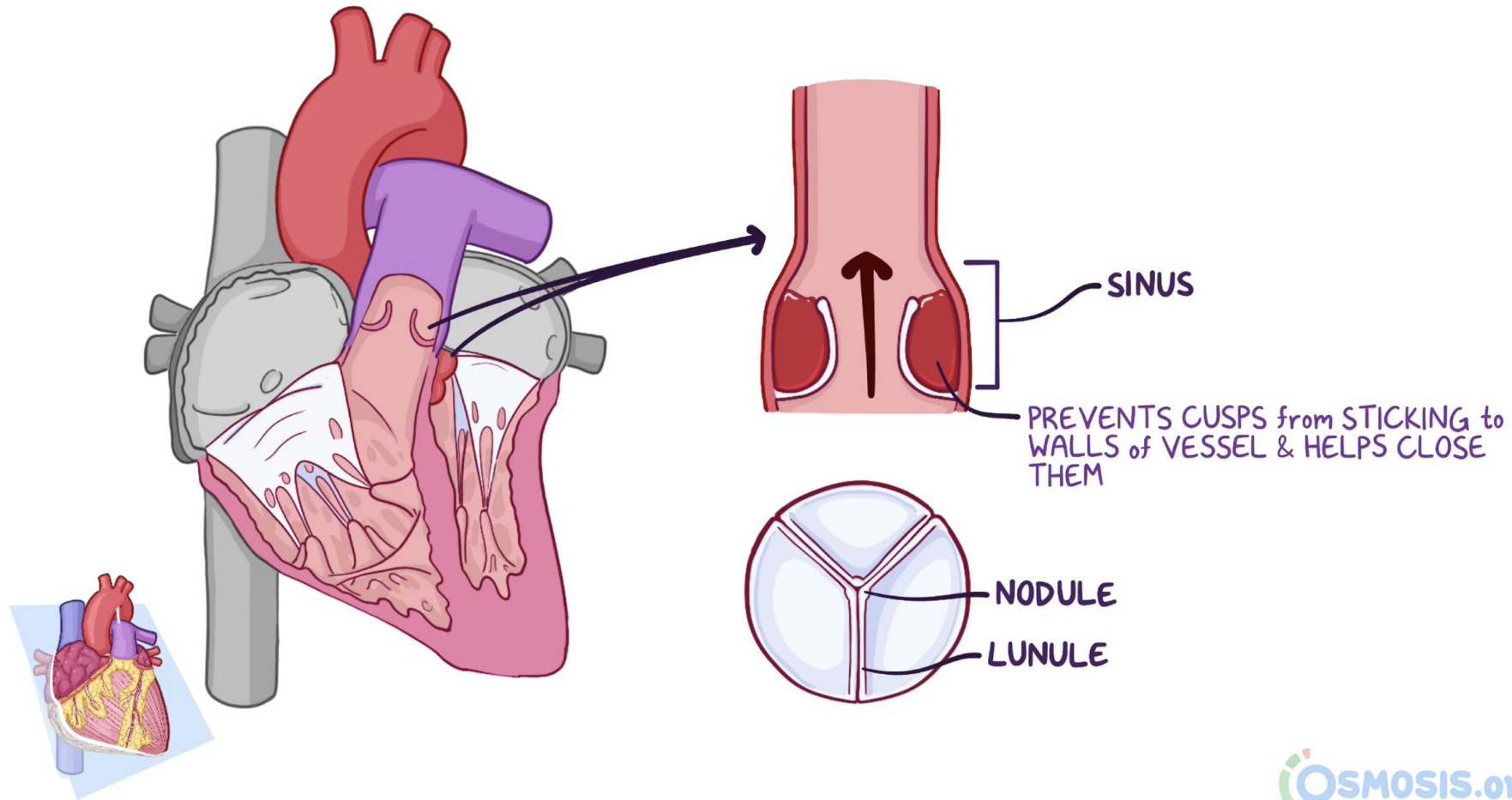
# Internal View of Left Ventricle



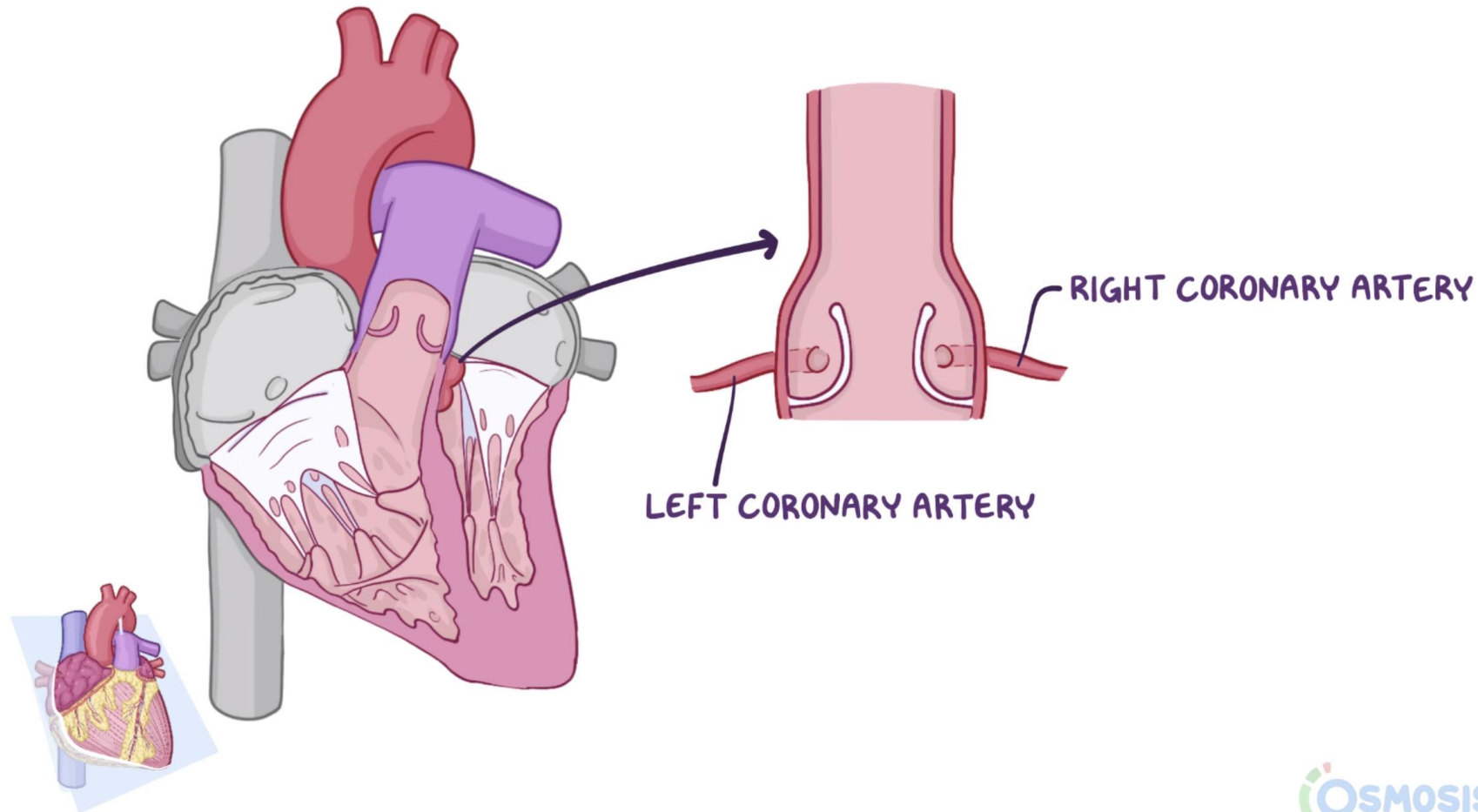
# Semilunar Valves



# Semilunar Valves



# Origin of the Coronary Arteries



## Clinical Application: Valvular Heart Disease

- **Stenosis** – narrowing and failure of a valve to open fully, slowing blood flow from a chamber
- **Insufficiency** (aka regurgitation or incompetence) - failure of the valve to close, allowing blood to flow back into the chamber it was ejected from
- **Murmurs** – audible vibrations caused by the turbulent blood flow of valvular stenosis or insufficiency
- Damaged or defective heart valves can be replaced surgically with **prosthetic valves**



### STENOSIS

\* NARROWING of the VALVULAR ORIFICE



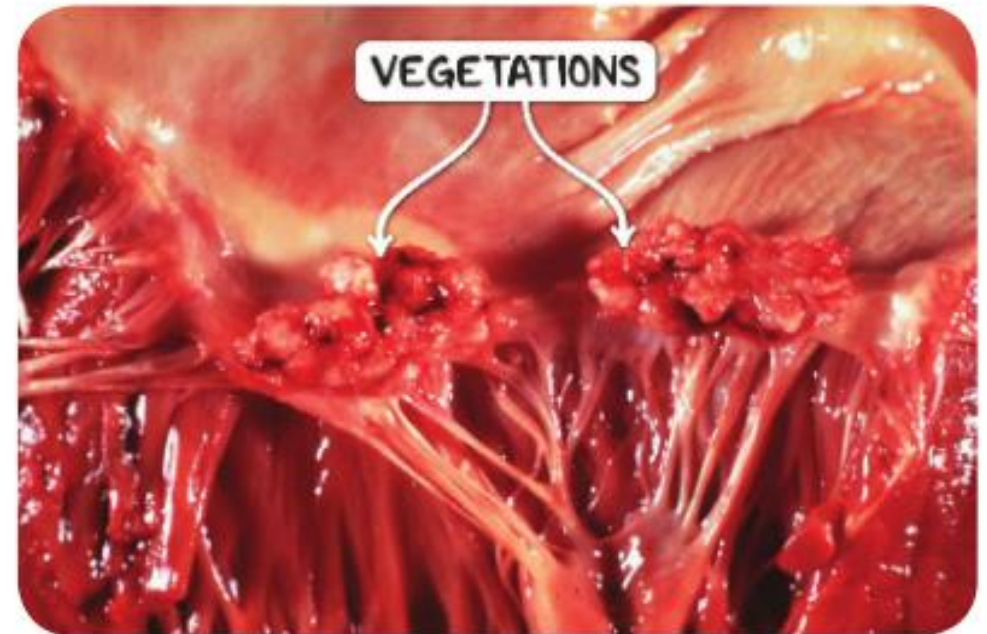
### INSUFFICIENCY or REGURGITATION

\* LEAFLETS FAIL to CLOSE CORRECTLY



## Clinical Application: Infective Endocarditis

- Infection caused by **bacteria that enter the bloodstream** and settle in the **endocardium** (inner lining of the heart), heart valves or blood vessels
- Tissue that experiences **turbulent blood flow** most at risk (valves), **mitral and aortic valve** most commonly affected
- Can be caused by **poor oral hygiene**, toothbrushing causing **minor injury** in the mouth, **dental procedures**
- **Symptoms:** fever, heart murmur, tachycardia, cough, aching joints/muscles



**Figure 12.2** Bacterial vegetations on the mitral valve in endocarditis.

# Clinical Application: Cardiac Patients and Antibiotic Prophylaxis

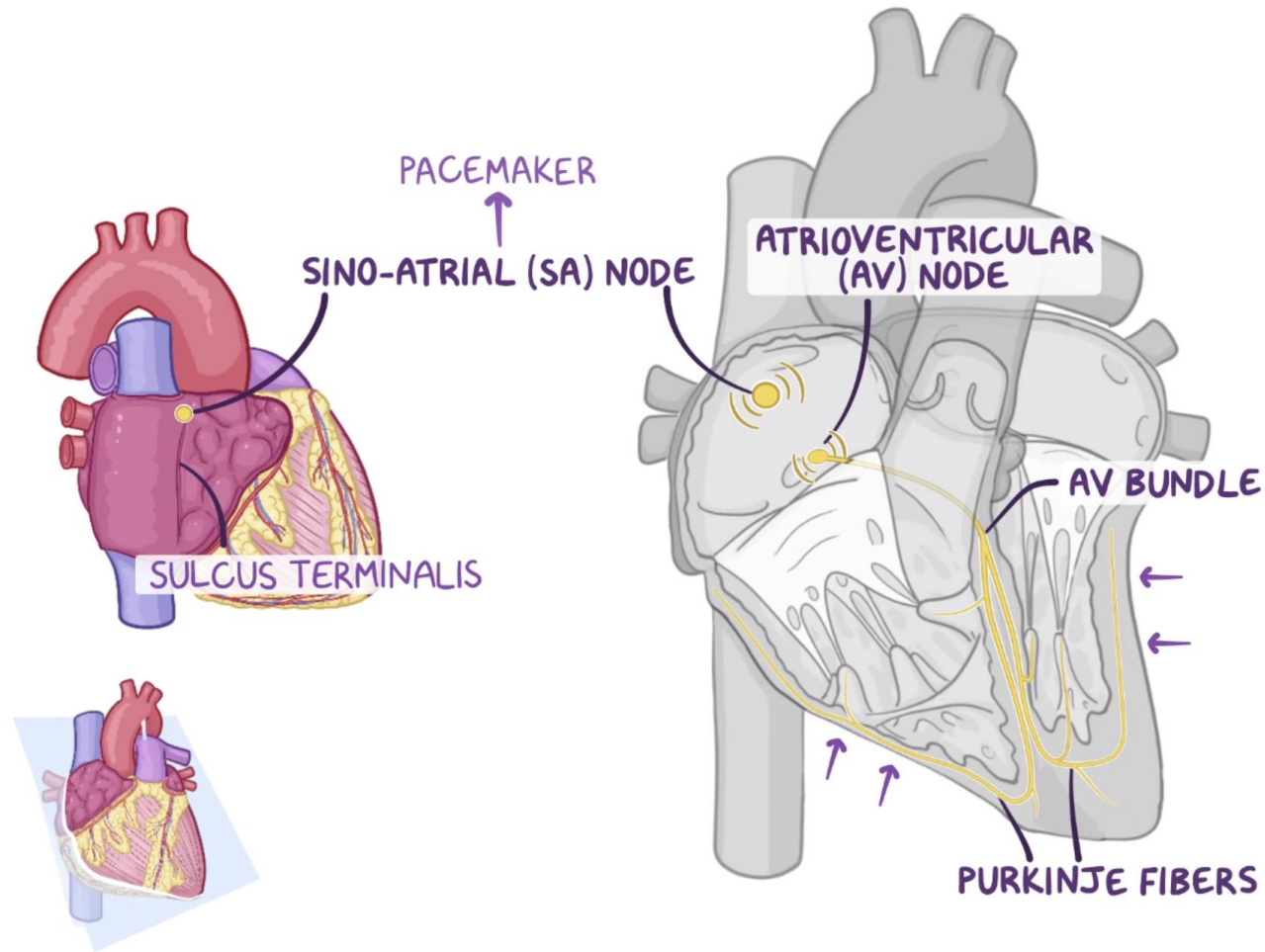
## ADA

American  
Dental  
Association®

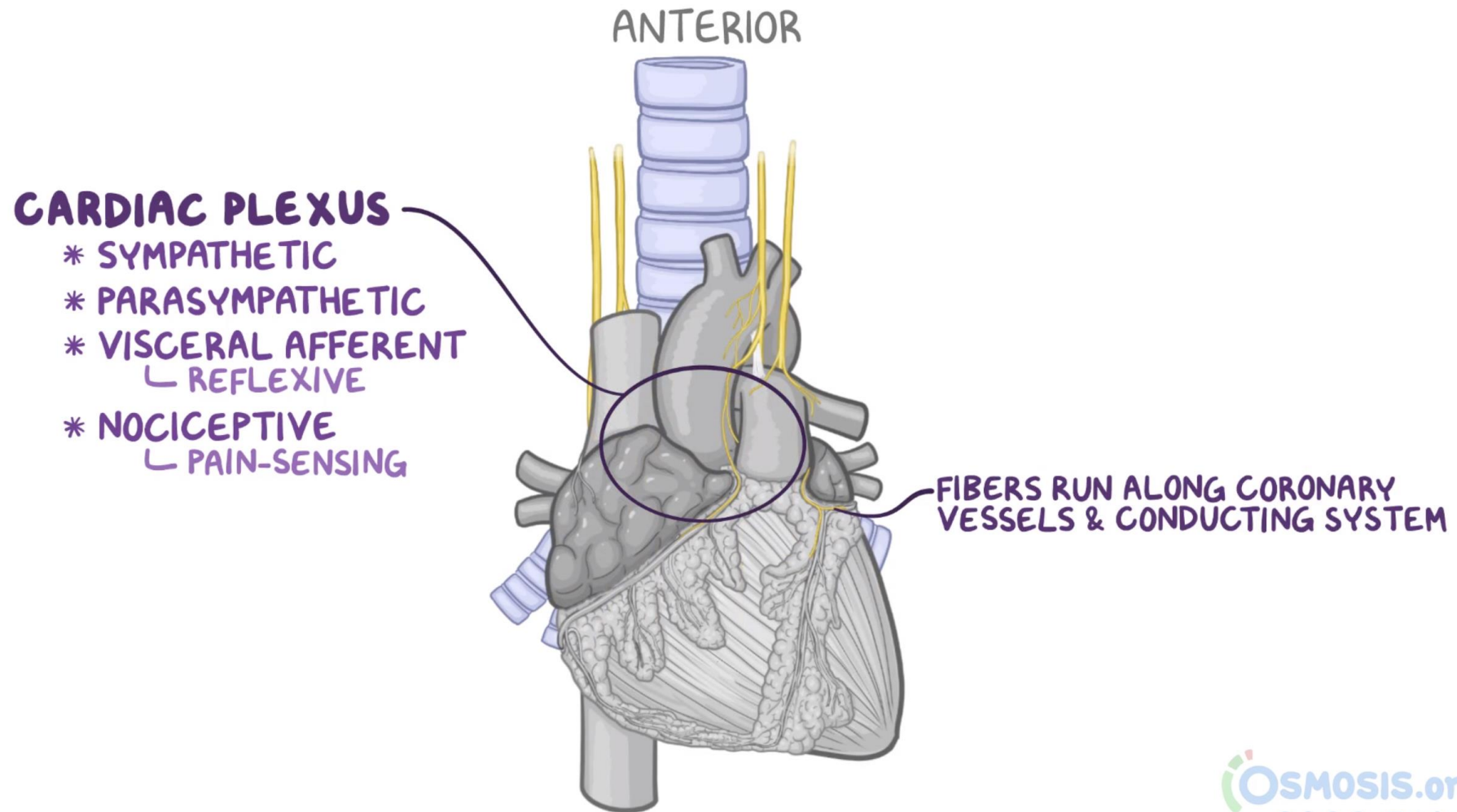
**The current infective endocarditis/valvular heart disease guidelines state that use of preventive antibiotics before certain dental procedures is reasonable for patients with:**

- **prosthetic cardiac valves**, including transcatheter-implanted prostheses and homografts;
- prosthetic material used for cardiac valve repair, such as annuloplasty rings and chords;
- **a history of infective endocarditis**;
- a cardiac transplant with **valve regurgitation** due to a structurally abnormal valve;
- the following **congenital (present from birth) heart disease**:
  - unrepaired cyanotic congenital heart disease, including palliative shunts and conduits
  - any repaired congenital heart defect with residual shunts or valvular regurgitation at the site of or adjacent to the site of a prosthetic patch or a prosthetic device

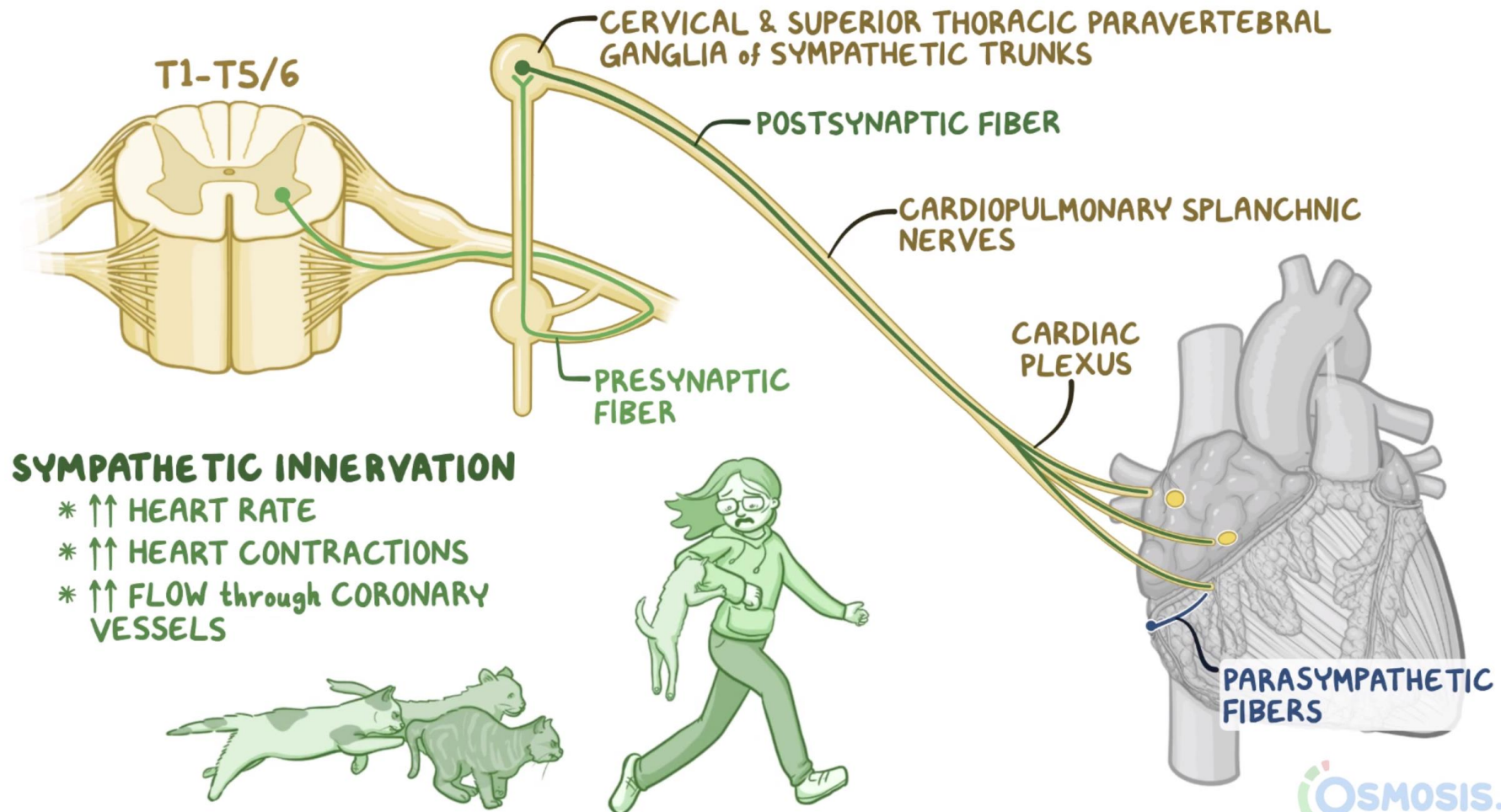
# Conducting System of the Heart



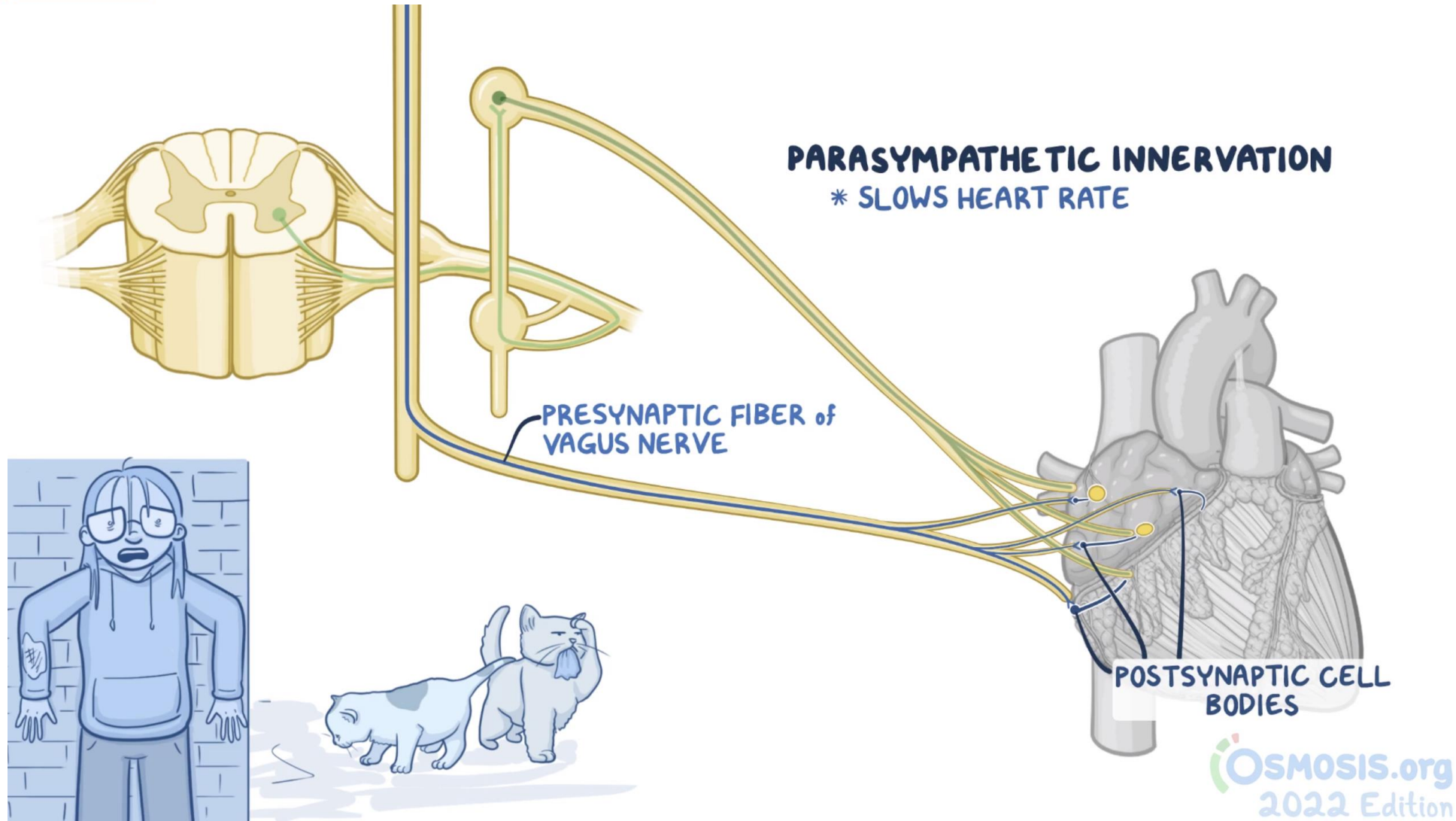
# Innervation of the Heart



# Sympathetic Innervation of Heart

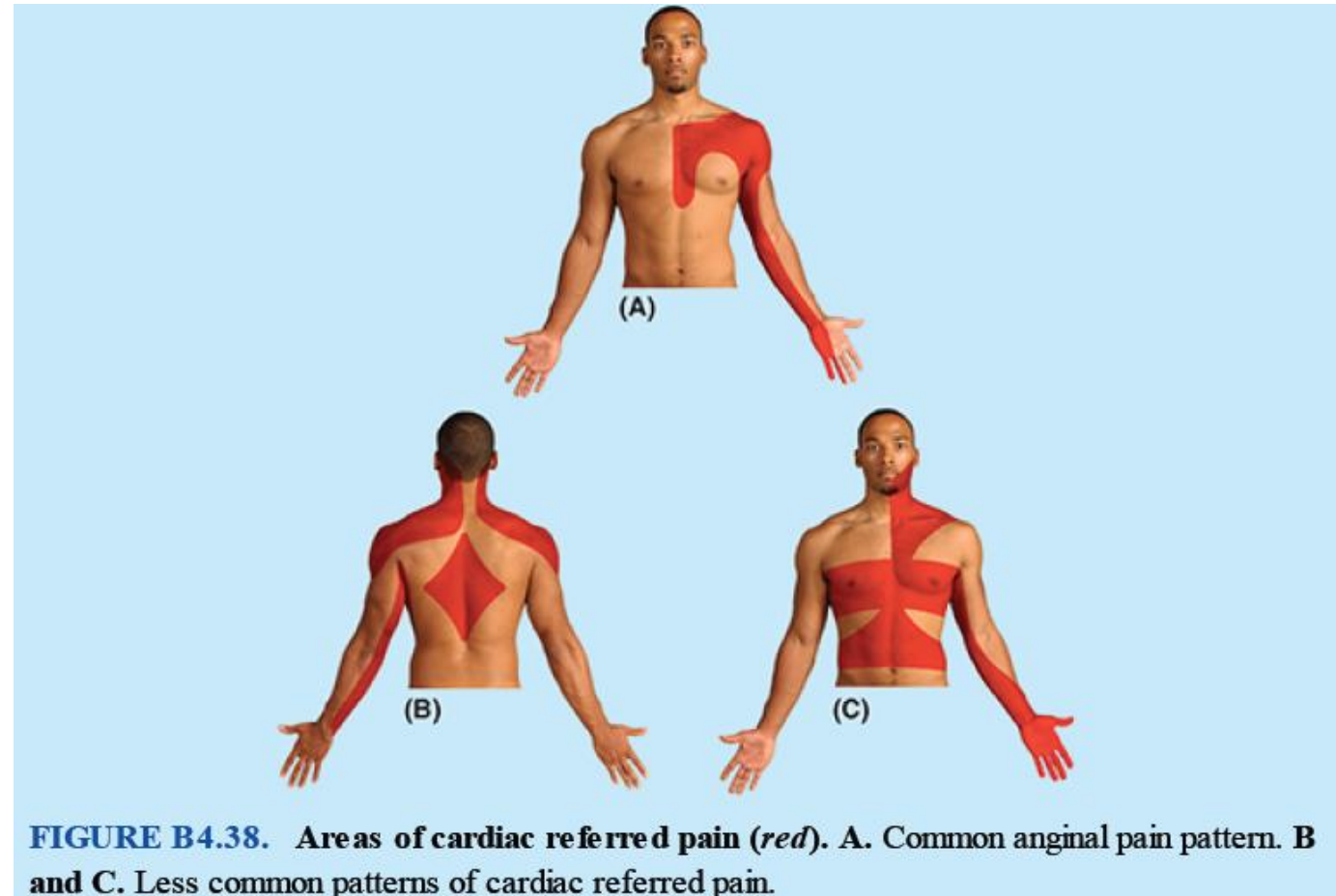


# Parasympathetic Innervation of Heart



## Clinical Application: Cardiac Referred Pain

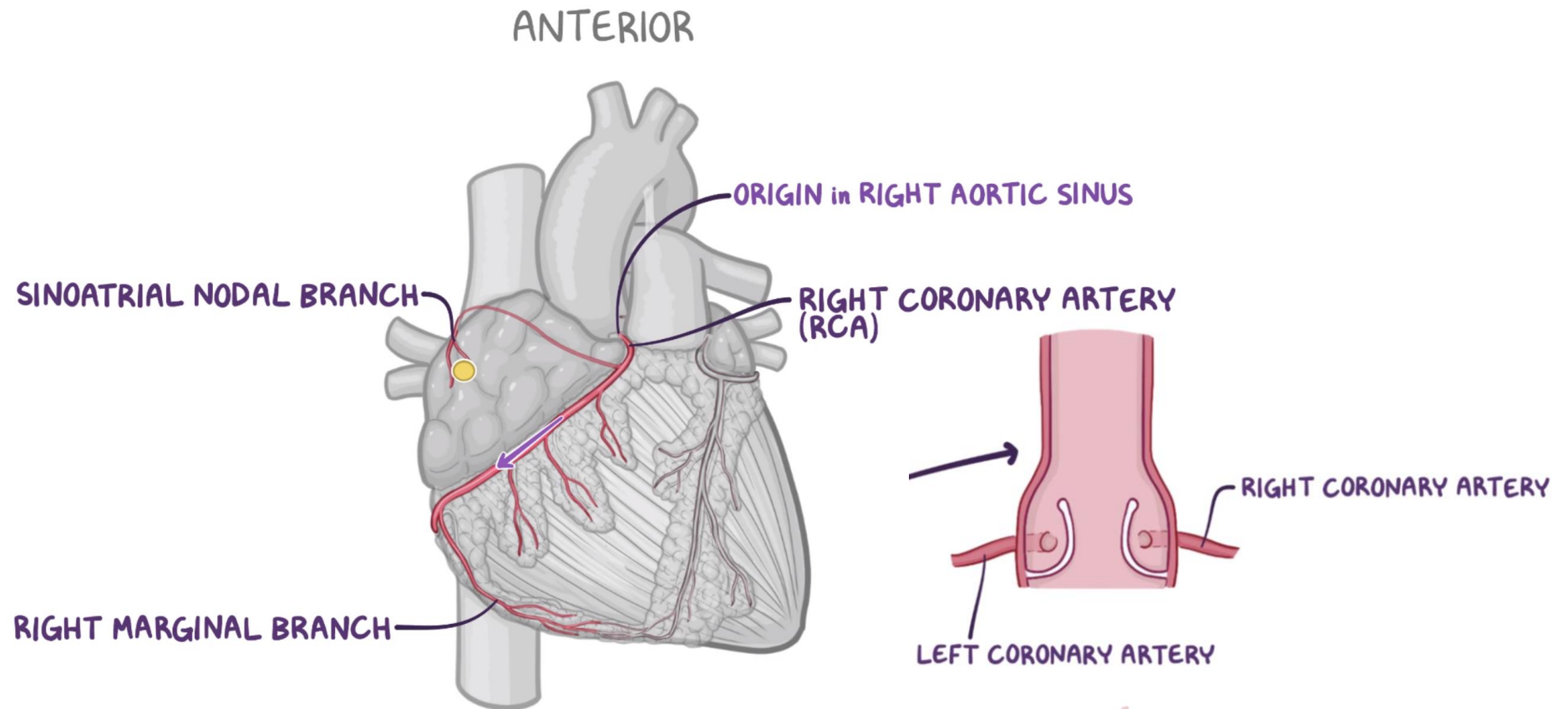
- **Ischemia** (lack of oxygen) as in angina pectoris or a myocardial infarction can stimulate pain fibers in the heart
- Pain fibers run centrally in middle and inferior cervical branches and especially in the thoracic cardiac branches of the sympathetic trunk
- Axons enter spinal cord segments **T1 through T4/5** especially on the **left side**
- Presents as referred pain to the superficial somatic area supplied by the spinal nerves involved
- Most commonly the left shoulder, arm and chest





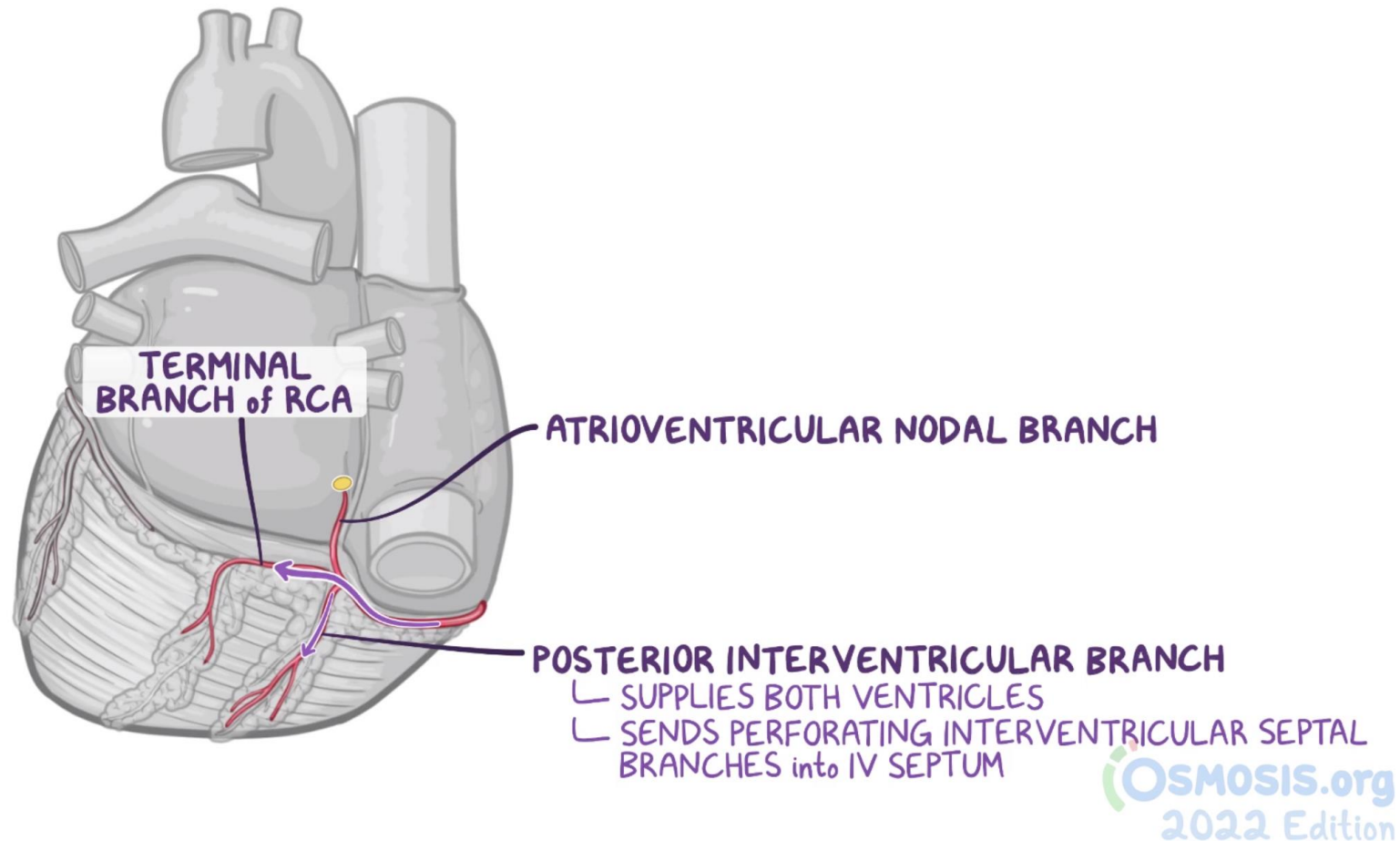
# ANATOMY OF THE CORONARY CIRCULATION

# Right Coronary Artery: Anterior View

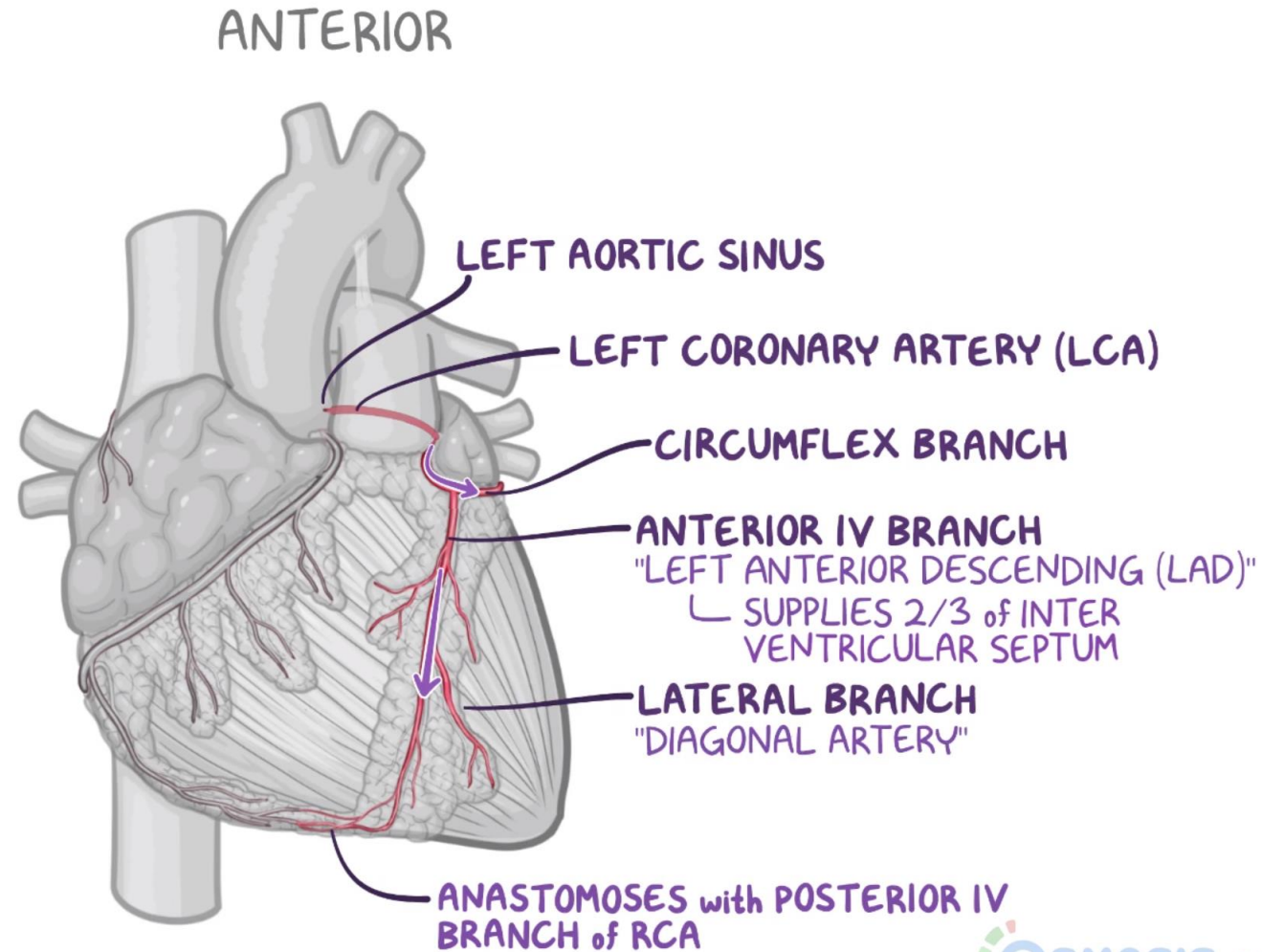


# Right Coronary Artery: Posterior View

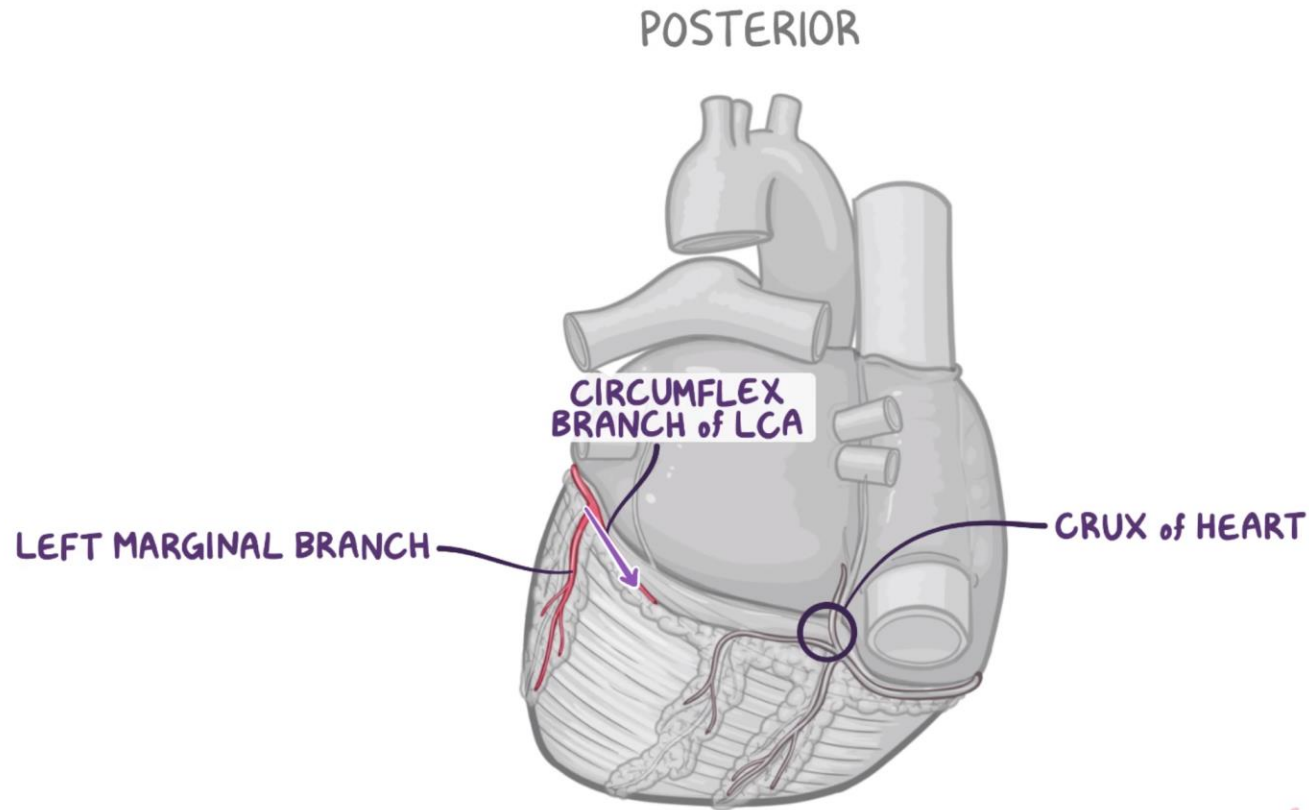
POSTERIOR



# Left Coronary Artery: Anterior View



# Left Coronary Artery: Posterior View

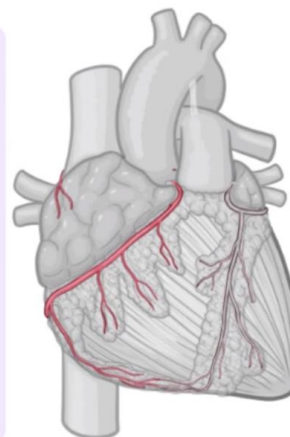


# Areas Supplied by Coronary Arteries

USUALLY

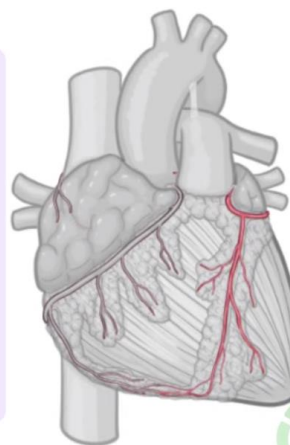
## RCA

- ~ DIAPHRAGMATIC SURFACE of the HEART
- ~ RIGHT ATRIUM
- ~ MOST of RIGHT VENTRICLE
- ~ DIAPHRAGMATIC PART of LEFT VENTRICLE
- ~ POSTERIOR 1/3 of IVS
- ~ SA NODE
- ~ AV NODE



## LCA

- ~ LEFT ATRIUM
- ~ MOST of LEFT VENTRICLE
- ~ PART of RIGHT VENTRICLE
- ~ ANTERIOR 2/3 of IVS
  - └ including AV BUNDLE
- ~ SA NODE in SOME

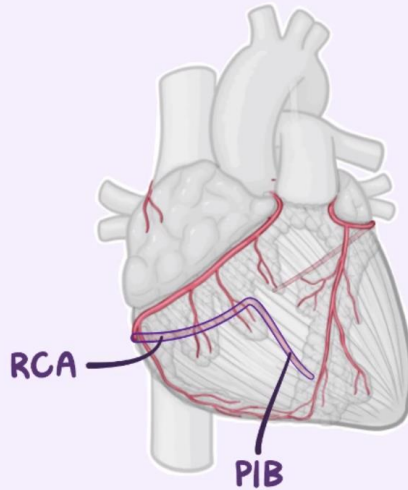


# Anatomical Variation

## DOMINANCE of CORONARY ARTERIAL SYSTEM

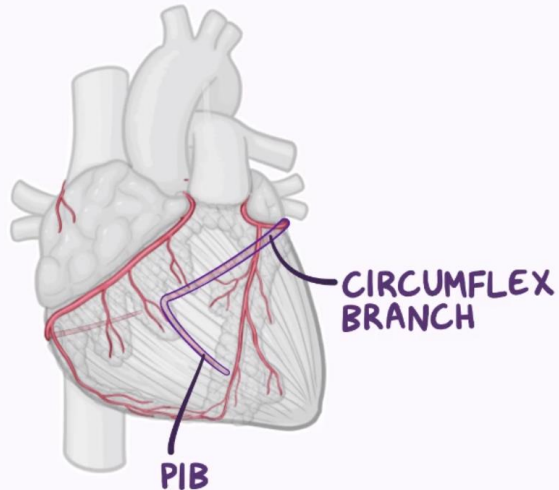
\* whether RIGHT or LEFT CORONARY ARTERY gives off POSTERIOR INTERVENTRICULAR BRANCH (PIB)

67-85%

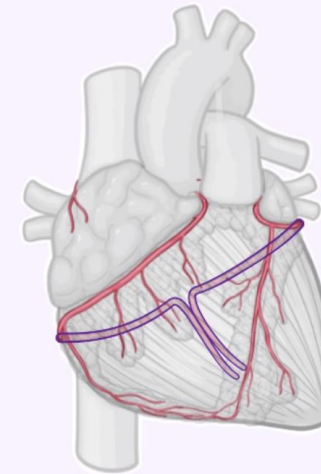


\* RCA & LCA EACH SUPPLY  
50%

8-15%

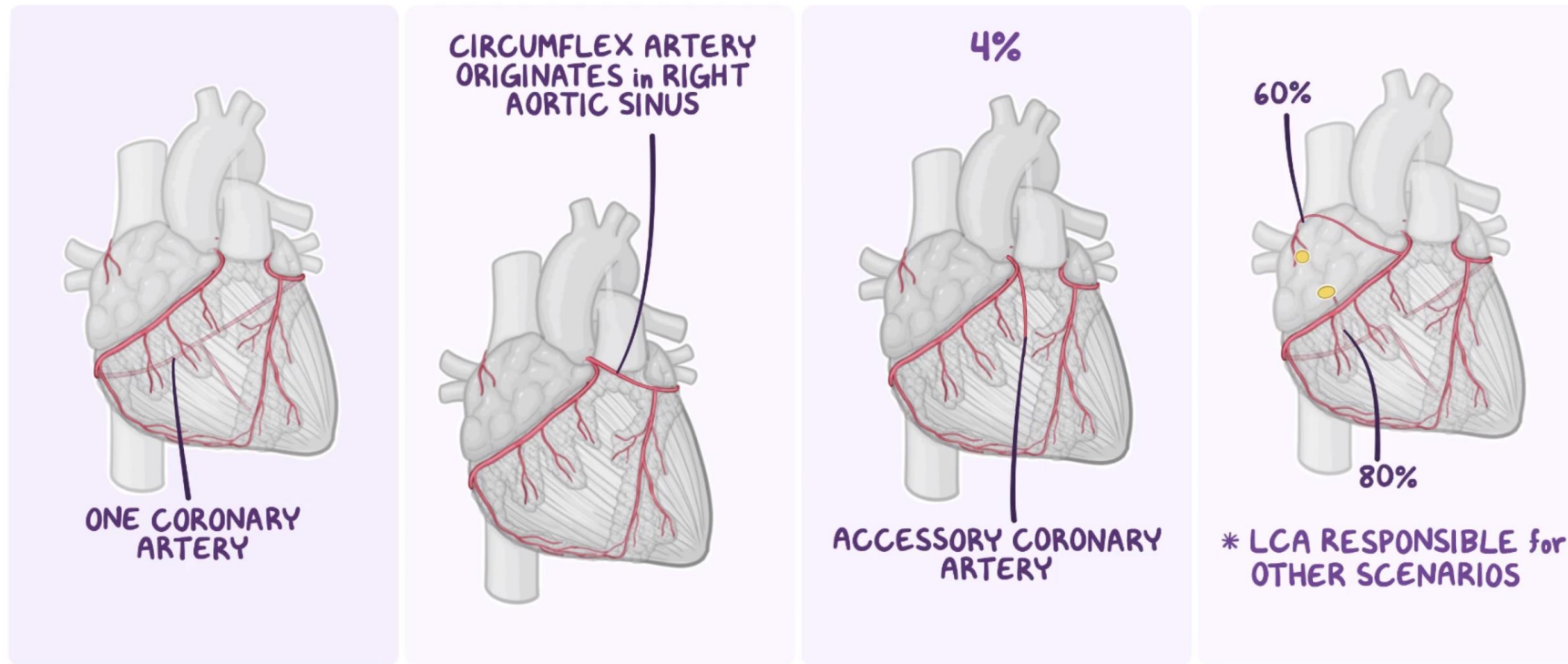


7-18%



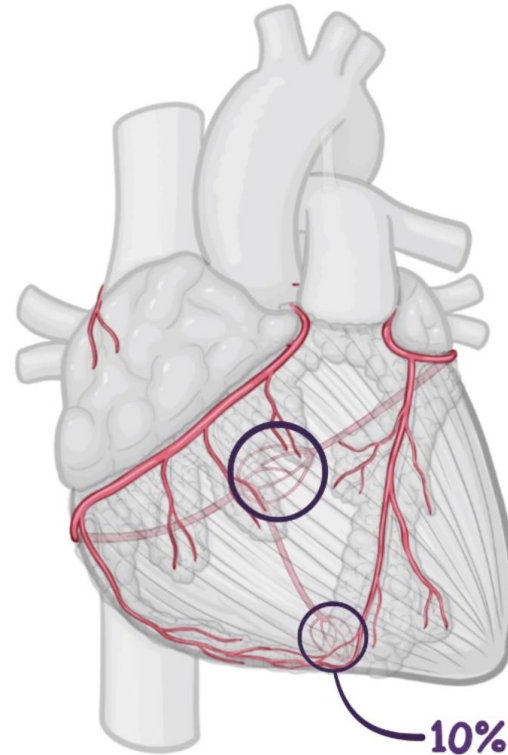
\* CODOMINANCE

# Anatomical Variation



# Collateral Circulation

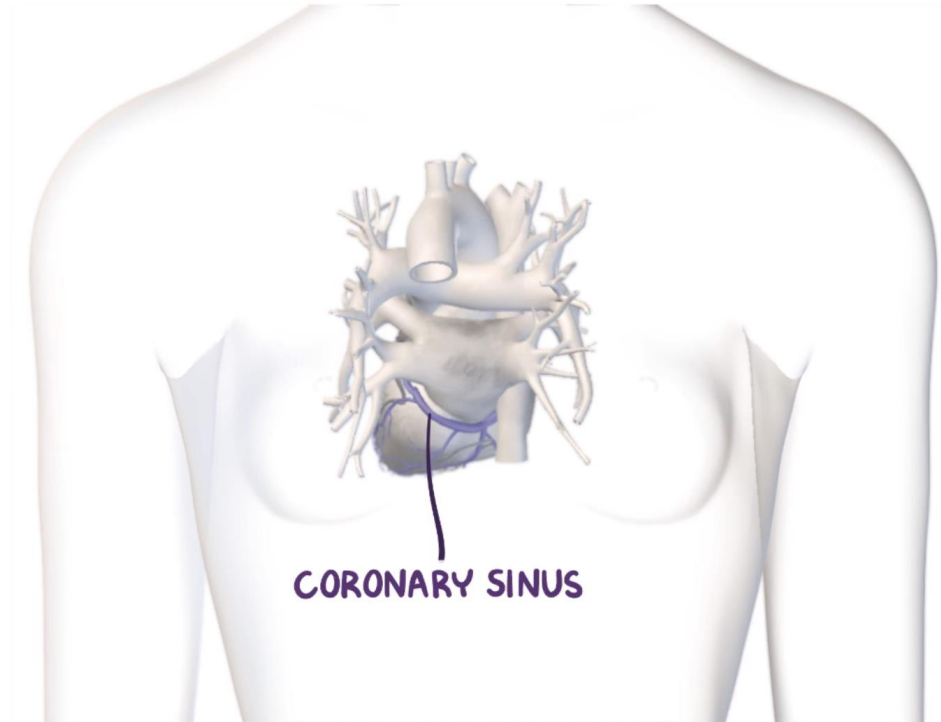
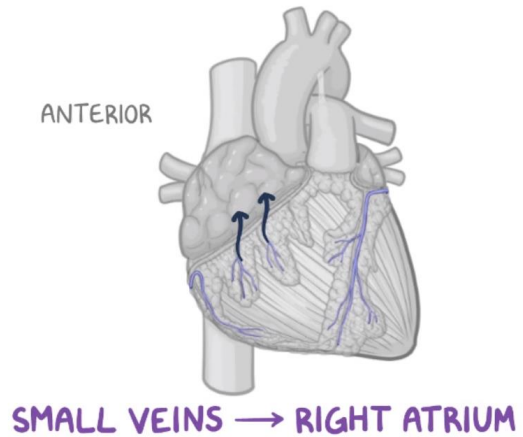
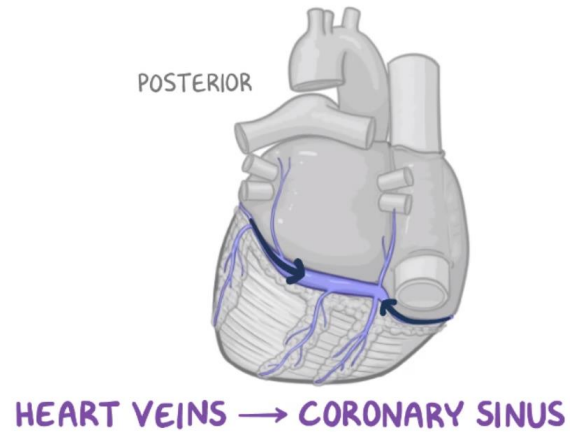
## COLLATERAL CIRCULATION



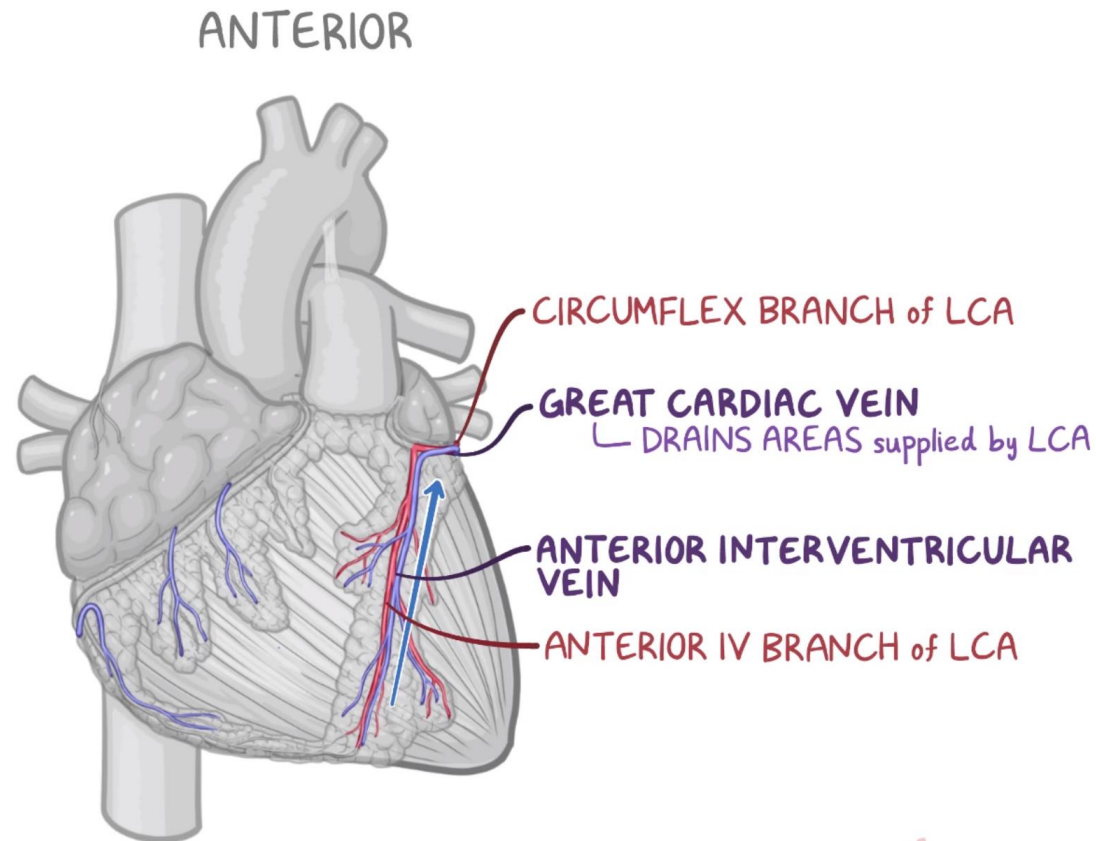
10%

↳ POTENTIAL to DEVELOP for ALL HEARTS

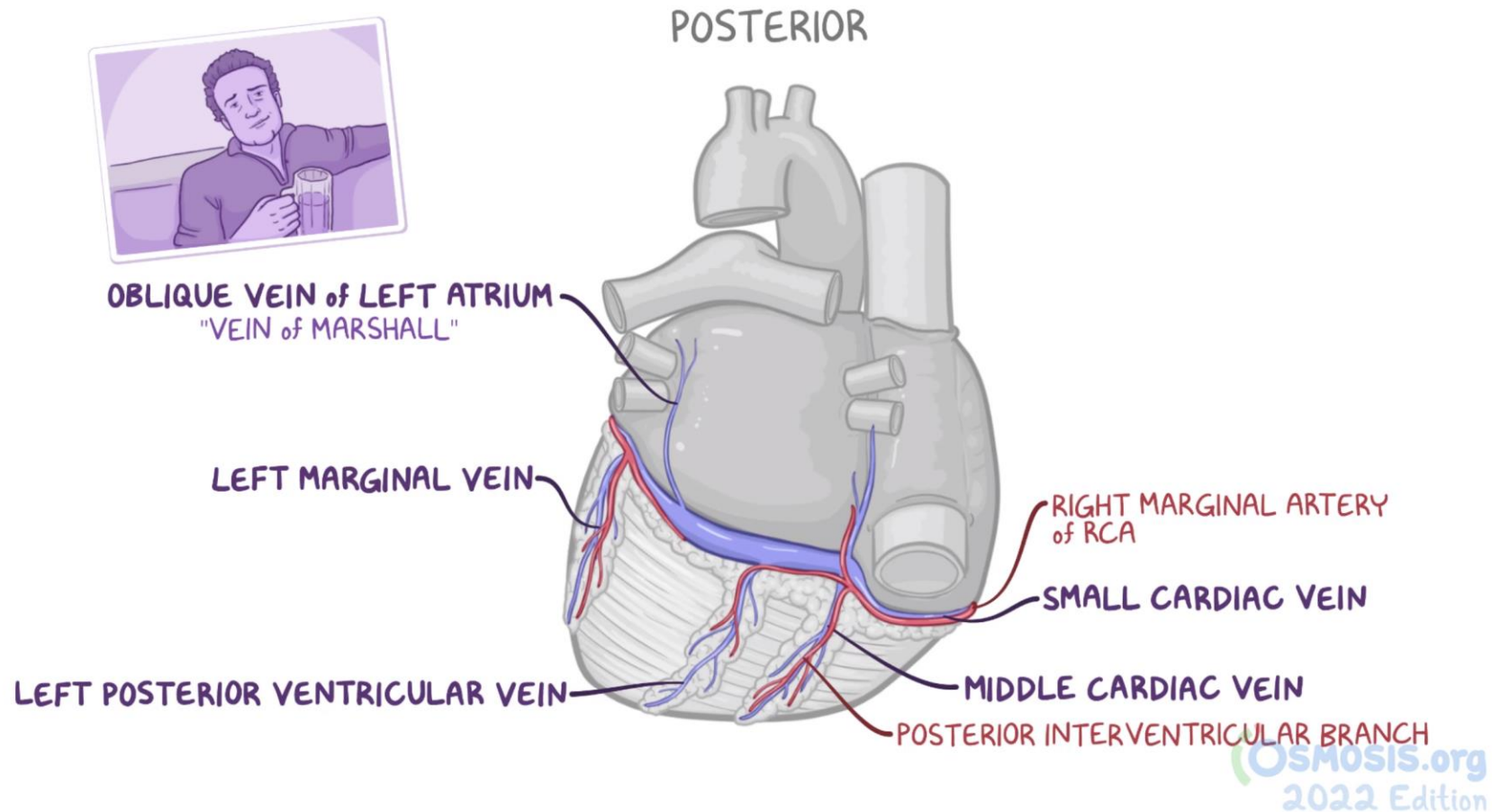
# Coronary Venous Drainage



# Coronary Venous Drainage

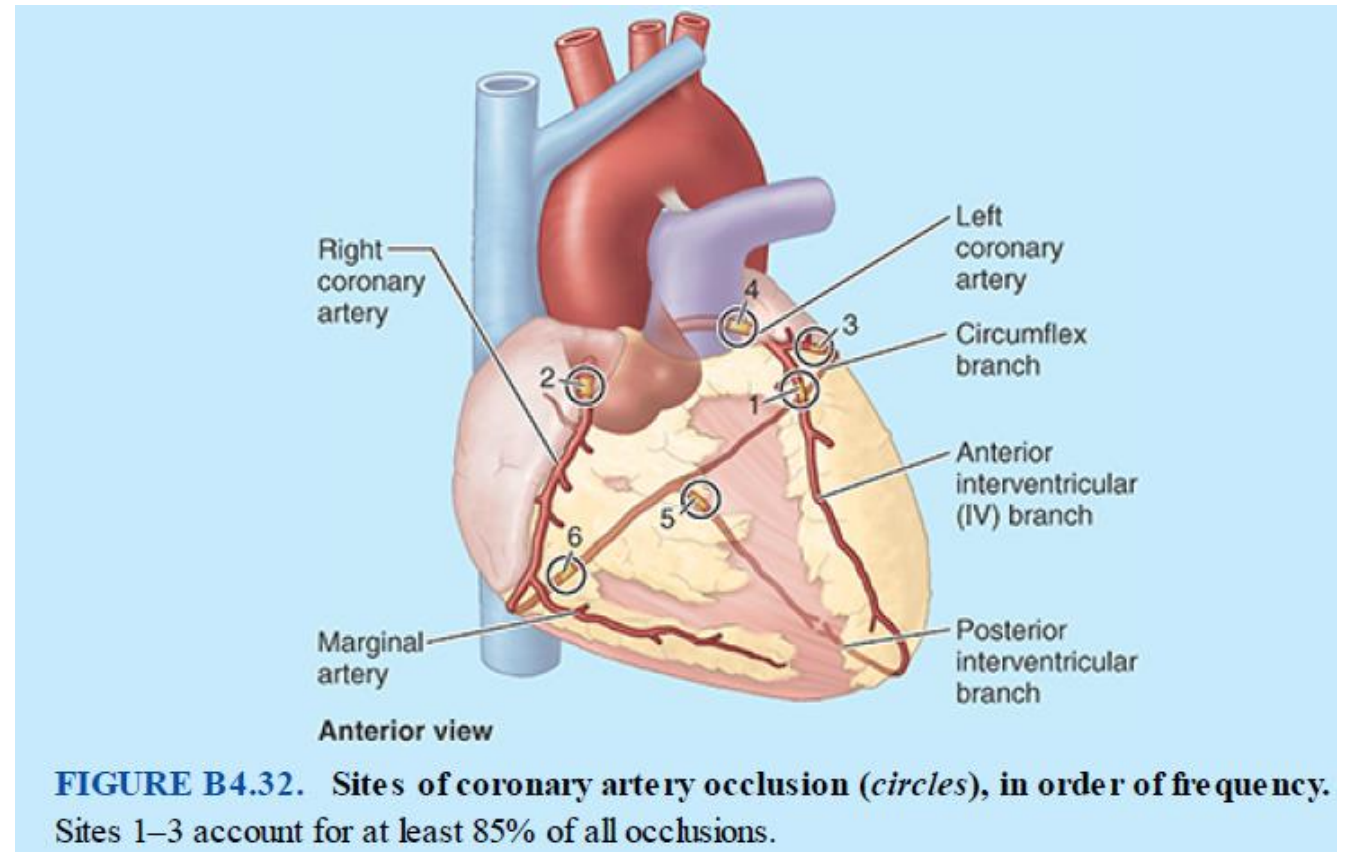


# Coronary Venous Drainage

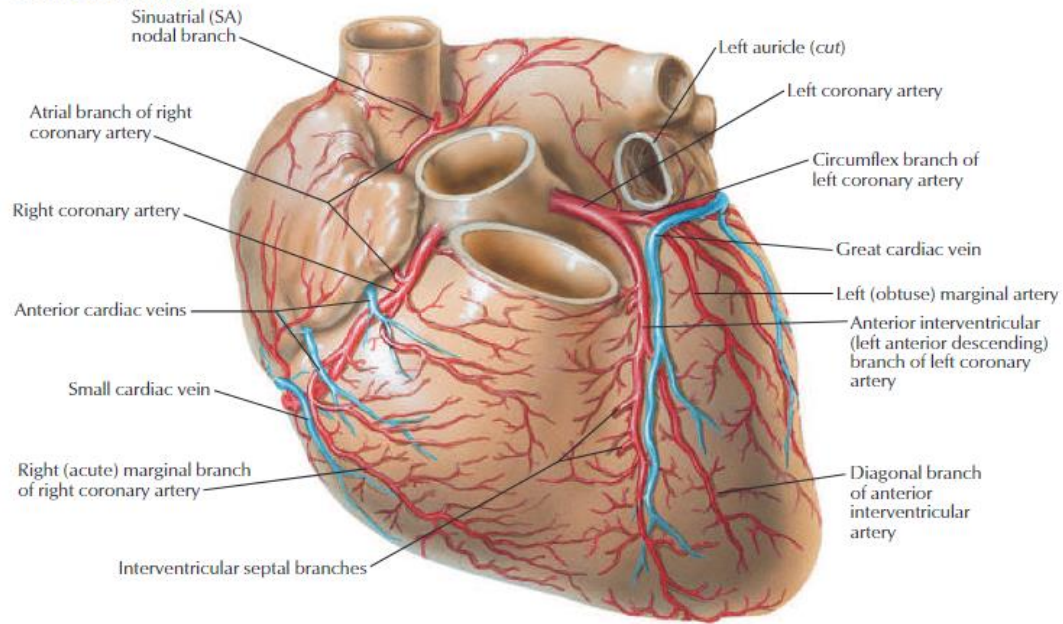


## Clinical Application: Myocardial Infarction

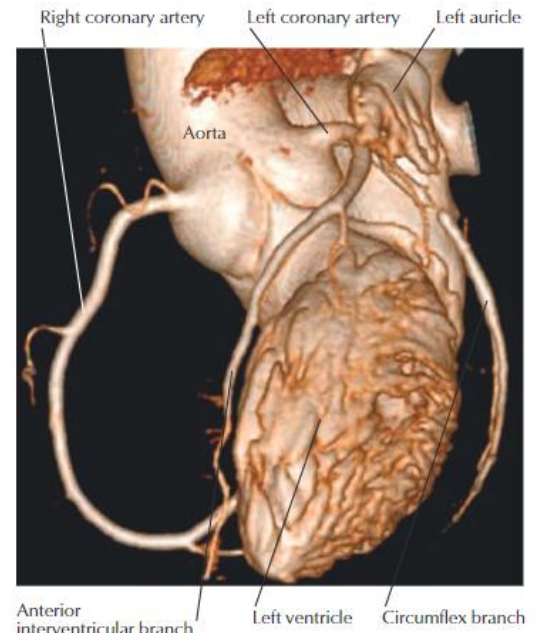
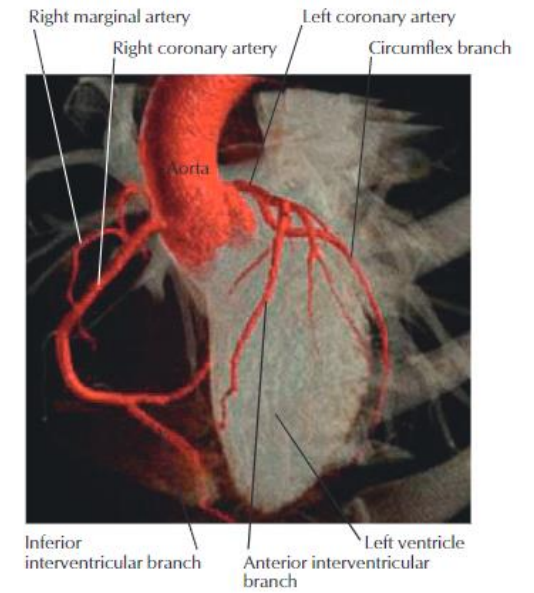
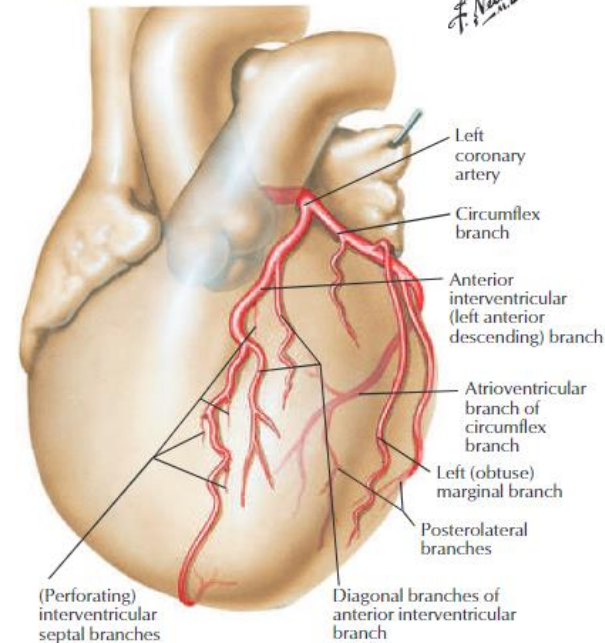
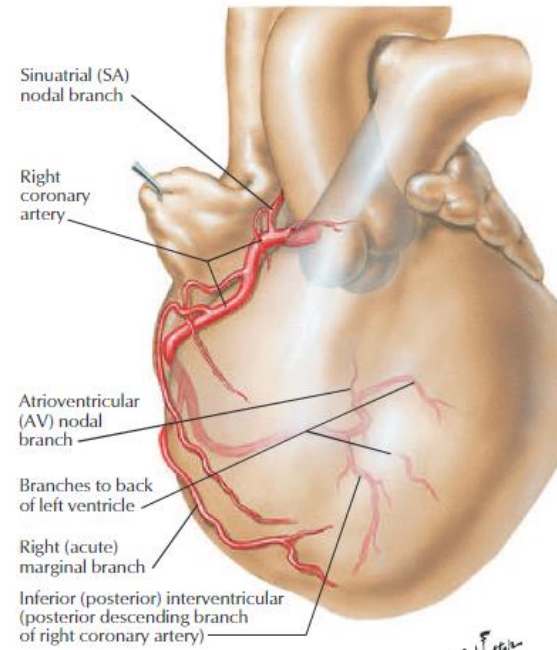
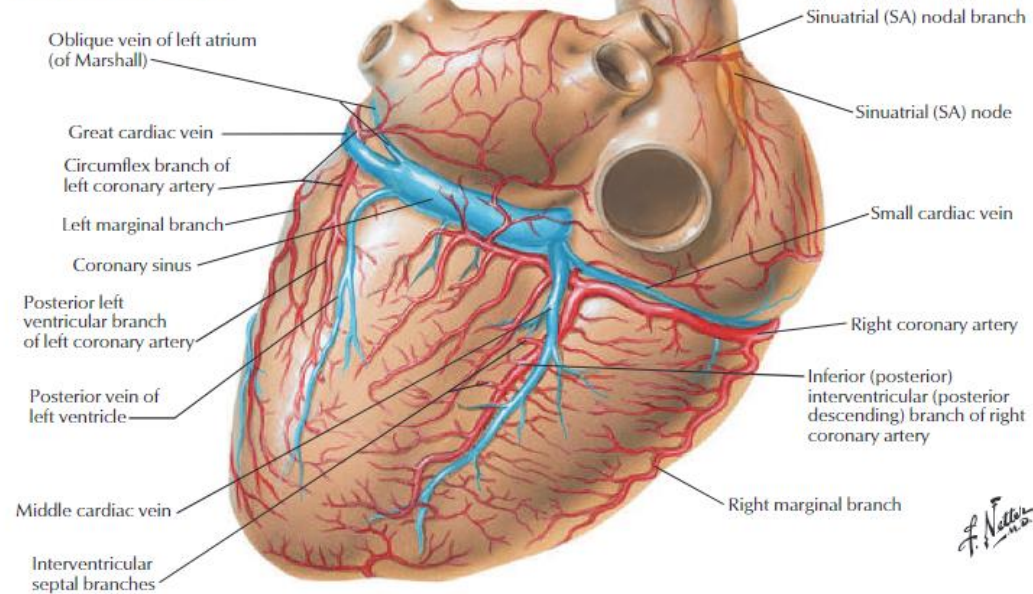
- Sudden occlusion of coronary artery leads to **lack of blood supply** and oxygen to region of **myocardium**, leading to necrosis (tissue death)
- Three most common sites from greatest to least:
  - 1) Anterior Interventricular Branch of Left Coronary Artery (LAD)
  - 2) Right Coronary Artery
  - 3) Circumflex Branch of Left Coronary Artery



## Sternocostal surface



## Diaphragmatic surface



# Breathing Into Stress



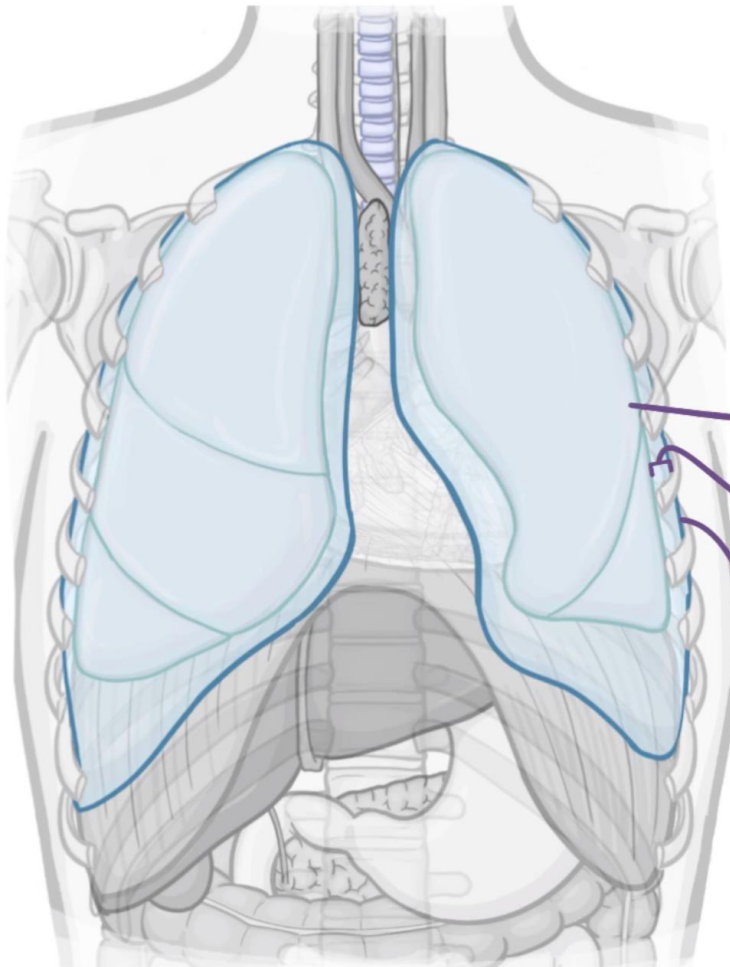
# Lungs: Learning Outcomes

- Describe the parts of the pleura, pleural lines of reflection, and recesses
- Identify the surfaces and borders of the lungs
- Describe the differences in structure and surface features of the right and left lungs
- Identify the impressions and the positions of various organs and vessels on the lungs
- Identify and describe the branching of the tracheobronchial tree
- Describe the pulmonary functional and nutritive circulation
- Describe the general innervation and lymphatic drainage of the lungs and pleura
- Discuss the clinical application of the anatomy in a pneumothorax, lung cancer, foreign body aspiration, pulmonary embolism and referred pain



# ANATOMY OF THE PLEURA

# The Pleura



## PLEURA

↳ THIN MEMBRANE that COVERS EACH LUNG

### VISCERAL PLEURA

↳ INTIMATELY ADHERES to LUNG

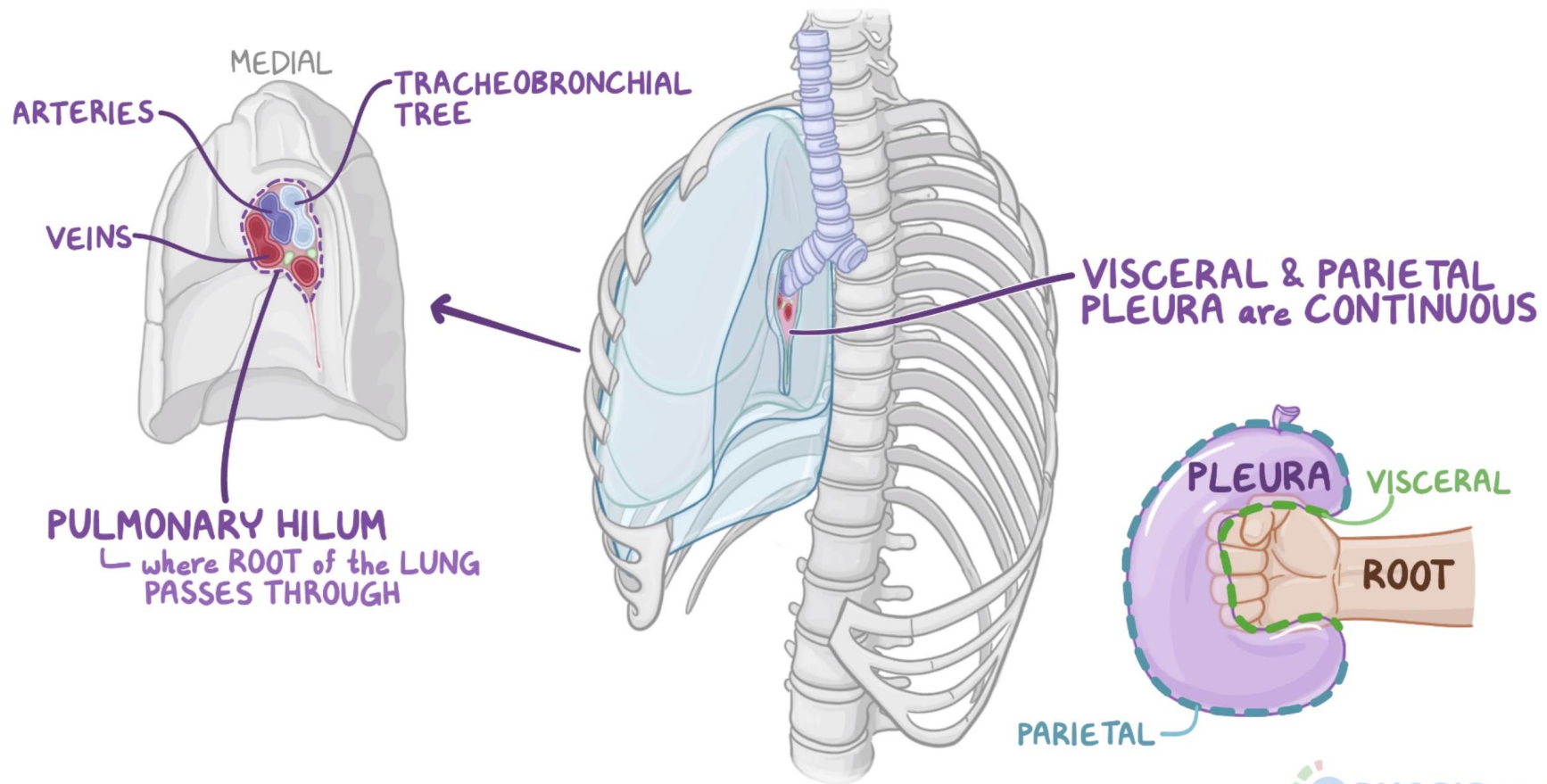
### PLEURAL CAVITY

↳ FILLED with FLUID

### PARIETAL PLEURA

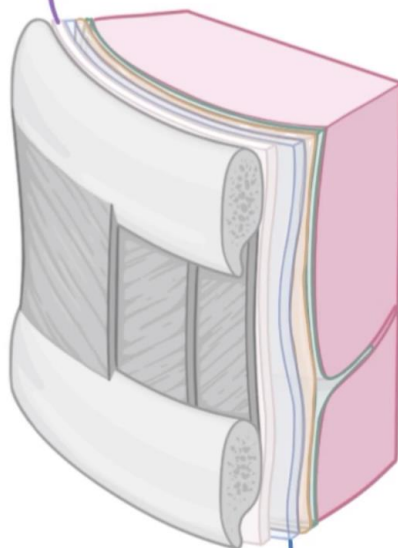
↳ LINES PULMONARY CAVITY

# Pulmonary Hilum



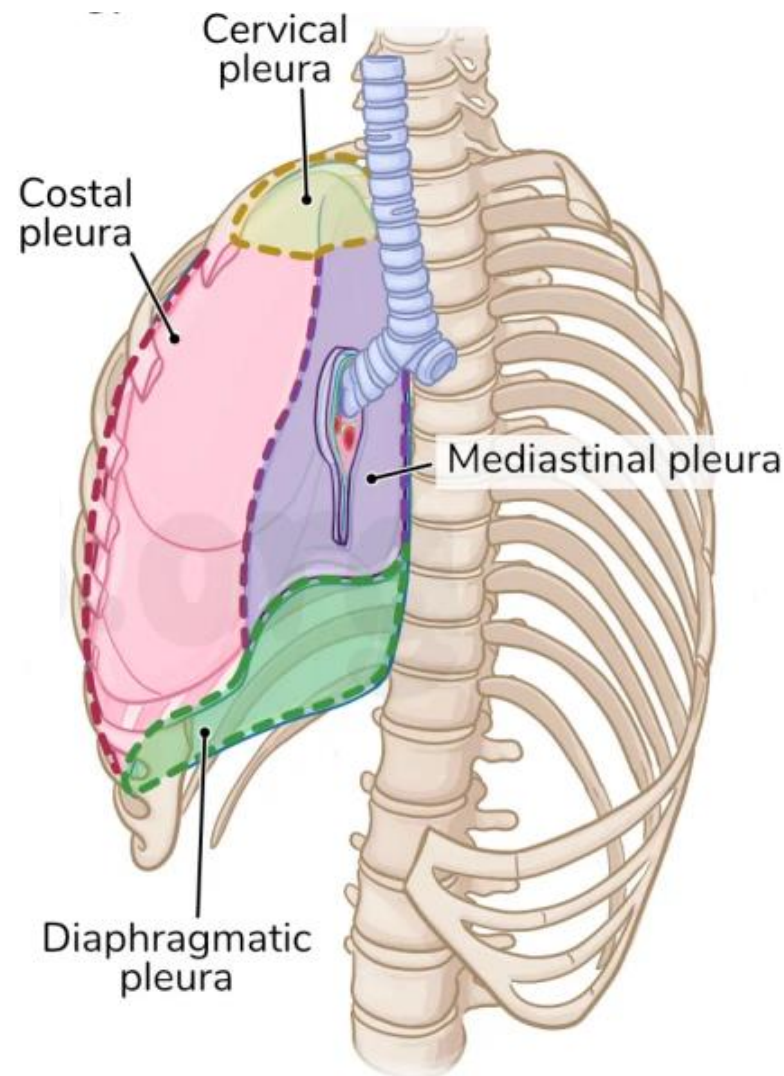
# 4 Parts of Parietal Pleura

ENDOTHORACIC FASCIA

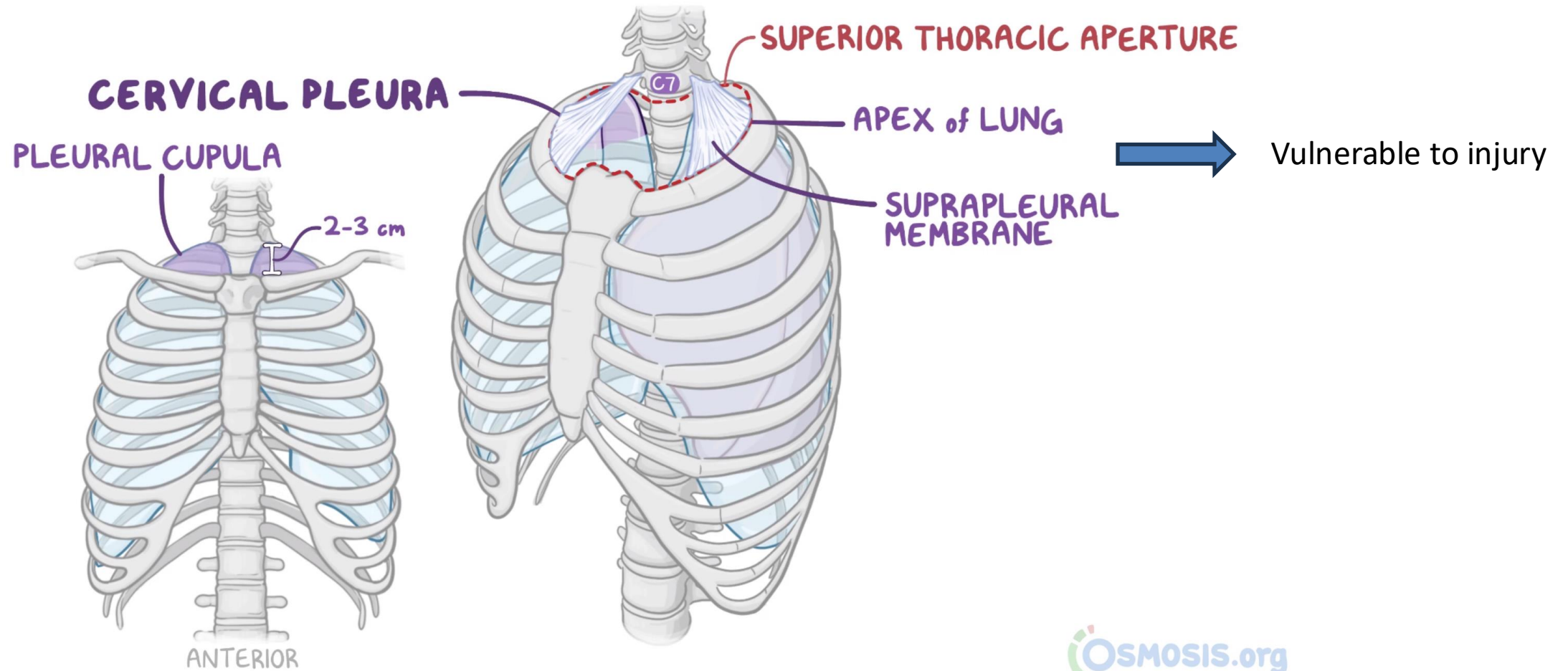


PARIETAL PLEURA

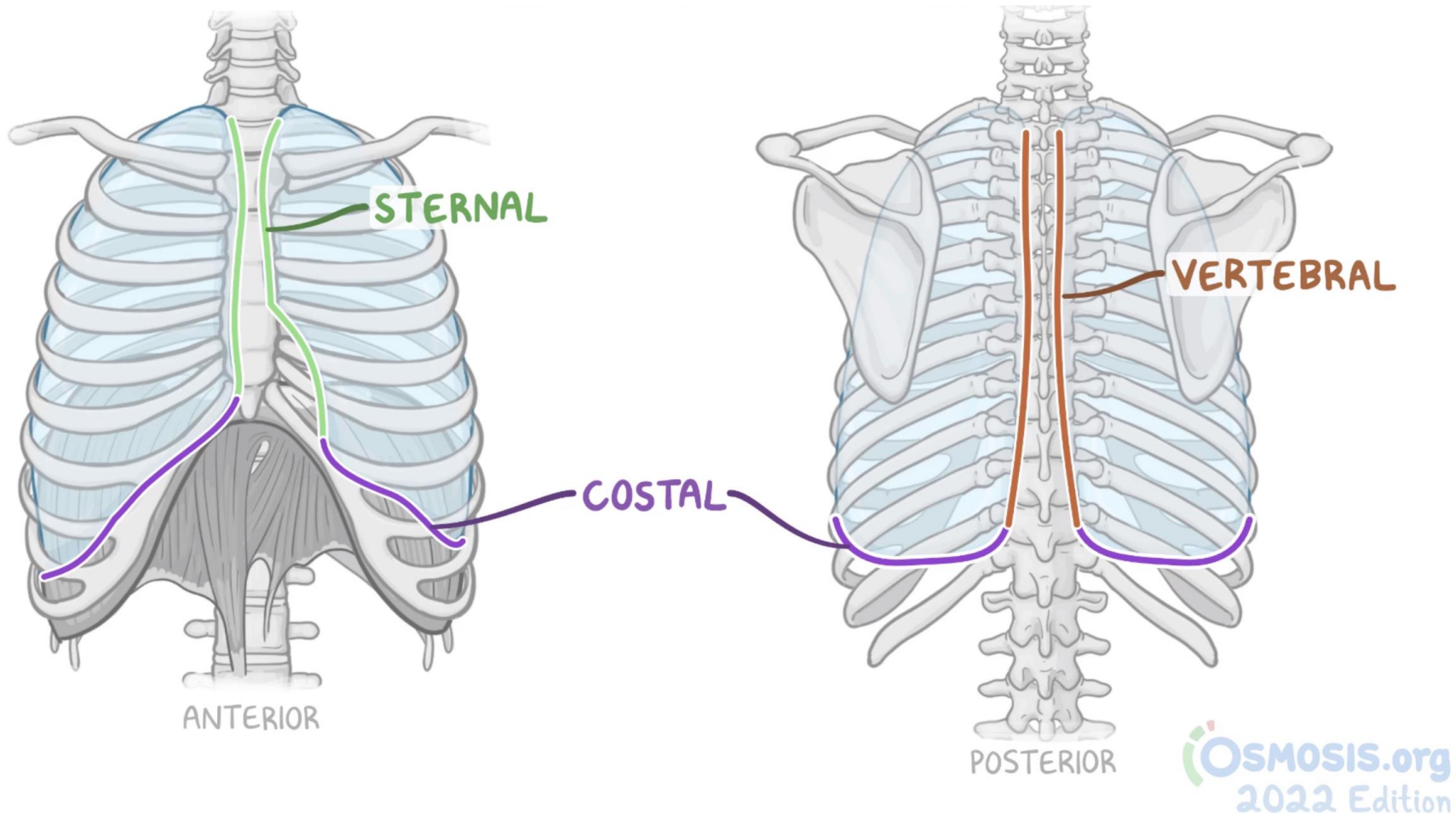
- └ THICKER than VISCERAL
- └ LINES PULMONARY CAVITIES
- └ ADHERES to:
  - ~ THORACIC WALL
  - ~ MEDIASTINUM
  - ~ DIAPHRAGM



# Cervical Pleura and Pleural Cupula



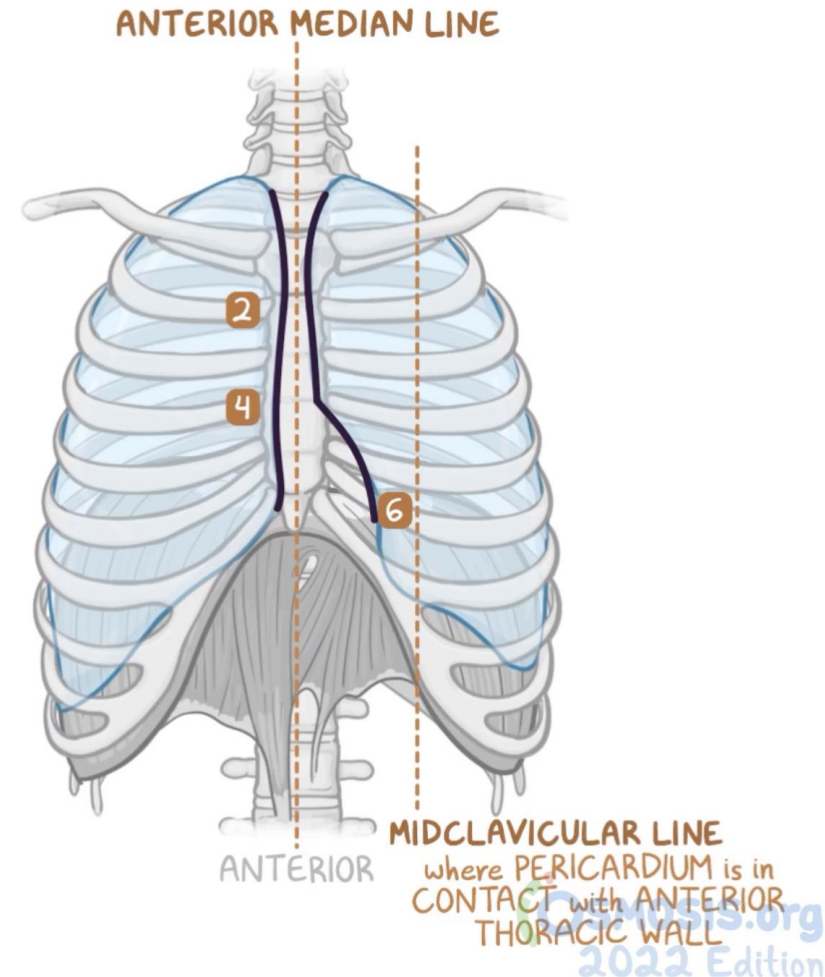
# Lines of Pleural Reflection



# Sternal Lines of Reflection

## STERNAL LINES

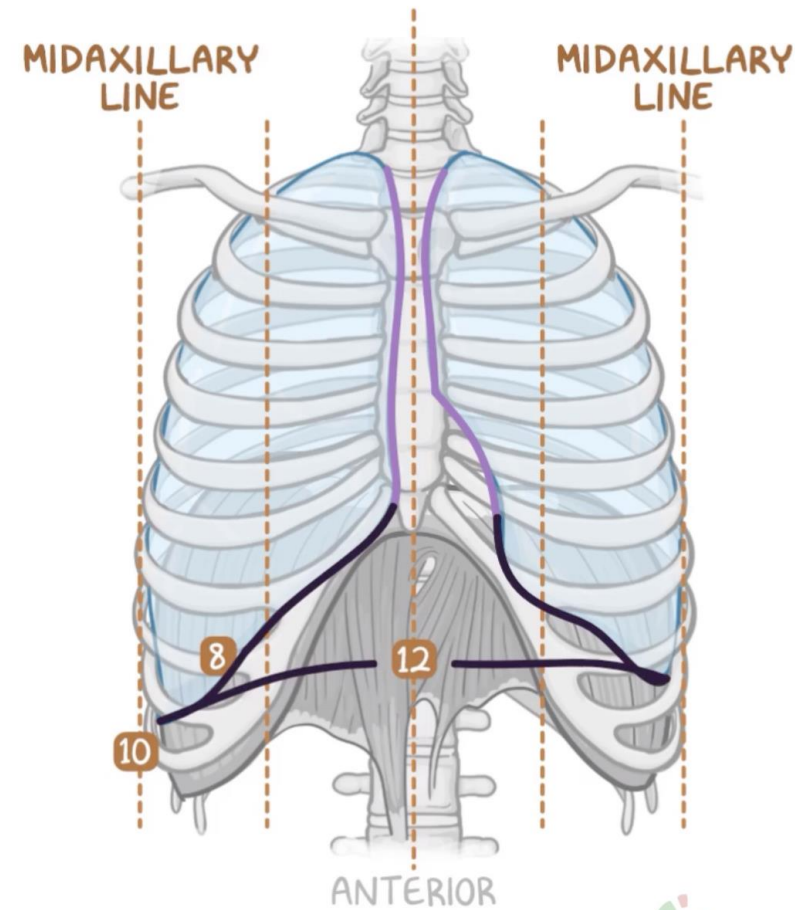
- └ OCCUR ANTERIORLY
- └ where COSTAL PLEURA is CONTINUOUS with MEDIASTINAL PLEURA



# Costal Lines of Reflection

## COSTAL LINES

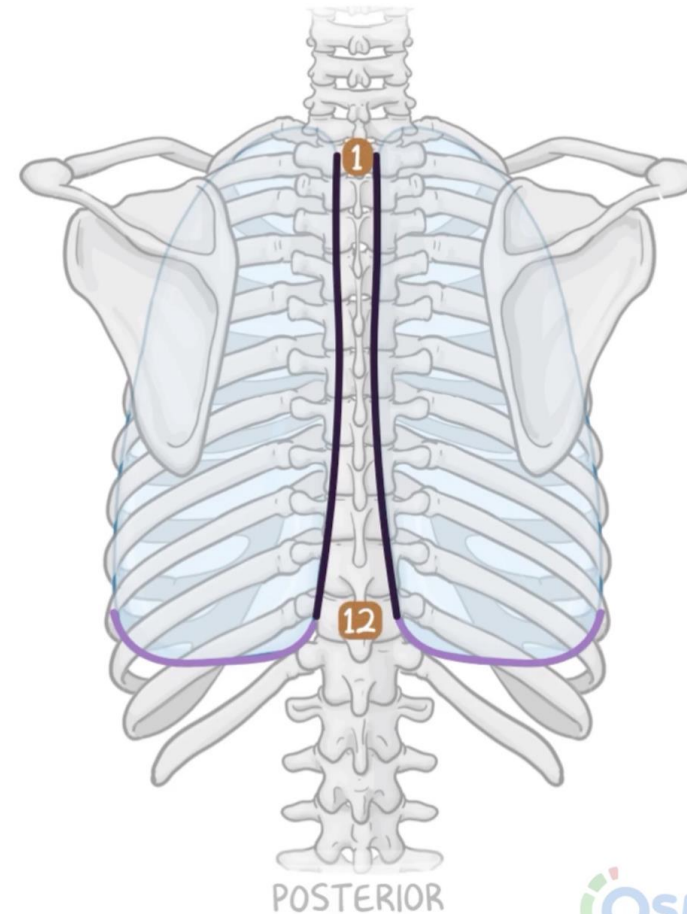
↳ where COSTAL PLEURA is CONTINUOUS with DIAPHRAGMATIC PLEURA INFERIORLY

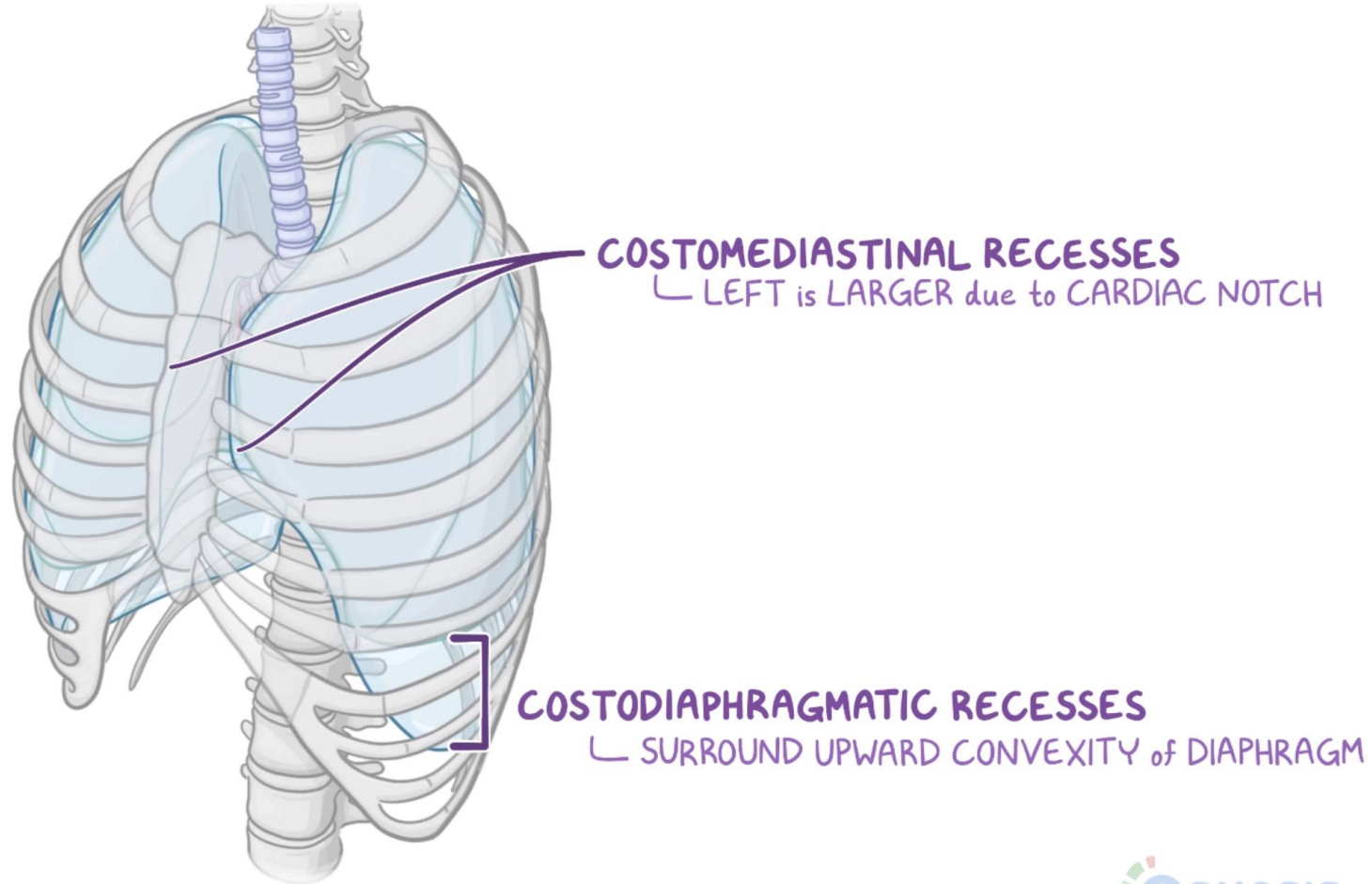


# Vertebral Lines of Reflection

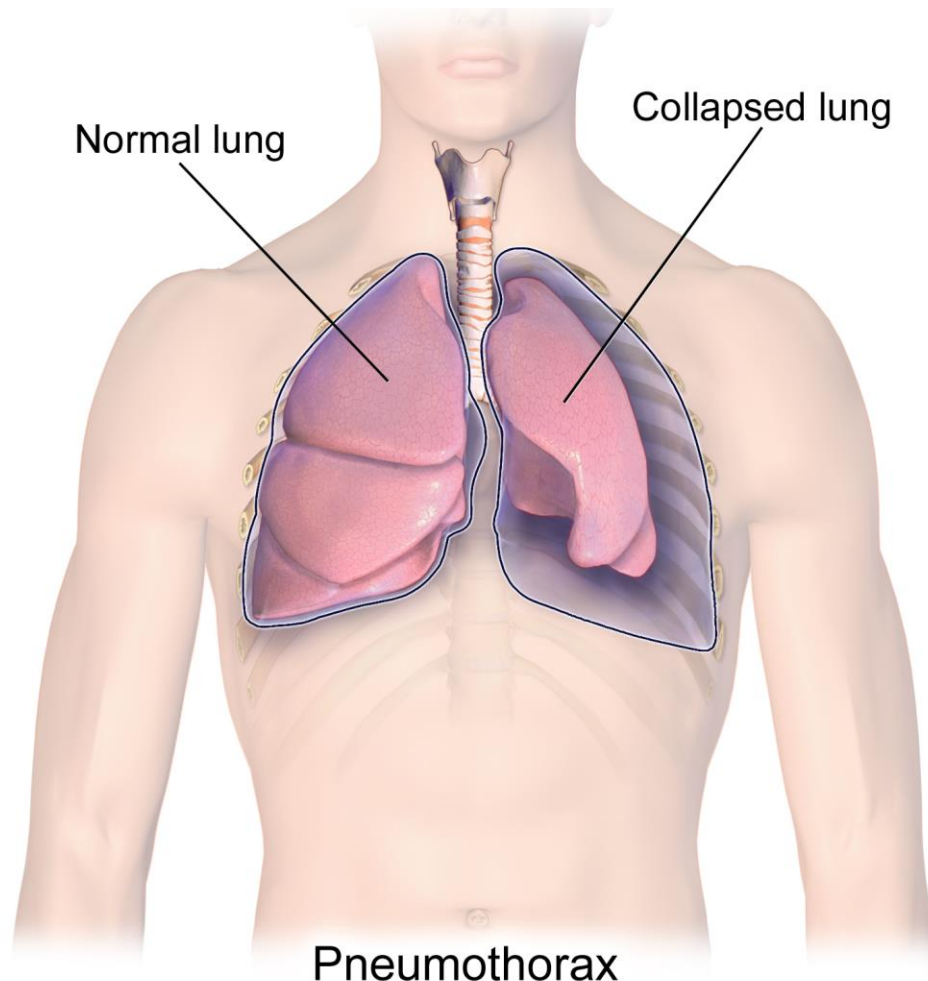
## VERTEBRAL LINES

↳ where COSTAL PLEURA is CONTINUOUS  
with MEDIASTINAL PLEURA  
POSTERIORLY





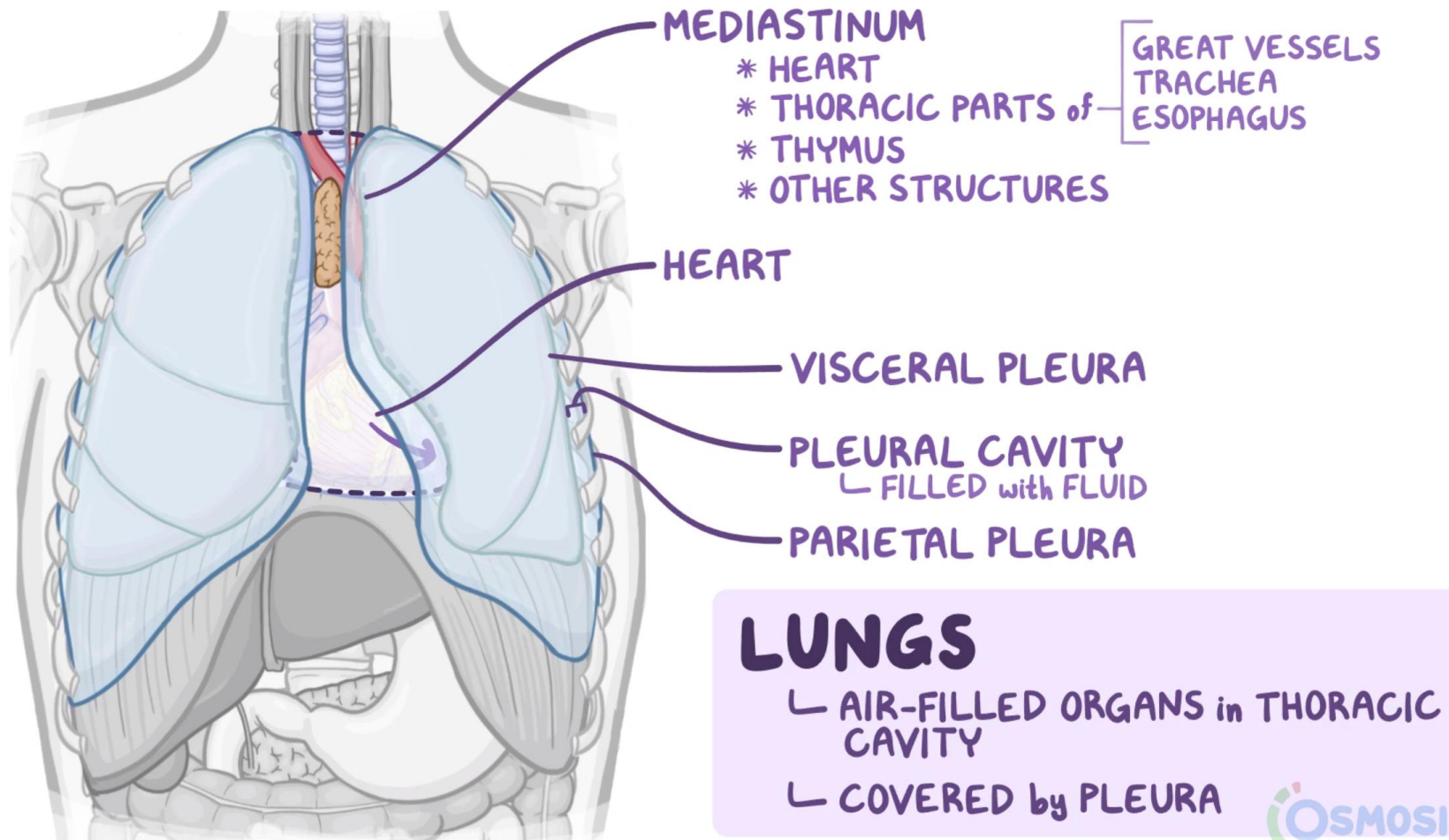
# Clinical Application: Pneumothorax



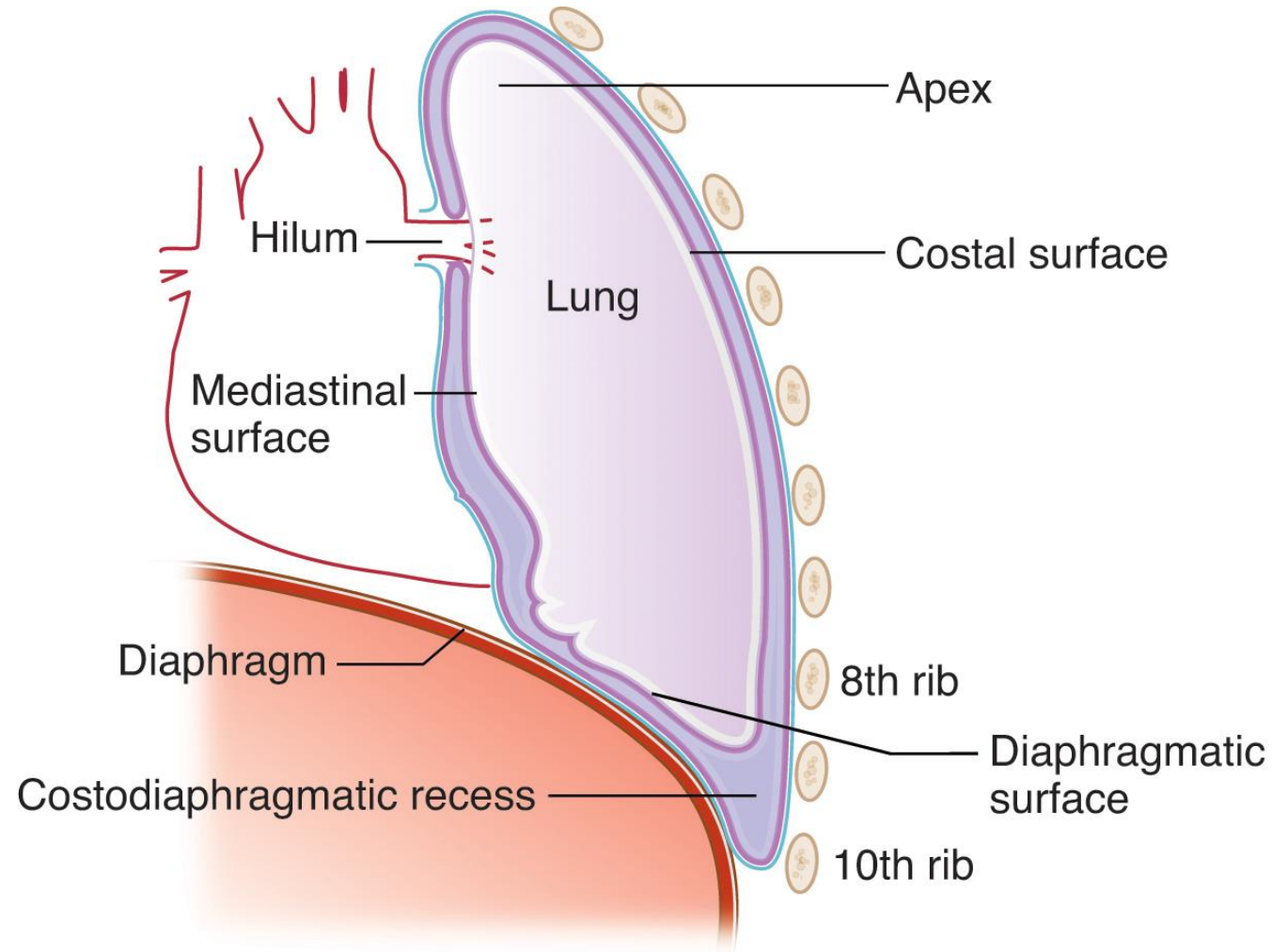
- Air introduced into pleural cavity
- Lung collapses due to loss of **negative pressure** in the pleural cavity
- Injury: gunshot or knife wound, fractured rib, air blisters of the lung break open as in scuba diving or travel to high altitude
- Leads to **pneumothorax**
- **Symptoms:** shortness of breath, painful respiration
- Hemothorax – blood in pleural cavity
- Hydrothorax – excess fluid in pleural cavity



# ANATOMY OF THE LUNGS AND TRACHEOBRONCHIAL TREE

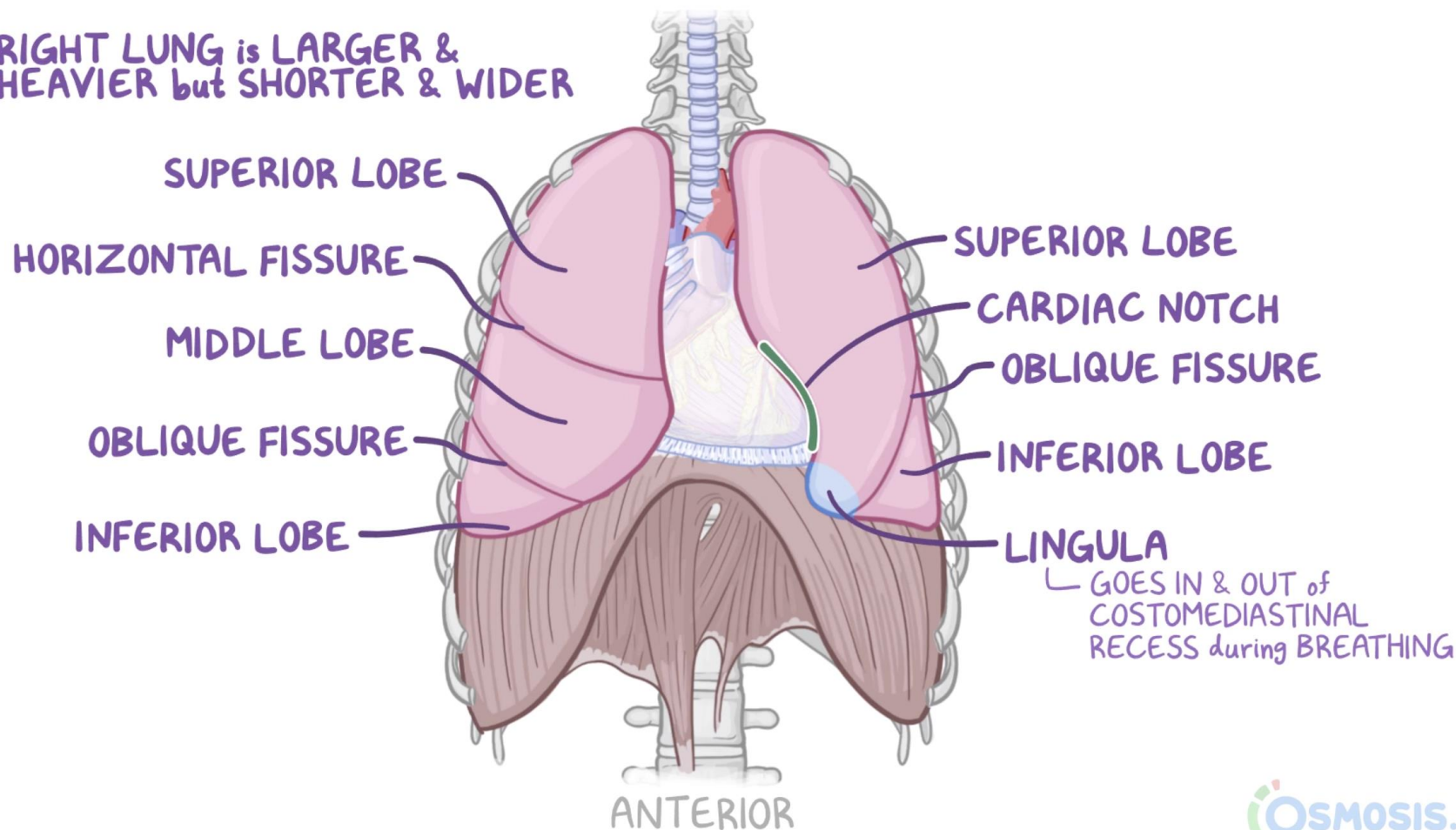


# Surfaces of the Lung

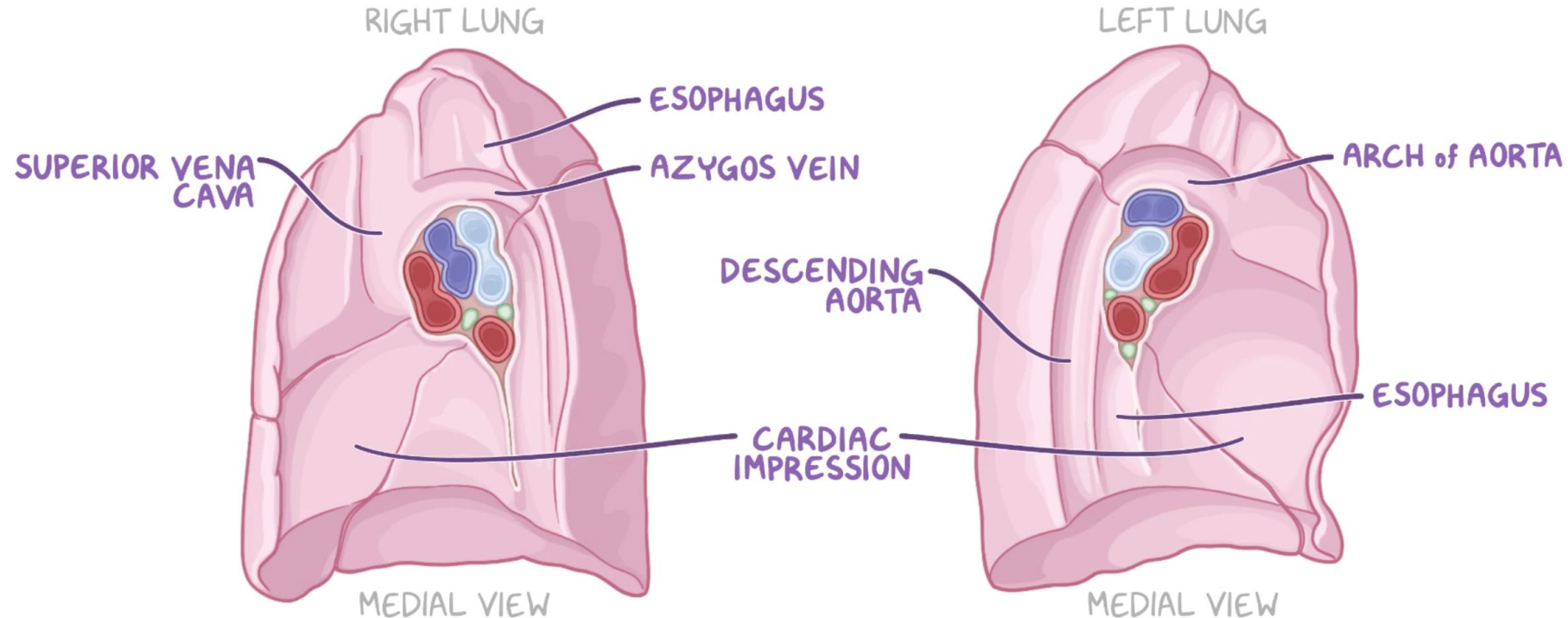


# Right vs. Left Lungs

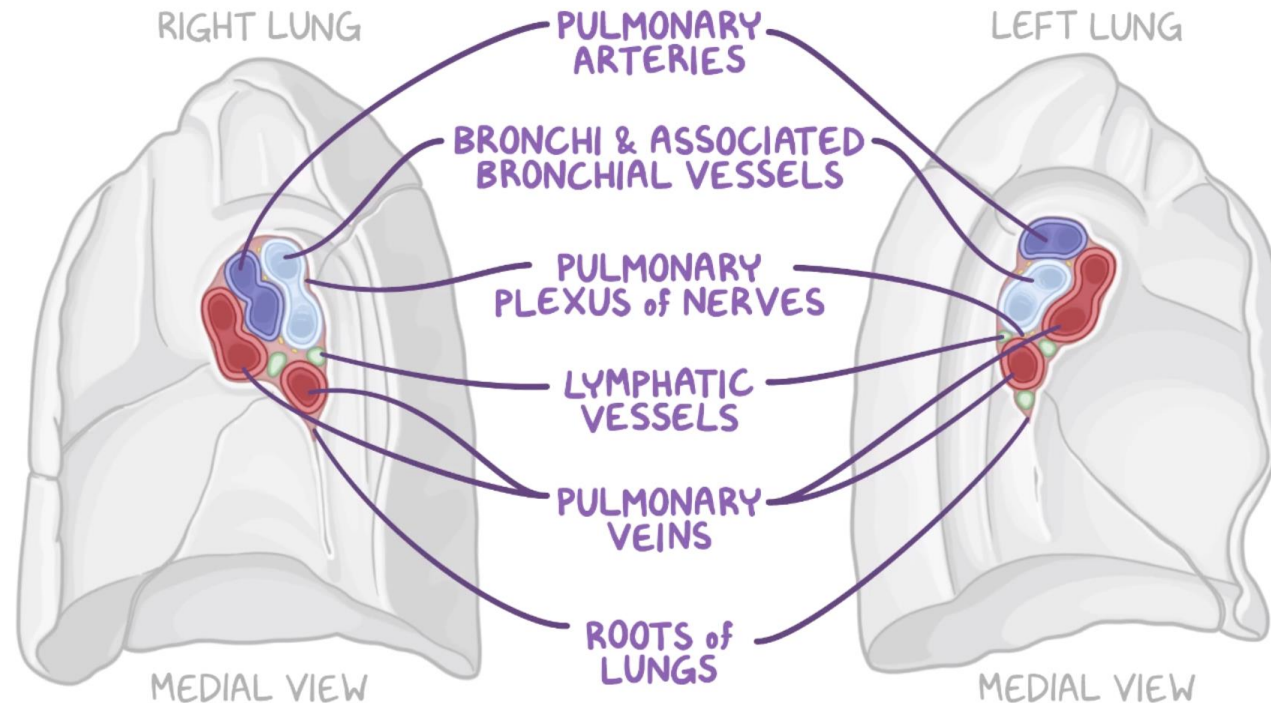
\* RIGHT LUNG is LARGER & HEAVIER but SHORTER & WIDER



# Impressions on the Lung

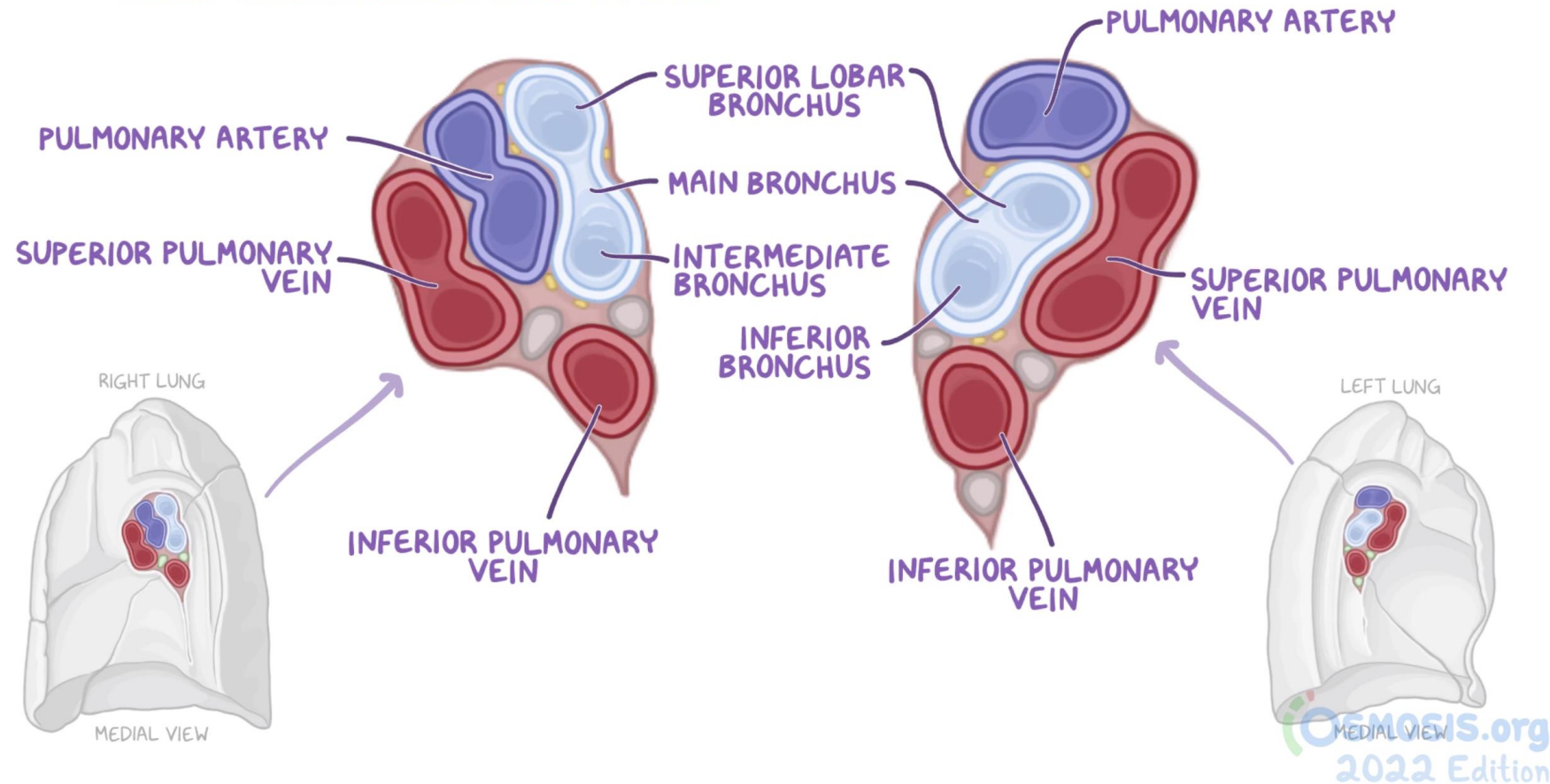


# Roots of the Lungs



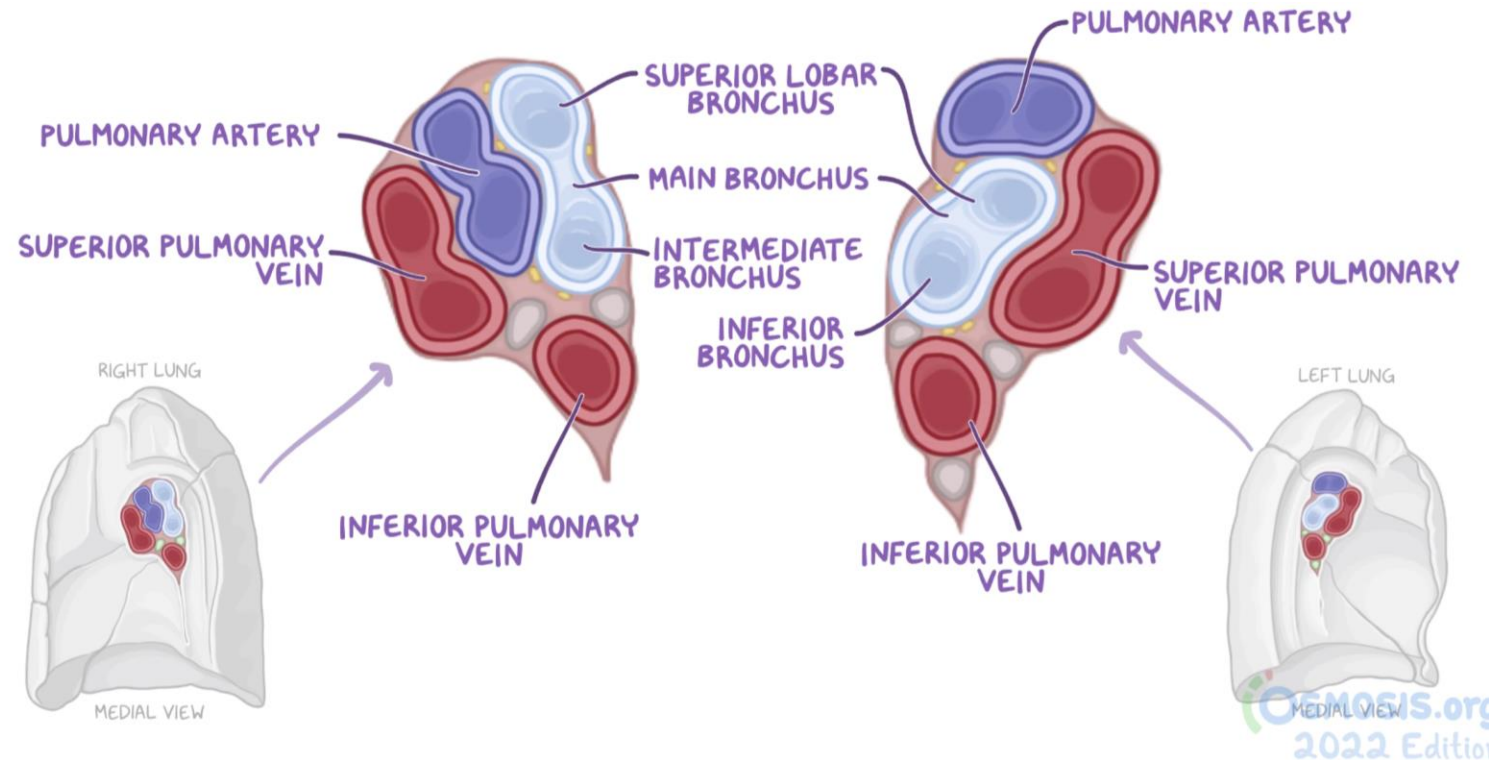
# Right vs. Left Root of Lung

\* IDENTIFY POSITION of MAIN BRONCHUS, PULMONARY ARTERY & VEIN  
 ↳ SLIGHT VARIATIONS from PERSON to PERSON

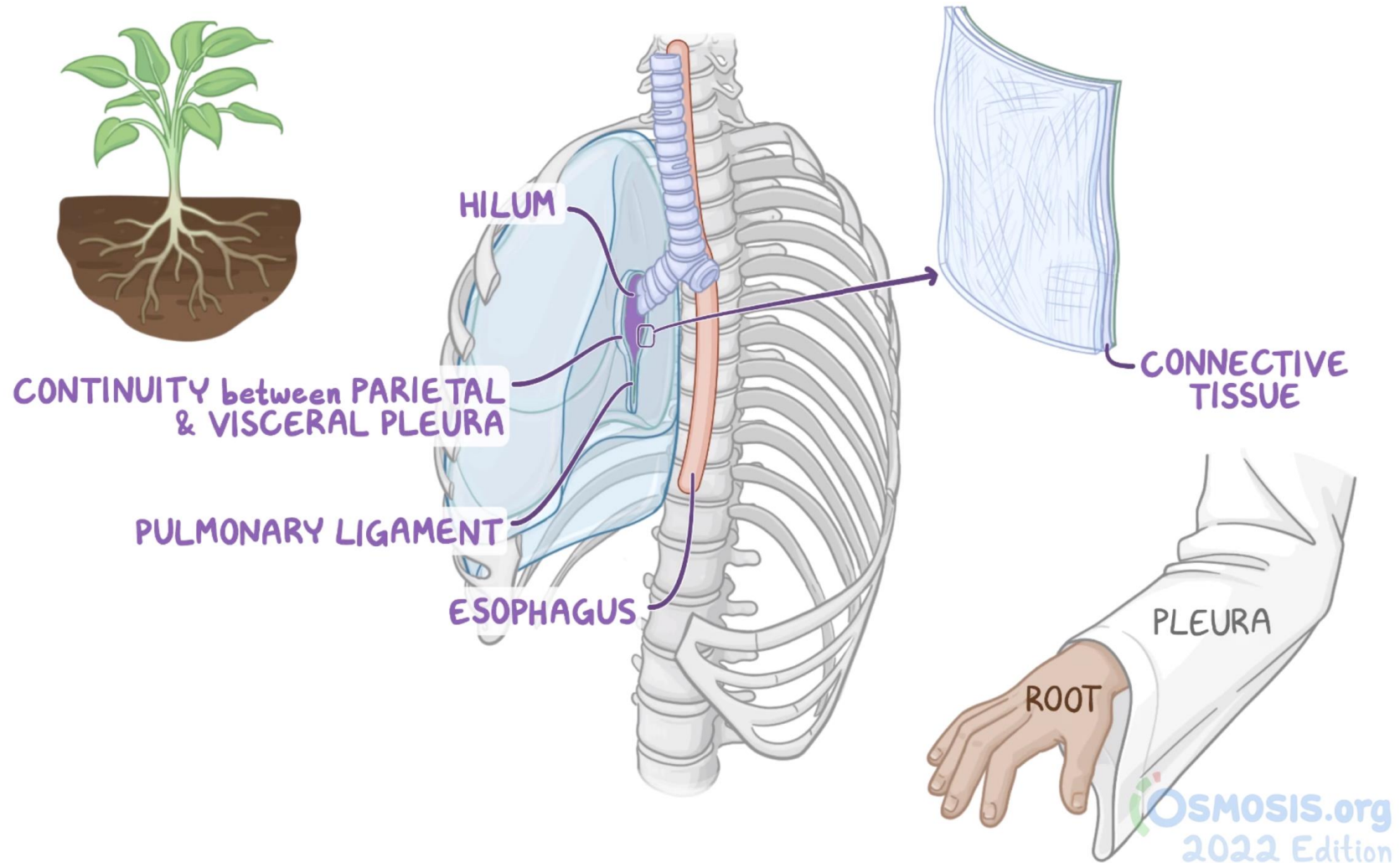


# RALS

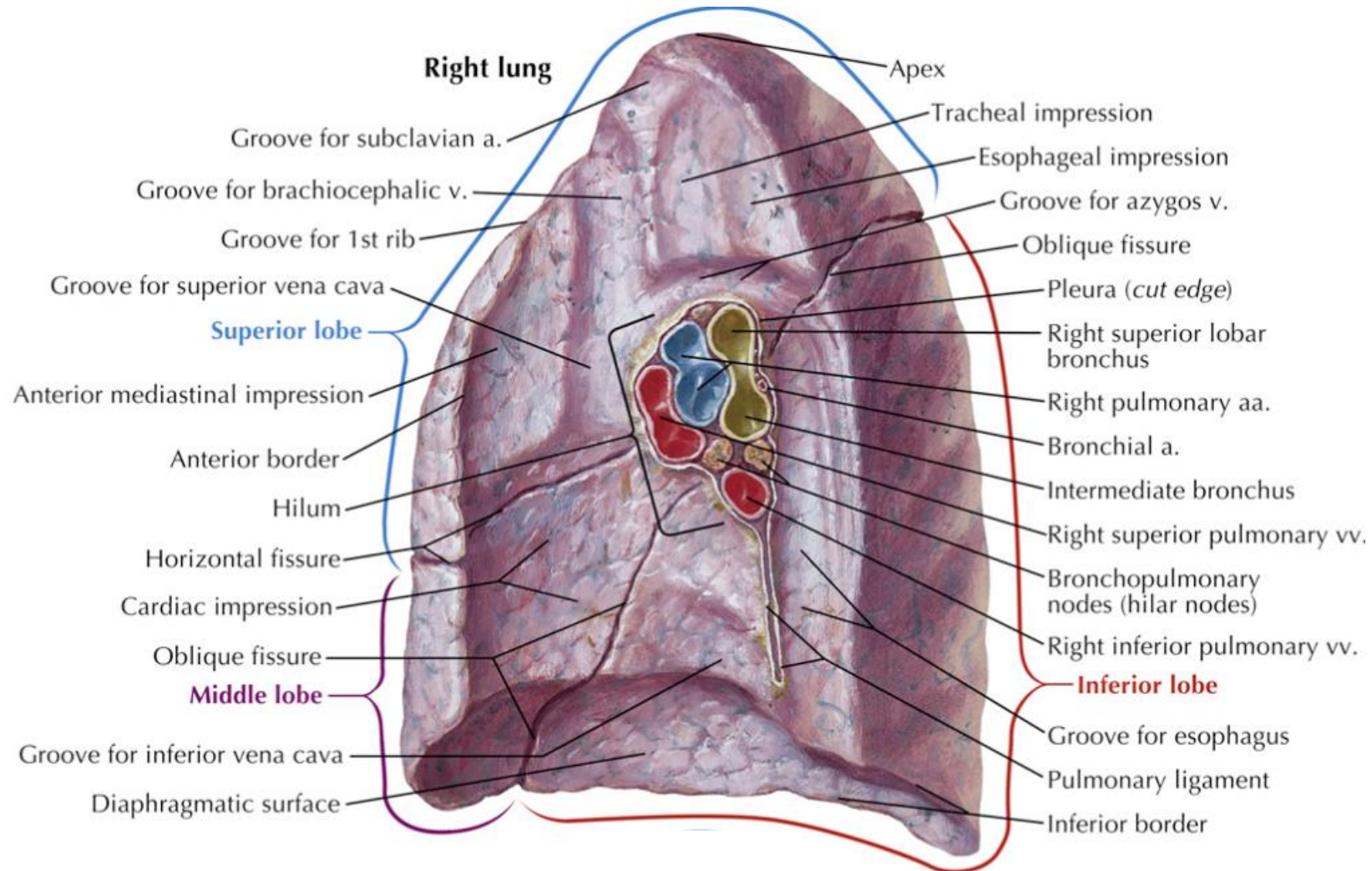
- Right Anterior – on the right lung the pulmonary artery is anterior to the bronchus
- Left Superior – on the left lung the pulmonary artery is superior to the bronchus



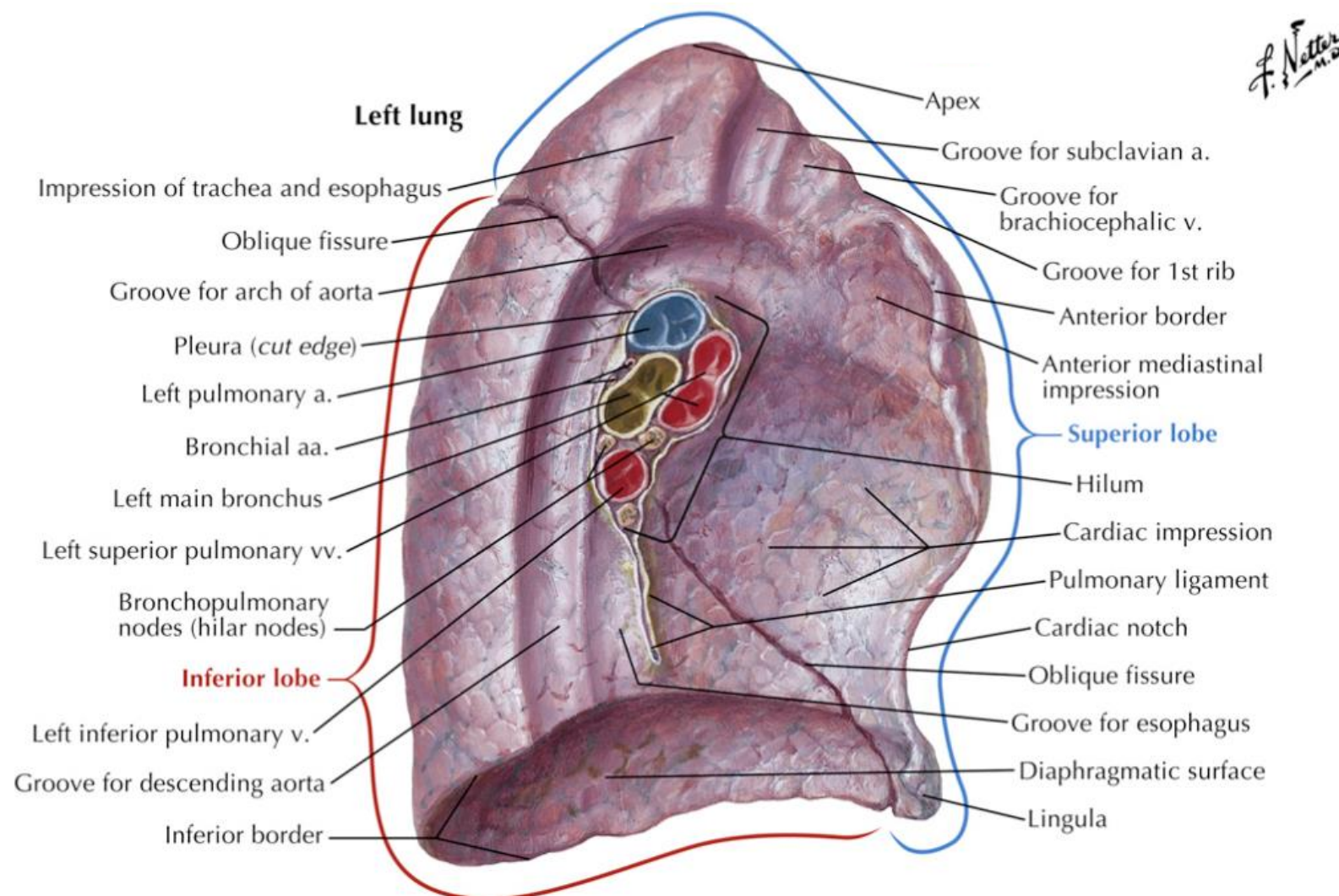
# Hilum of the Lung



# Embalmed Right Lung – Medial View

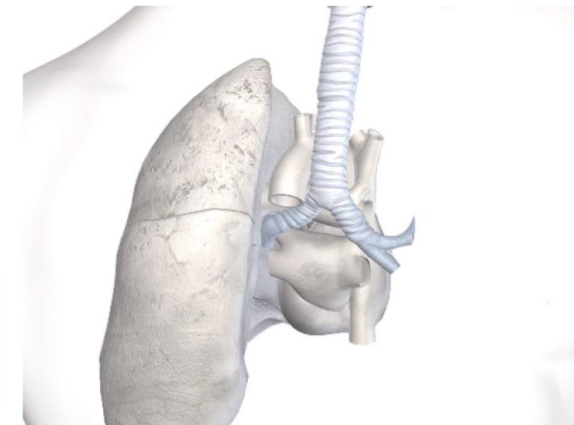
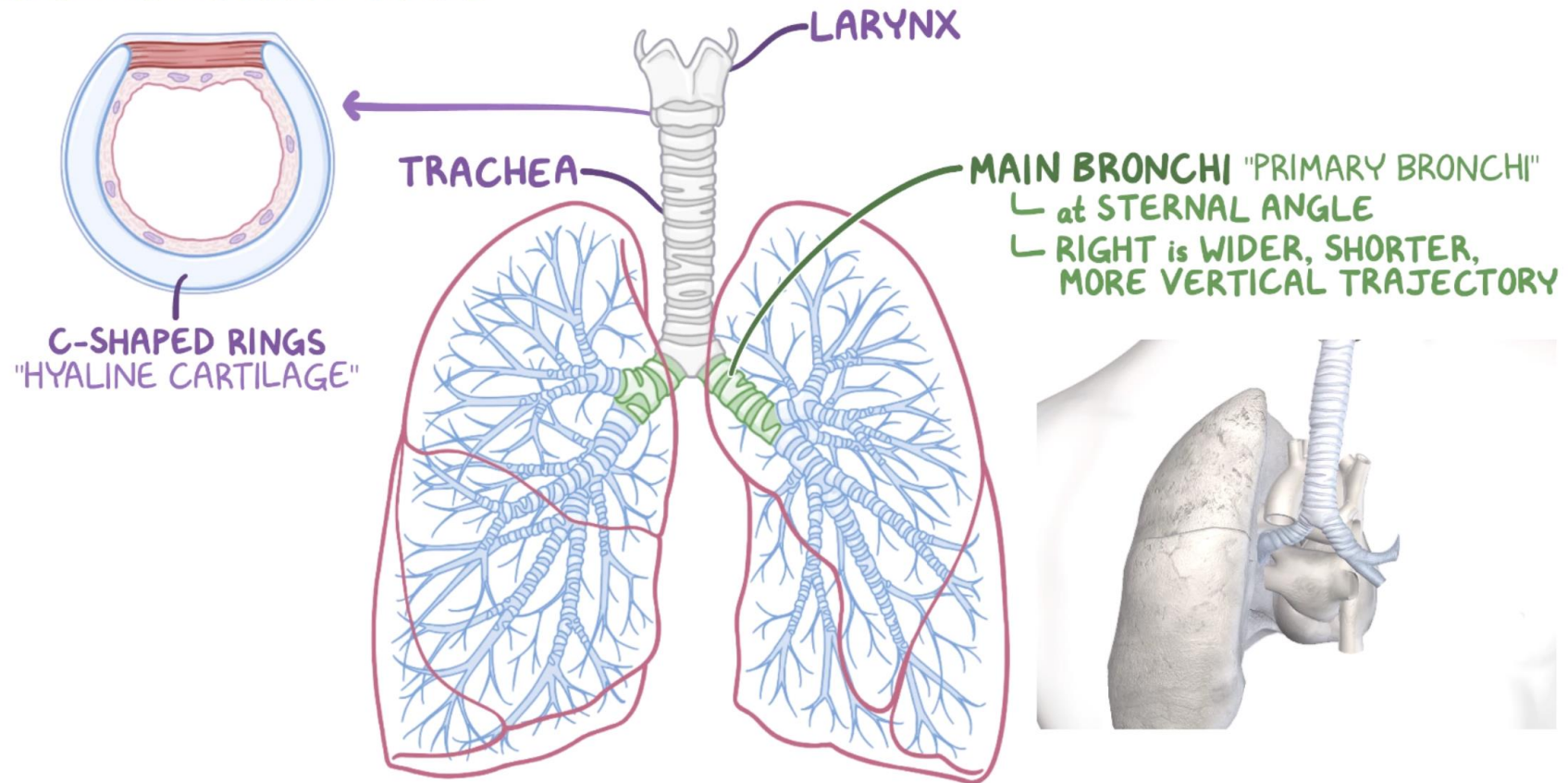


# Embalmed Left Lung – Medial View



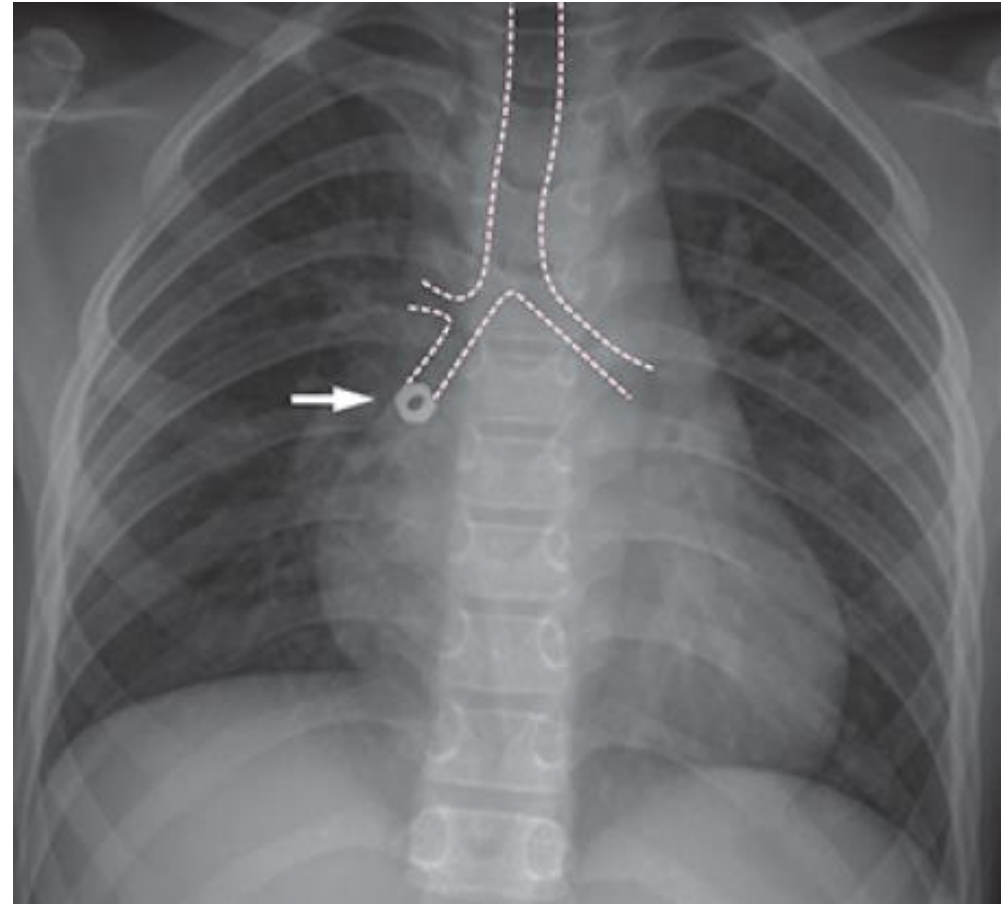
# Primary (Main) Bronchi

## TRACHEOBRONCHIAL TREE



## Clinical Application: Foreign Body Aspiration

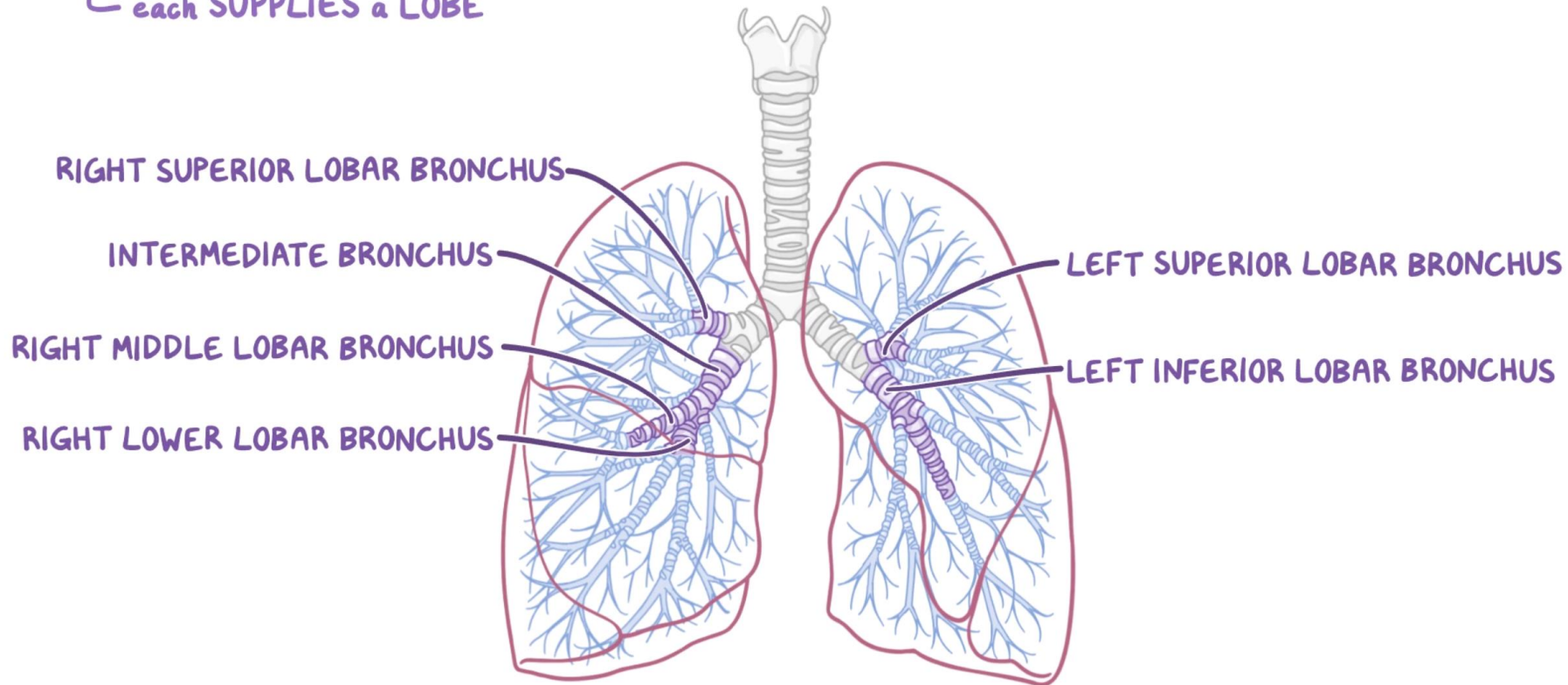
- **Right main bronchus** is wider, shorter and runs more vertically than the left main bronchus
- Right main bronchus more often the site of aspiration
- Common in dental procedures
- Moral of the Story: Use a rubber dam!



# Secondary Lobar Bronchi

## SECONDARY LOBAR BRONCHI

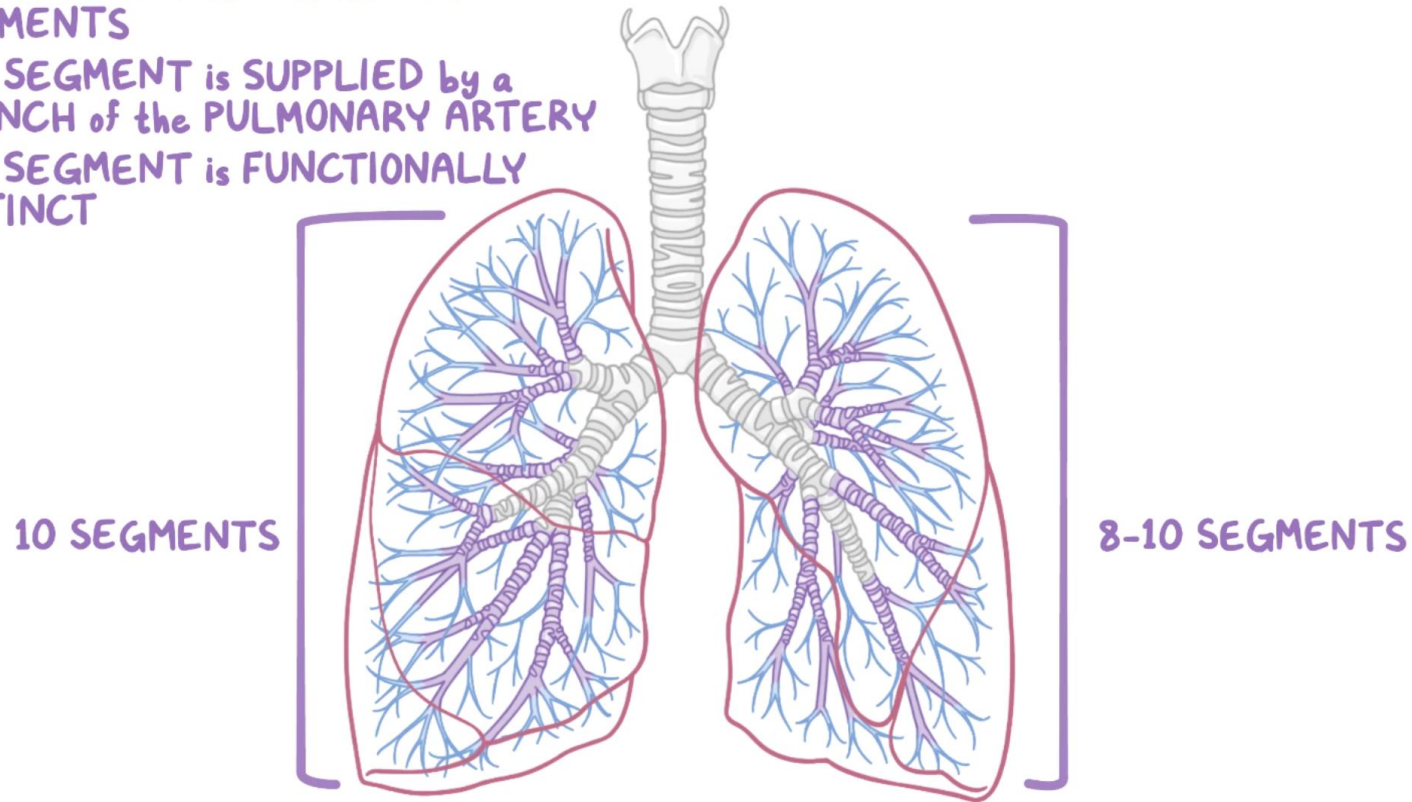
↳ each SUPPLIES a LOBE



# Tertiary Segmental Bronchi

## TERTIARY SEGMENTAL BRONCHI

- └ SUPPLY BRONCHOPULMONARY SEGMENTS
- └ each SEGMENT is SUPPLIED by a BRANCH of the PULMONARY ARTERY
- └ each SEGMENT is FUNCTIONALLY DISTINCT



### Clinical Application:

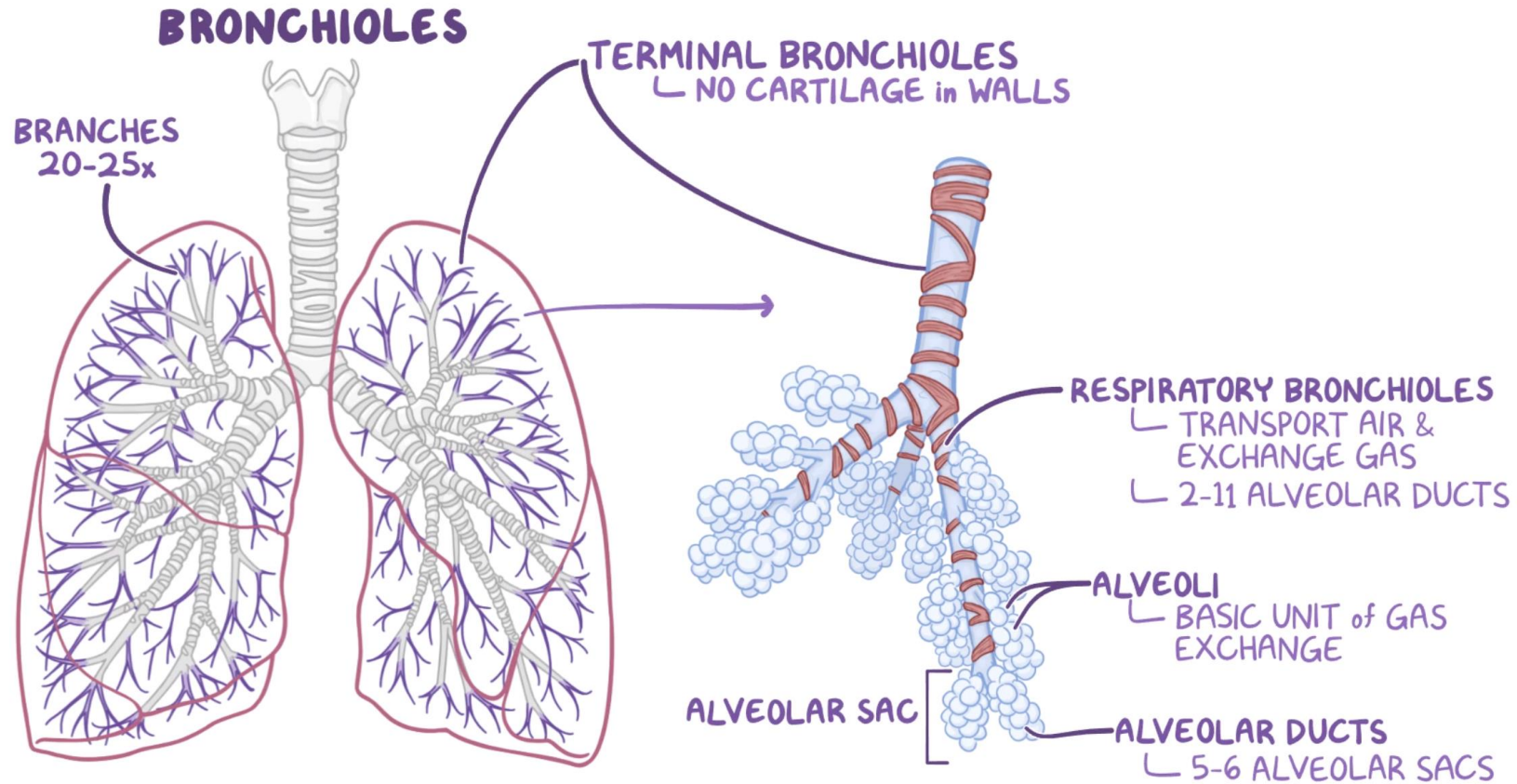
Bronchopulmonary segments are functionally distinct, which is important for surgeons who resect the lungs, as in the treatment of lung cancer.

If tumors or abscesses are located in a single segment, they can be surgically removed (**segmentectomy**).

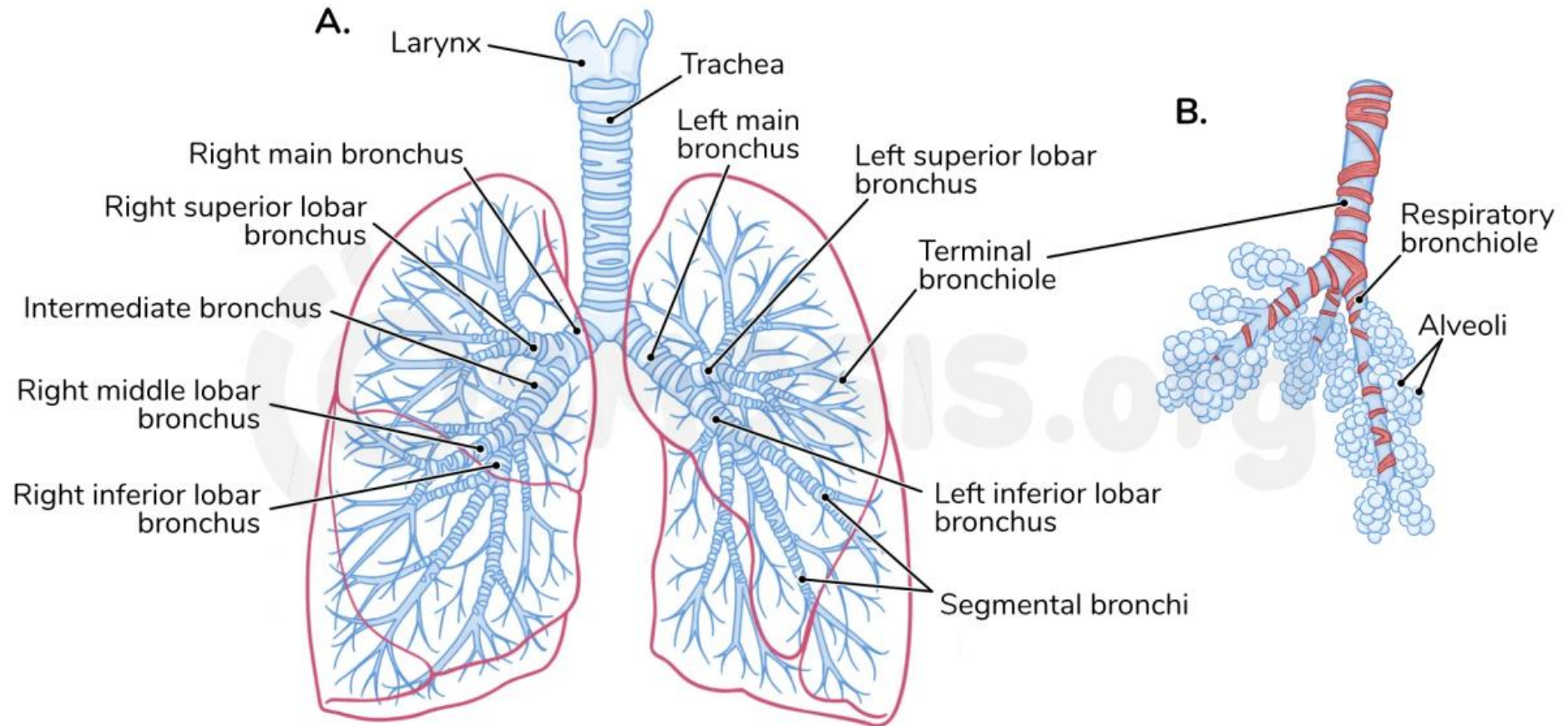
**Lobectomy** – entire lobe removed

**Pneumonectomy** – entire lung removed

# Conducting Bronchioles, Terminal Bronchioles, Respiratory Bronchioles, Alveoli

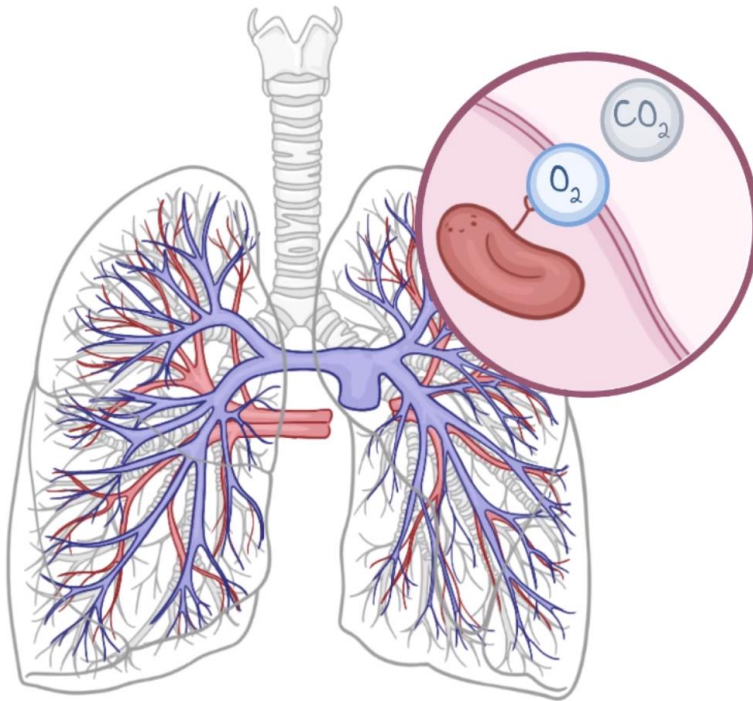


# Branching of Tracheobronchial Tree



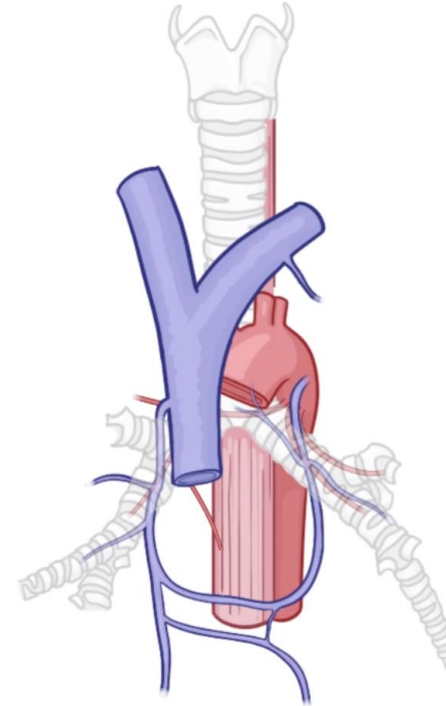
# Functional vs. Nutritive Circulation

## FUNCTIONAL CIRCULATION



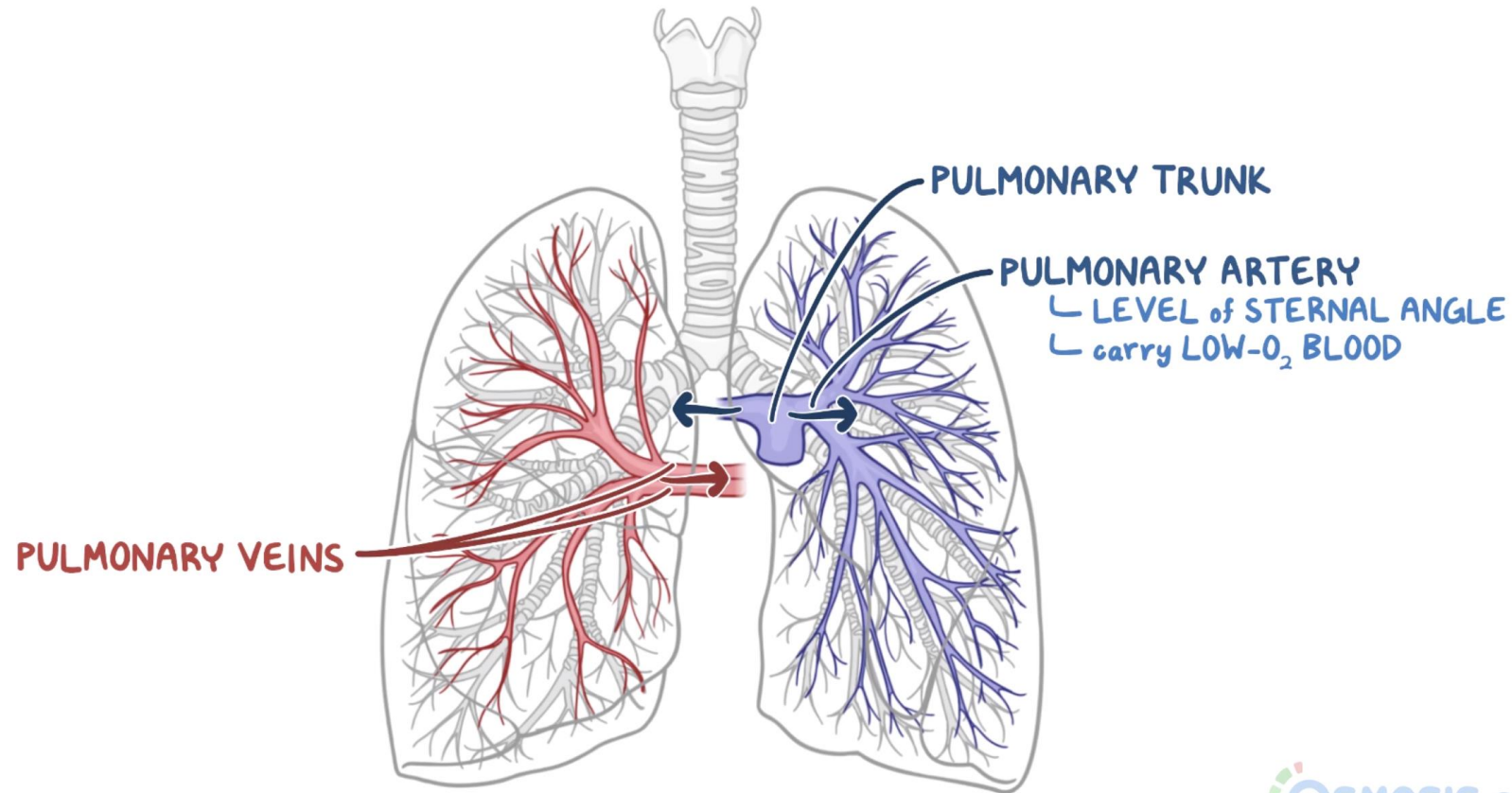
- \* LOW- $O_2$  BLOOD COMES from HEART
- \* EXCHANGE  $CO_2$  &  $O_2$

## NUTRITIVE CIRCULATION



- \* OXYGENATED BLOOD NOURISHES LUNGS

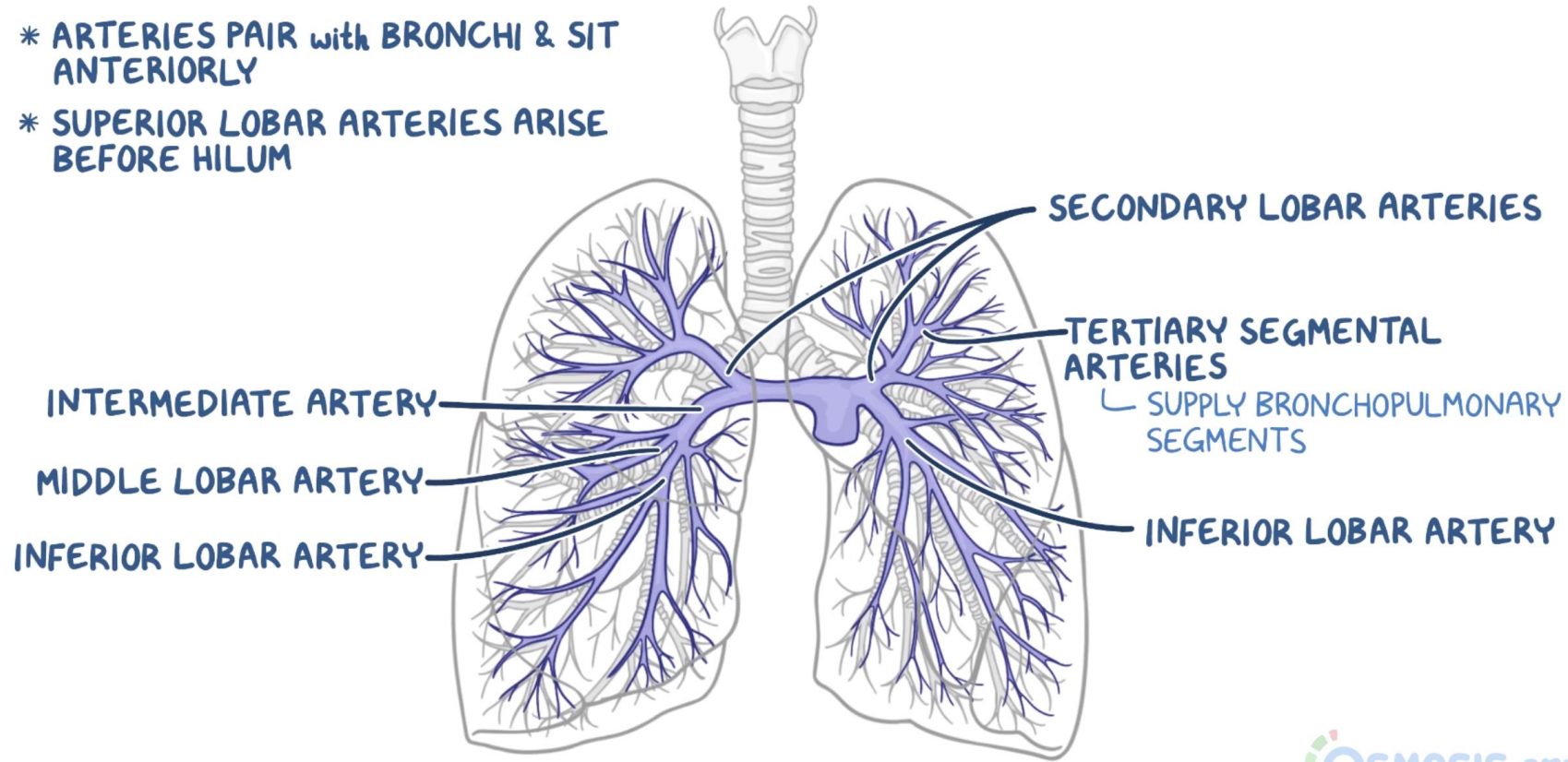
# Functional Circulation



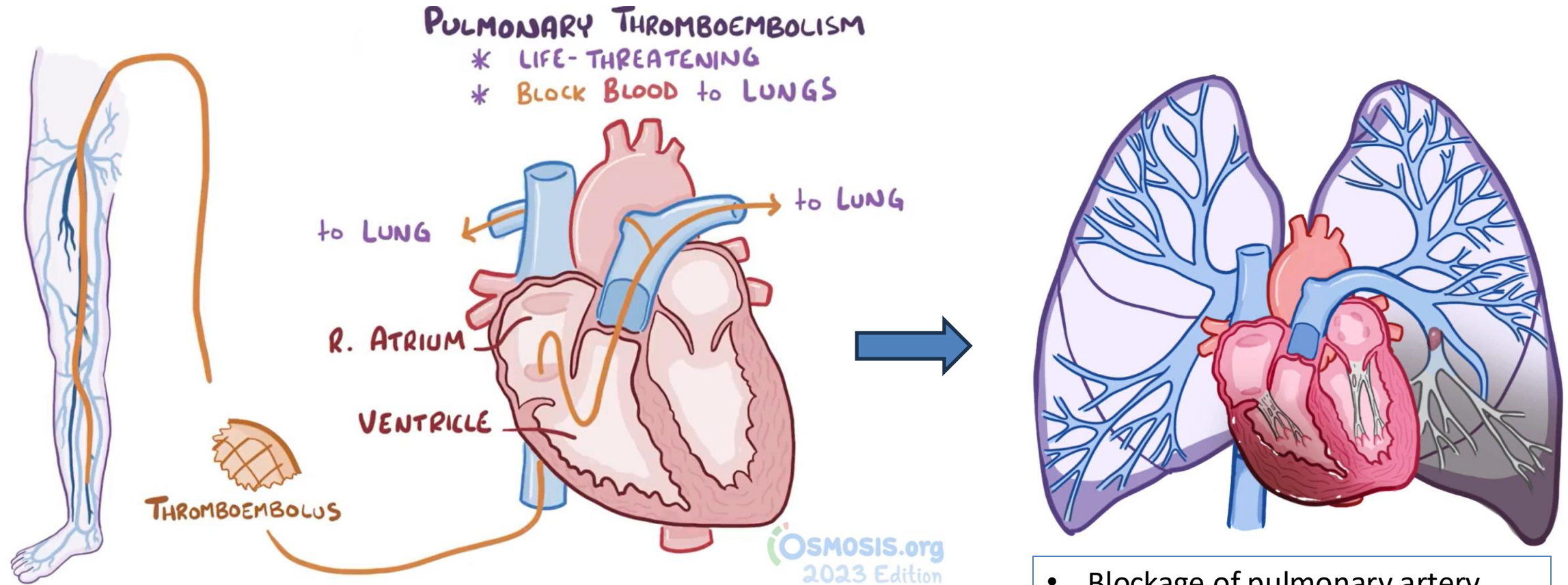
# Pulmonary Arteries Pair with Bronchi

## PULMONARY ARTERIES

- \* ARTERIES PAIR with BRONCHI & SIT ANTERIORLY
- \* SUPERIOR LOBAR ARTERIES ARISE BEFORE HILUM



# Clinical Application: Pulmonary Embolism



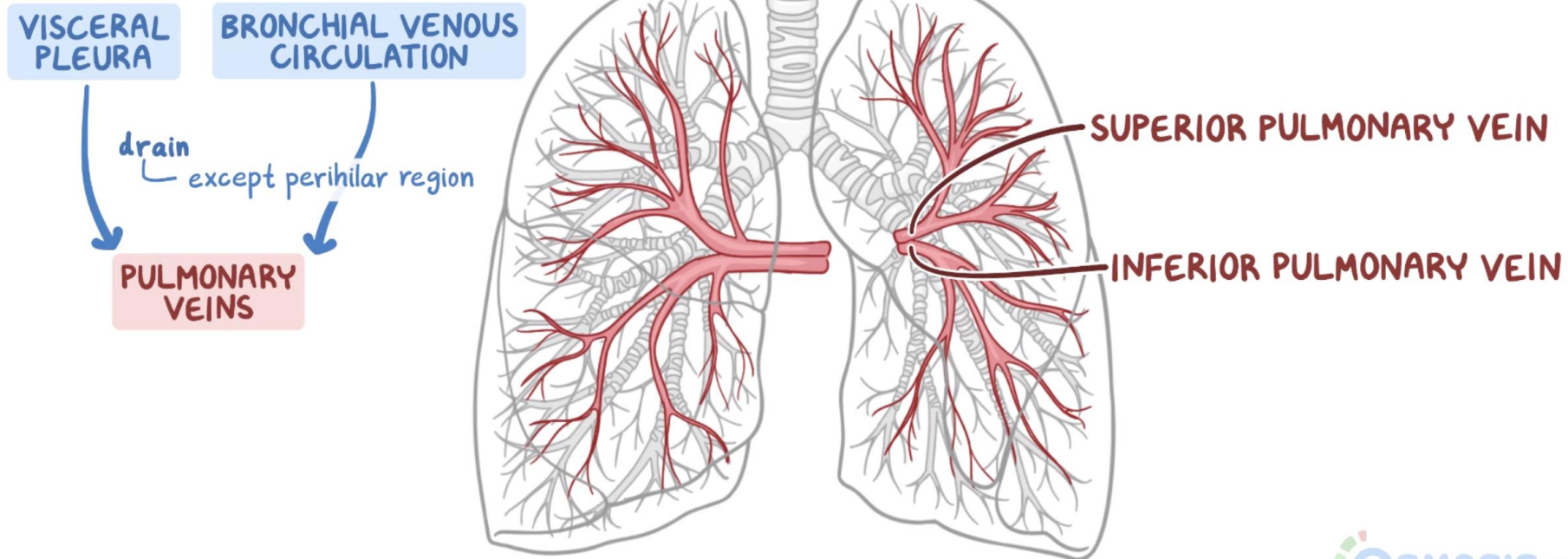
- Blockage of pulmonary artery
- Acute respiratory distress
- Right heart dilation and failure

# Pulmonary Veins Follow Their Own Path

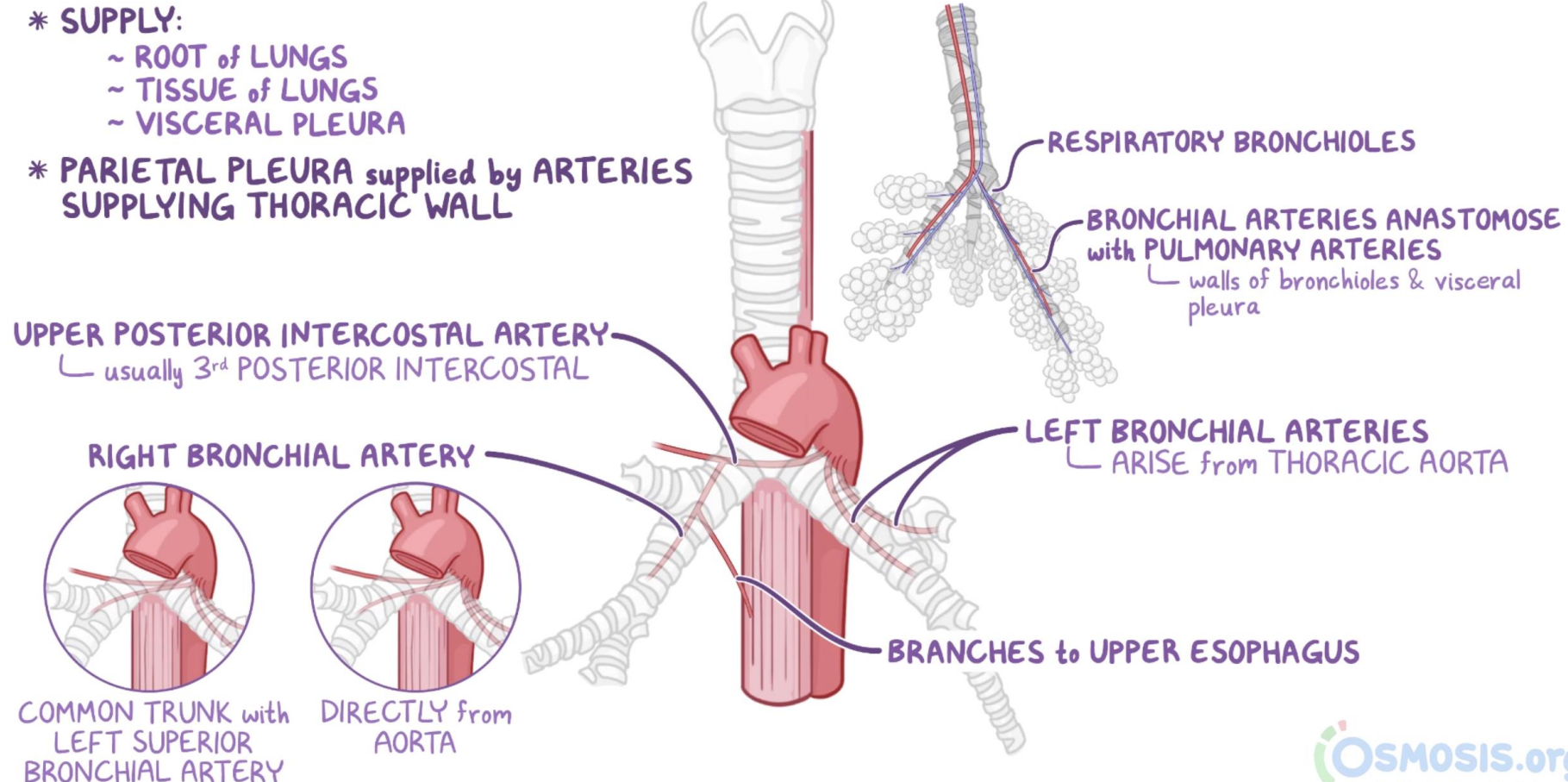
## PULMONARY VEINS

- \* CARRY O<sub>2</sub>-RICH BLOOD to LEFT ATRIUM
- \* DON'T BUDDY UP with BRONCHI/ARTERIES
- \* RECEIVE BLOOD from ADJACENT BRONCHOPULMONARY SEGMENTS

- \* SMALL VOLUME of ↓ O<sub>2</sub> enters LARGE VOLUME of OXYGENATED BLOOD



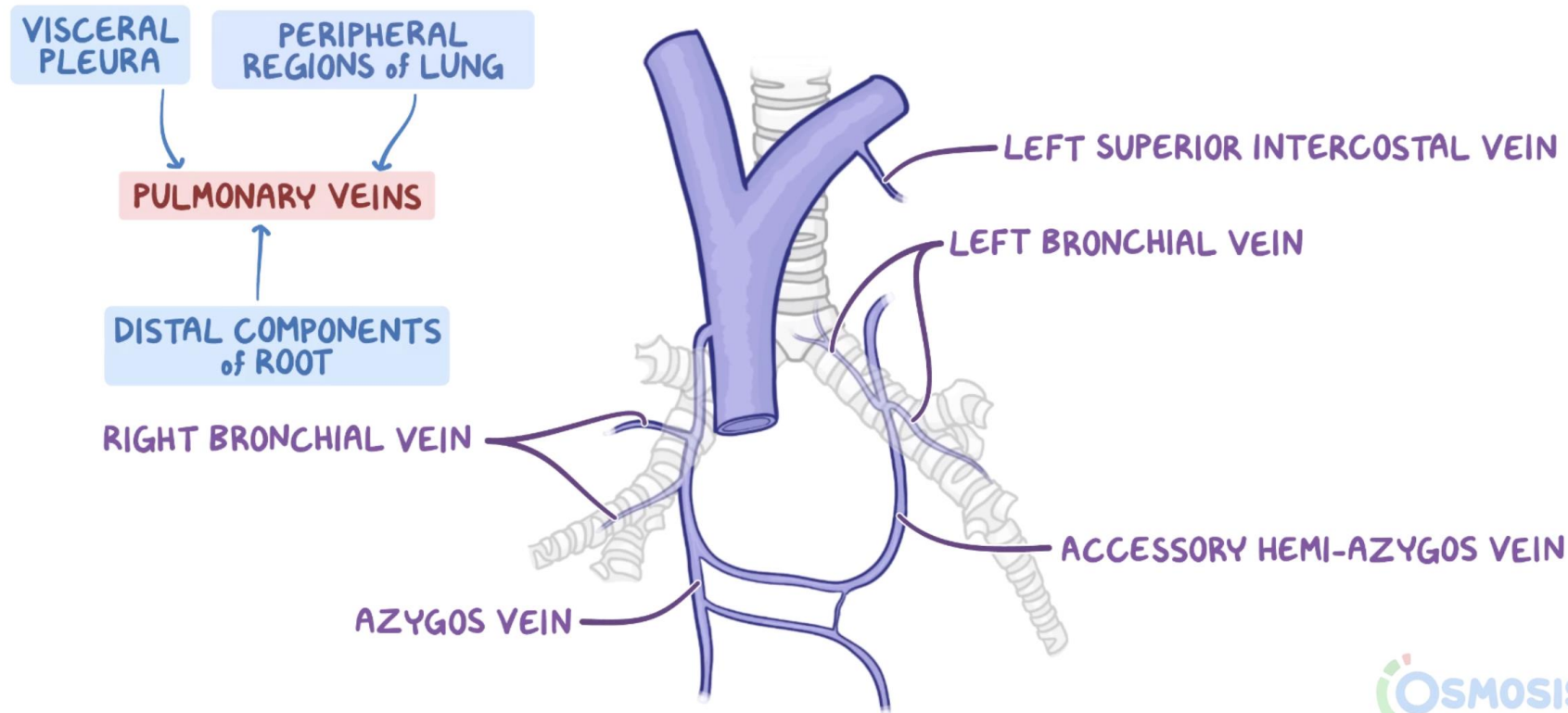
# Nutritive Circulation: Arterial Supply



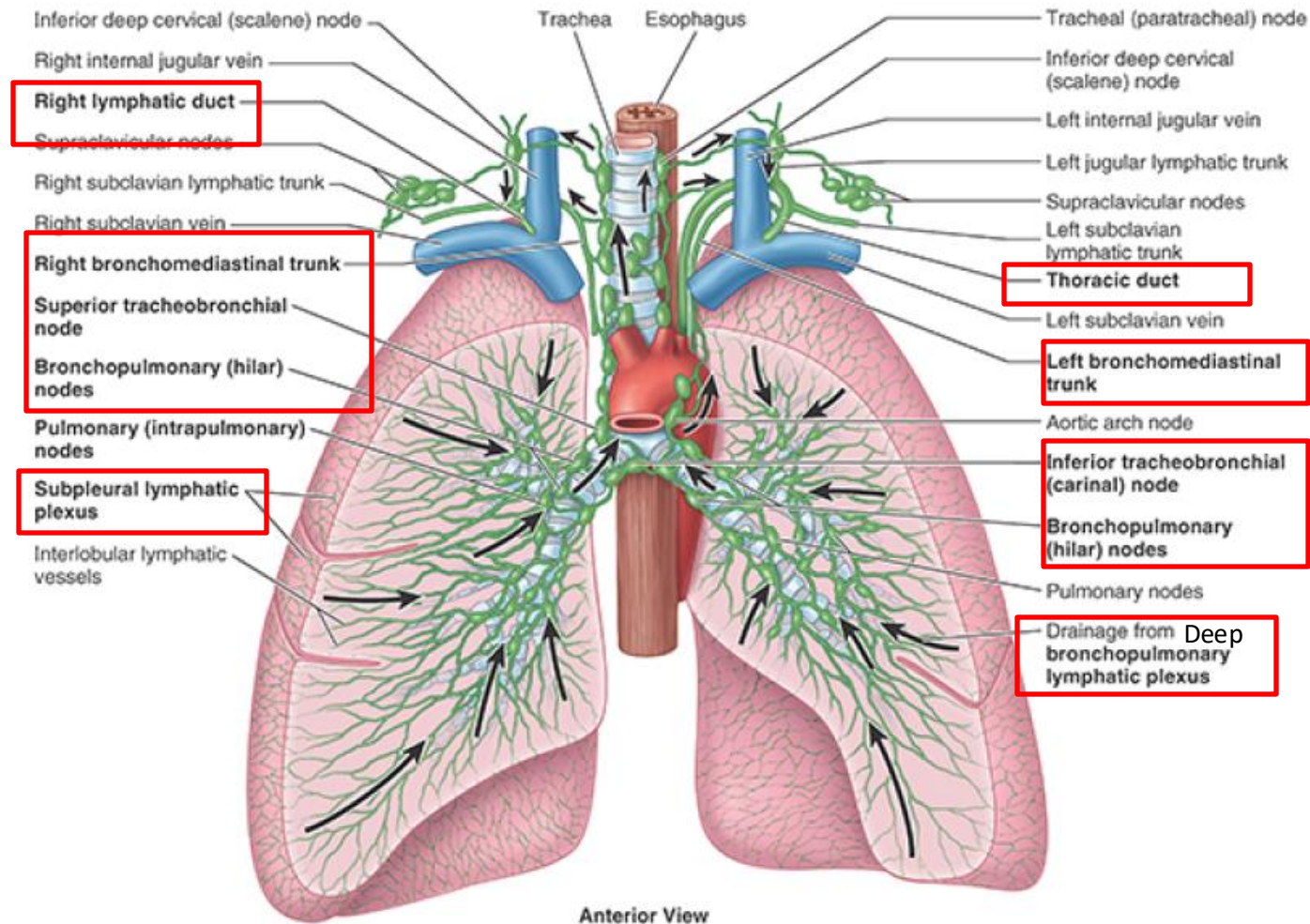
# Nutritive Circulation: Venous Drainage

## NUTRITIVE CIRCULATION

- \* DRAIN BLOOD NEAR PROXIMAL PART of ROOT
- \* DRAIN BLOOD from ESOPHAGEAL VEINS

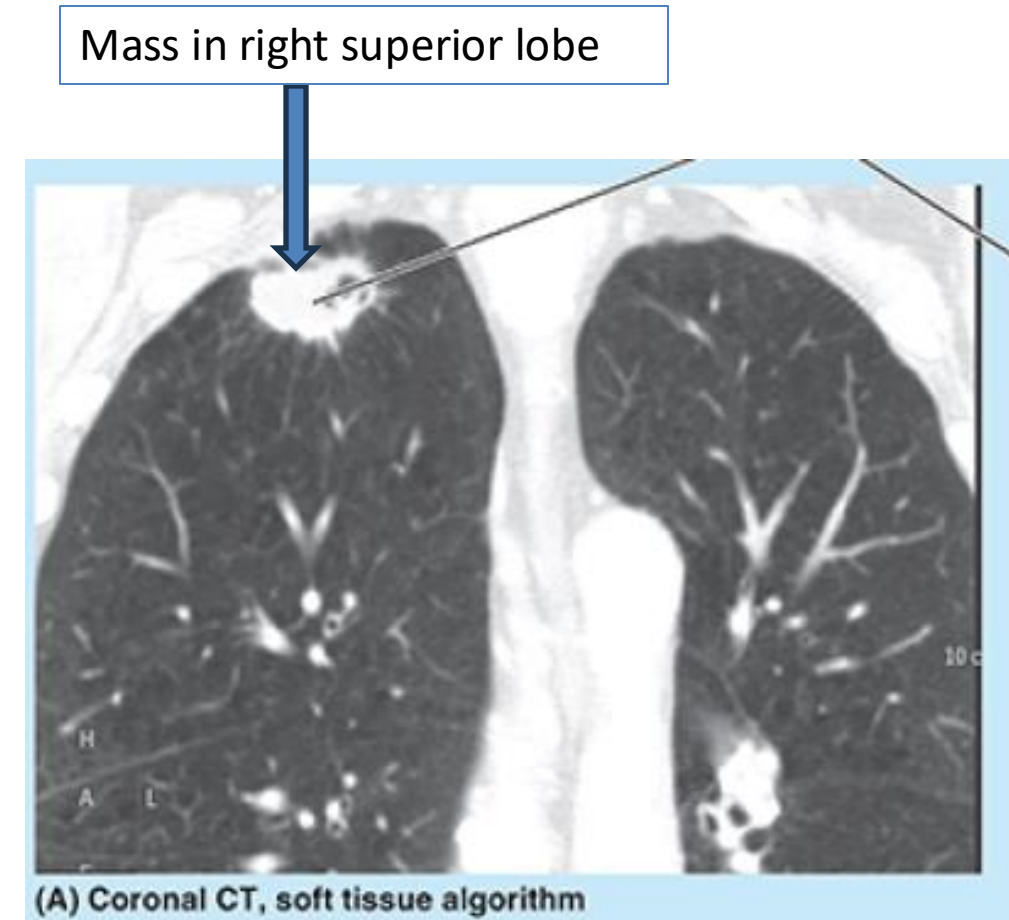


# Lymphatic Drainage of Lungs



# Clinical Application: Lung Cancer

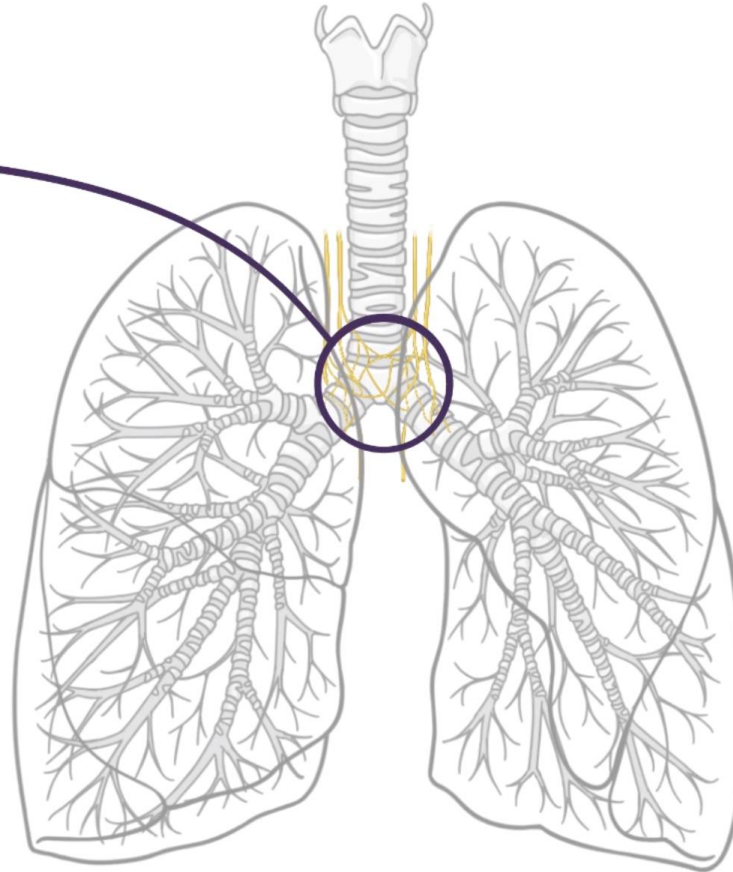
- Primary cause: cigarette smoking
- Most common primary site: mucosa of primary bronchi
- **Lymphogenous metastasis** to bronchopulmonary lymph nodes and to other thoracic lymph nodes
- **Hematogenous metastasis** via blood stream commonly to the brain, bones, lungs and suprarenal glands
- **Detection:** supraclavicular lymph node enlargement; persistent, productive cough with blood (hemoptysis); cancer cells in sputum (saliva)
- Tumors can grow and cause injury to other mediastinal structures:
  - Phrenic nerve injury – paralysis of one half of diaphragm
  - Recurrent laryngeal nerve injury – hoarseness, aphonia
  - Superior vena cava – venous congestion, headaches, dizziness, redness, edema



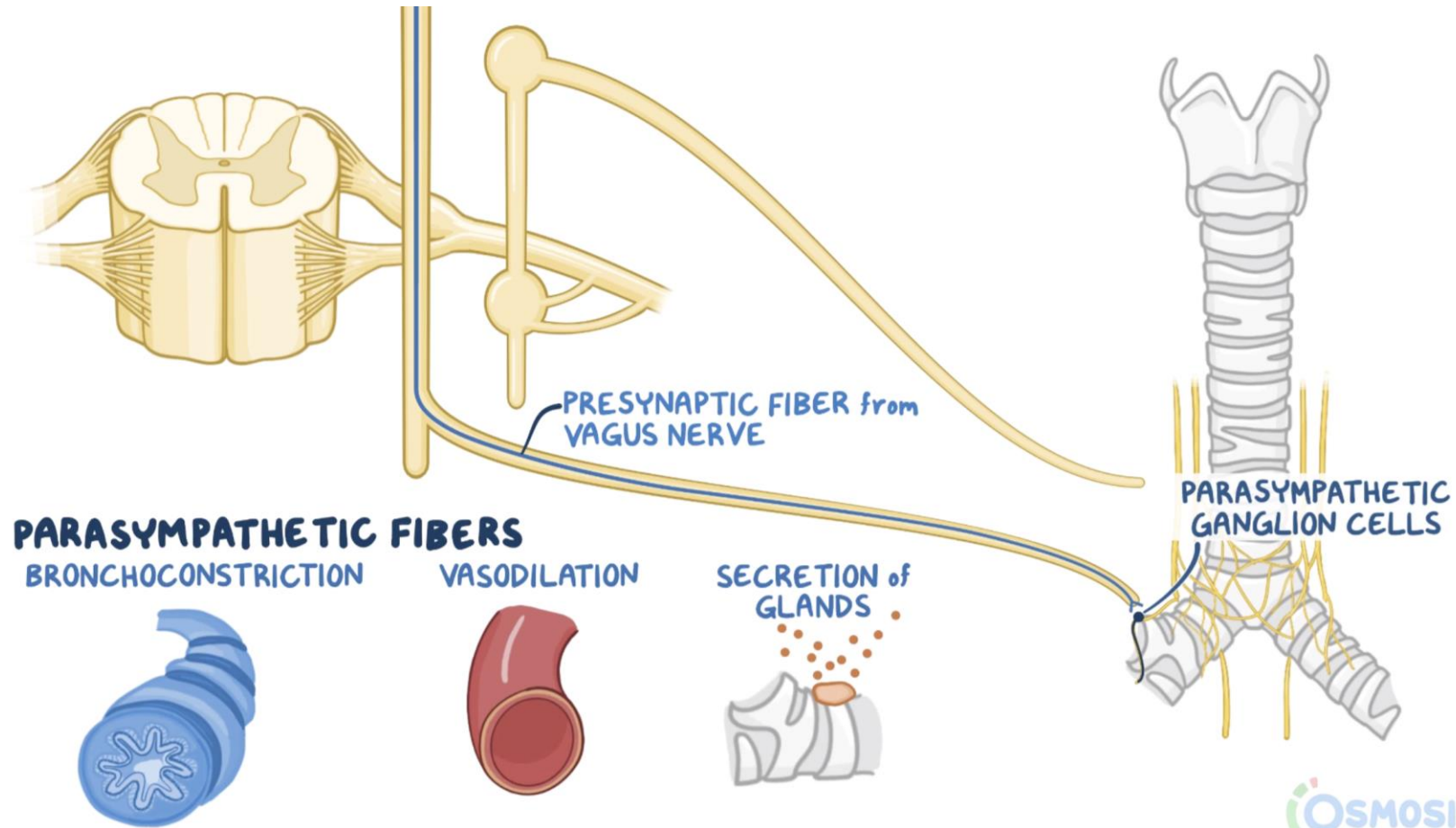
# Pulmonary Plexus

## PULMONARY PLEXUS

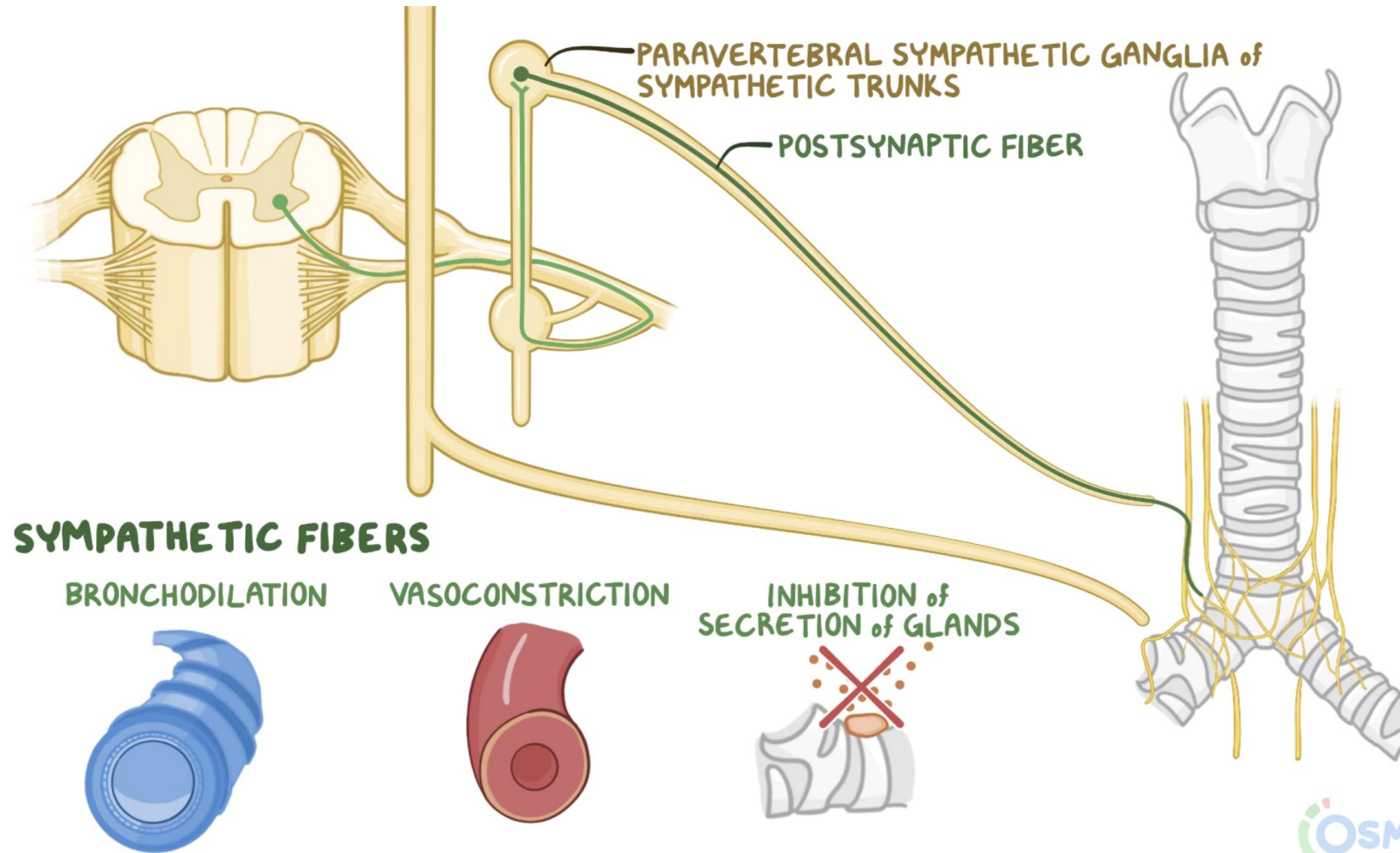
- \* ANTERIOR & POSTERIOR to ROOT of LUNGS
- \* PARASYMPATHETIC FIBERS
- \* SYMPATHETIC FIBERS
- \* VISCERAL AFFERENT FIBERS



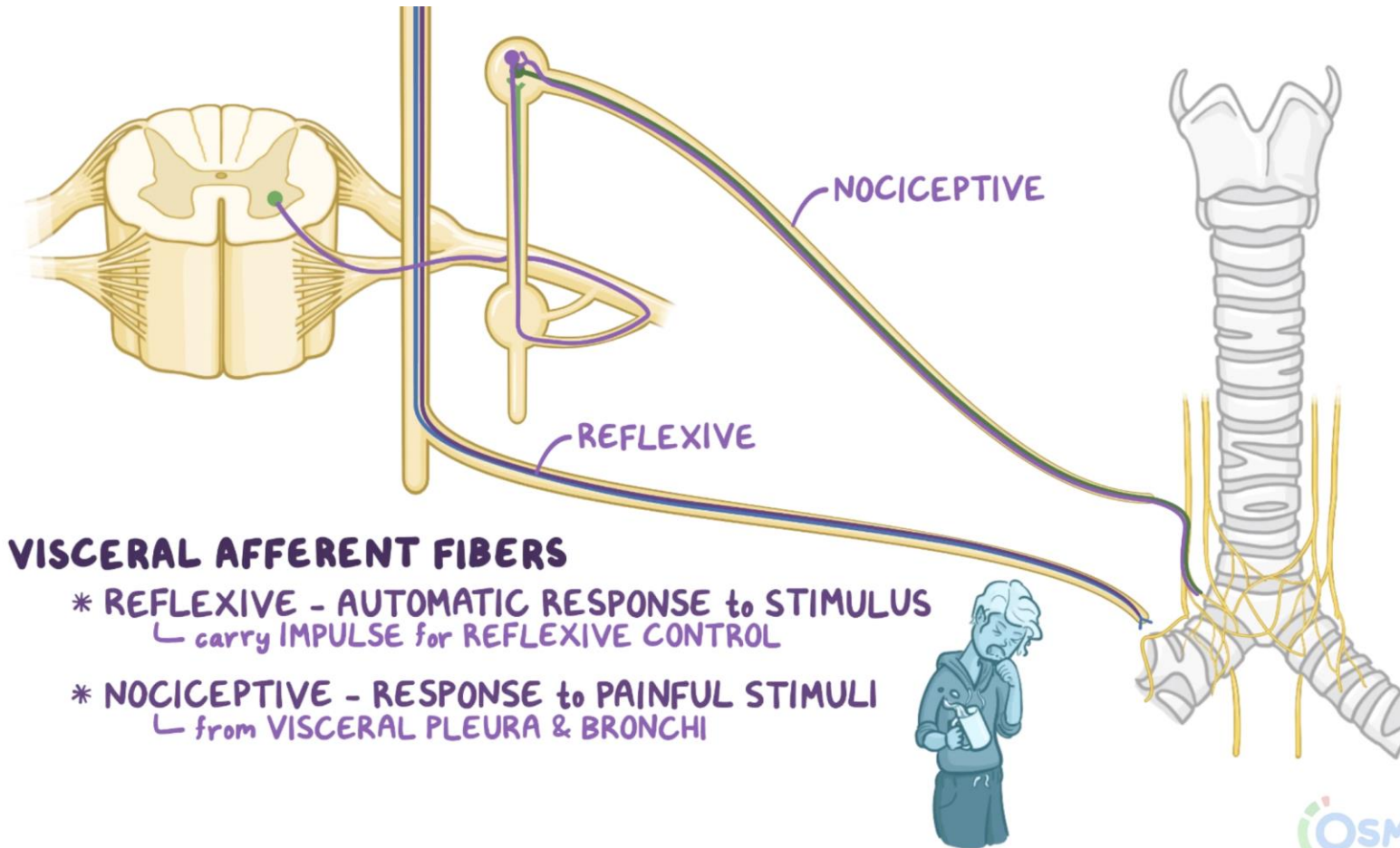
# Parasympathetic Innervation



# Sympathetic Innervation



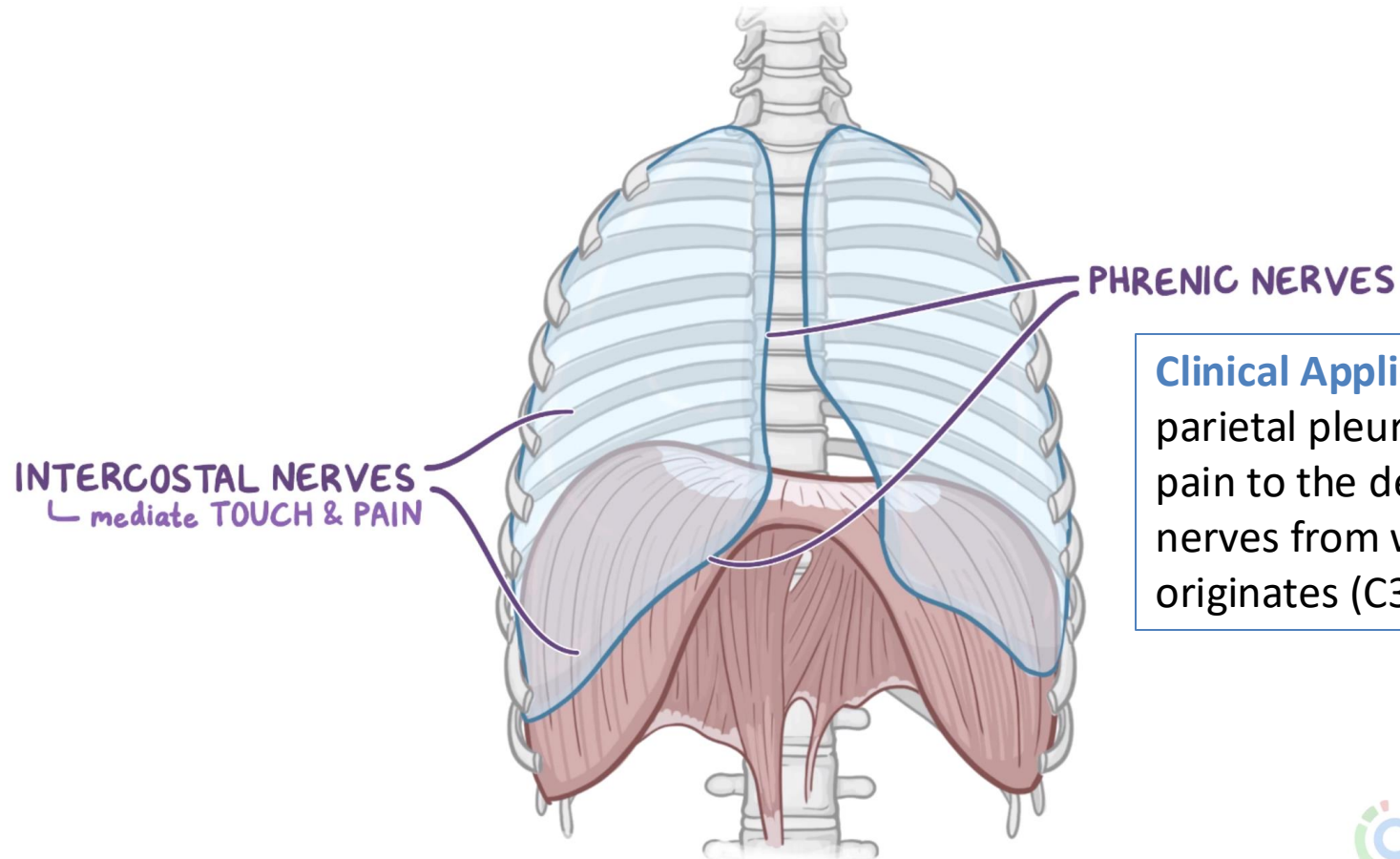
# Visceral Afferent Fibers



## VISCERAL AFFERENT FIBERS

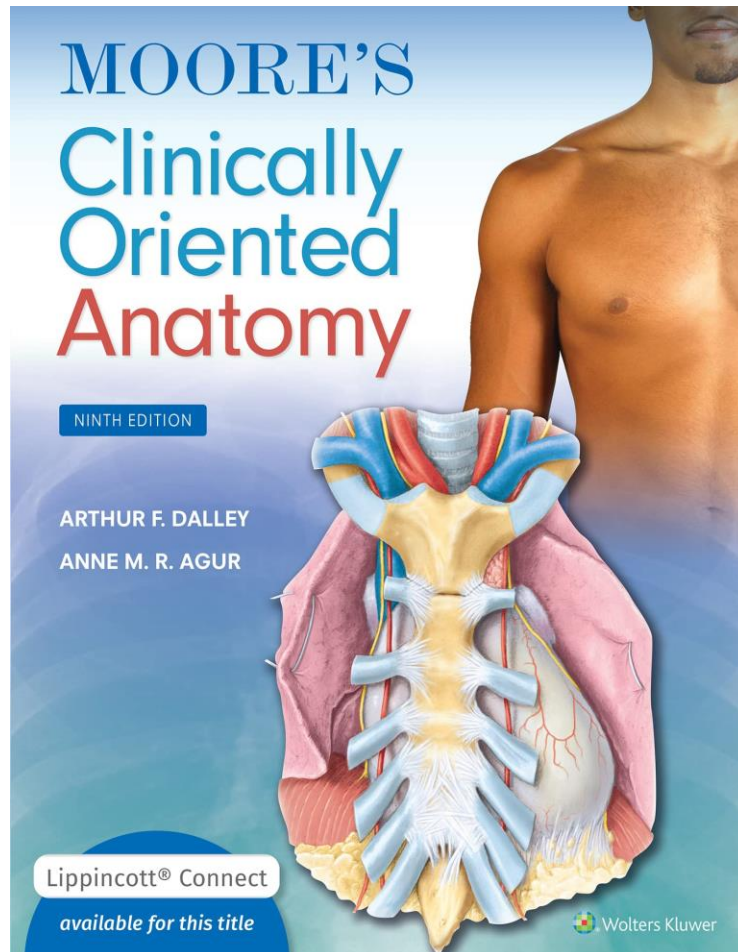
- \* REFLEXIVE - AUTOMATIC RESPONSE to STIMULUS  
↳ carry IMPULSE for REFLEXIVE CONTROL
- \* NOCICEPTIVE - RESPONSE to PAINFUL STIMULI  
↳ from VISCERAL PLEURA & BRONCHI

# Innervation to Parietal Pleura



**Clinical Application:** Irritation of the parietal pleura can also cause referred pain to the dermatomes of the spinal nerves from which the phrenic nerve originates (C3-5)

# Reading: Moore's Clinically Oriented Anatomy



## CHAPTER 4

- Pages 332-354
- Pages 362-390

# Sources

- American Dental Association
- American Heart Association
- Dalley, Arthur F.; Agur, Anne M.R. *Moore's Clinically Oriented Anatomy*. Philadelphia, PA: Wolters Kluwer, 2023.
- Ellis, Harold; Mahadeva, Vishy. *Clinical Anatomy: Applied Anatomy for Students and Junior Doctors*. Hoboken, NJ: John Wiley & Sons Ltd, 2019.
- Kaplan
- Netter, Frank H. *Atlas of Human Anatomy*. Philadelphia, PA: Elsevier, 2019.
- Osmosis.org

THANK YOU!

Questions, comments or feedback?

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